

Process Costing

24 problems · workbook pages 8 to 16

What this module actually is

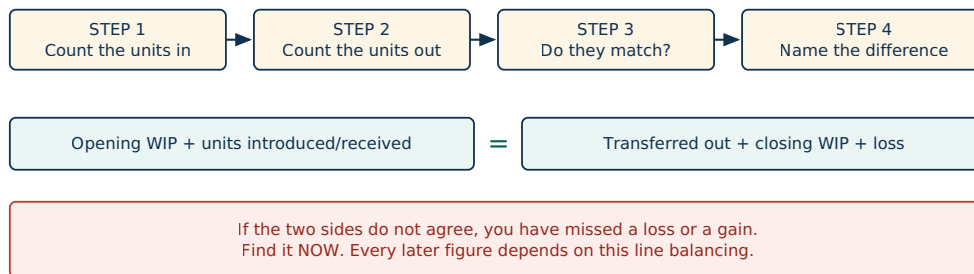
Process costing is used where a product is made in a **continuous chain of stages** — sugar, chemicals, oil, paper, cement. You cannot point at one unit and say “this one cost ₹40”, because every unit goes through the same pipe. So we do the only thing possible: **total the cost of a process and divide by the units it produced**. That single sentence is the whole module.

Everything else in these 24 problems is one of four complications layered on top of that idea:

Complication	What changes	Problems
Nothing — plain chain	Total each process, divide by units, carry the total forward to the next process.	1-4
Some units are lost	Part of the input disappears (evaporation, spoilage). Expected loss is <i>normal</i> and its cost is absorbed by the good units. Anything worse is <i>abnormal loss</i> ; anything better is <i>abnormal gain</i> . Both are taken out of the process at the good-unit rate.	5-9
Two or more products emerge	One process throws out several saleable products. The joint cost has to be split between them — usually by working <i>backwards</i> from selling price.	10-14
Each process adds profit	A process sells to the next at cost + profit. The profit inside unsold stock is not real yet, so a <i>reserve for unrealised profit</i> must be removed.	15-16
Some units are half-finished	Closing work-in-progress is 60% done. You convert part-finished units into <i>equivalent</i> finished units before dividing.	17-24

The one habit that solves every problem in this module

Before you write a single figure, do this on your rough sheet. It takes forty seconds and it is the difference between a full-marks answer and a mess.



The units reconciliation. Write it for every process, every time.

THE THREE RATE FORMULAS — THIS IS ALL THE ARITHMETIC THERE IS

$$\text{Cost per unit (no loss)} = \frac{\text{Total cost of the process}}{\text{Units produced}}$$

$$\text{Cost per good unit (with loss)} = \frac{\text{Total cost} - \text{Scrap value of normal loss}}{\text{Input units} - \text{Normal loss units}}$$

Learn this one properly. Normal loss is removed from BOTH the top and the bottom. Abnormal loss and abnormal gain are then valued at this same rate — never at a different one.

$$\text{Cost per unit (part-finished stock)} = \frac{\text{Cost of that element}}{\text{Equivalent units of that element}}$$

Material, labour and overhead each get their own rate, because they are each at a different stage of completion.

Q1 Three processes, overhead on wages

Plain chain · p.8

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

There are three distinct processes A, B and C. During the four-week period 1,000 units are produced and the following information is available.

Particulars	Process A	Process B	Process C
Direct materials	2,000	1,000	—
Direct wages	1,500	700	800
Direct expenses	300	100	—

The indirect production cost were ₹4,500 and these are to be apportioned on the basis of direct wage cost. Prepare the process accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

Three things to notice, and they decide the whole answer:

- **“1,000 units are produced”** — one figure for all three processes. So there is **no loss anywhere**. 1,000 units go in, 1,000 come out of every process. That means no normal loss working, no abnormal loss — the easiest kind.
- **“apportioned on the basis of direct wage cost”** — the question hands you the basis. You do not have to guess it. Whenever a question gives you a lump overhead and a basis, your first working note is always the same: find the ratio, then split.
- **Process C has no materials and no direct expenses**. Do not panic and do not leave the row out — write a dash. An examiner wants to see that you looked and found nothing.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Add up the given basis across all processes (here: total direct wages).
2. Overhead rate = Total indirect cost ÷ Total basis. Keep it as a plain number or a percentage — whichever divides cleanly.
3. Multiply each process's basis by that rate. **Check the three pieces add back to the original lump sum** before you go on.
4. Open Process A. Debit its own four costs. Total it. That total transfers out to B.
5. Open Process B. **First debit is the transfer from A**, then B's own costs. Total, transfer to C.
6. Open Process C the same way. Its total goes to Finished Stock.
7. Divide each process total by 1,000 to show cost per unit. The last one is the answer the examiner is looking for.

WORKING NOTES

W1 Apportionment of indirect production cost

Total direct wages = 1,500 + 700 + 800 = **₹3,000**

Overhead rate = ₹4,500 ÷ ₹3,000 = **1.5 times the wages** (i.e. 150% of direct wages)

Process	Direct wages	× 1.5	Overhead absorbed
A	1,500	1,500 × 1.5	2,250
B	700	700 × 1.5	1,050
C	800	800 × 1.5	1,200
Total	3,000		4,500

Cross-check: 2,250 + 1,050 + 1,200 = 4,500 equals the indirect cost given. If this does not tally, stop and re-divide.

Process A Account

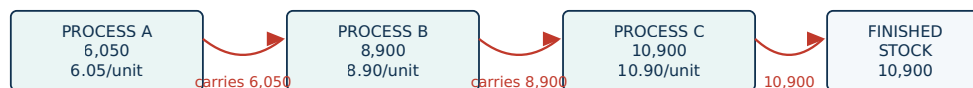
Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Direct materials	1,000	—	2,000	By Transfer to Process B 6,050 ÷ 1,000 units = ₹6.05 per unit	1,000	6.05	6,050
To Direct wages	—	—	1,500				
To Direct expenses	—	—	300				
To Indirect production cost W1: 1,500 × 1.5	—	—	2,250				
Total	1,000	—	6,050	Total	1,000	—	6,050

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A ← total of Process A a/c	1,000	6.05	6,050	By Transfer to Process C $8,900 \div 1,000 \text{ units} = ₹8.90 \text{ per unit}$	1,000	8.90	8,900
To Direct materials	—	—	1,000				
To Direct wages	—	—	700				
To Direct expenses	—	—	100				
To Indirect production cost W1: 700×1.5	—	—	1,050				
Total	1,000	—	8,900	Total	1,000	—	8,900

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B ← total of Process B a/c	1,000	8.90	8,900	By Transfer to Finished Stock $10,900 \div 1,000 = ₹10.90 \text{ per unit}$	1,000	10.90	10,900
To Direct materials	—	—	—				
To Direct wages	—	—	800				
To Direct expenses	—	—	—				
To Indirect production cost W1: 800×1.5	—	—	1,200				
Total	1,000	—	10,900	Total	1,000	—	10,900



Each box adds only its OWN cost; the arrow carries the whole running total forward.

The chain. Notice the cost per unit can only go up as you move right.

THE TRAP / WHERE MARKS ARE LOST

- Students write Process B's cost per unit as $2,850 \div 1,000 = ₹2.85$, using only B's own costs. Wrong. The unit leaving B has A's cost inside it too. **The transfer-in is a debit like any other.**
- Forgetting to show **units** in the account. Marks are given for the units column, and it is free — every figure is 1,000.
- Apportioning the overhead on the wrong basis (materials, or equally between three processes). The question said direct wage cost. Read it, underline it, use it.

FINAL ANSWER

Cost per unit — Process A	₹ 6.05
Cost per unit — Process B	₹ 8.90
Cost per unit — Process C (final)	₹ 10.90
Total cost of 1,000 units	₹ 10,900

State the final cost per unit in words as your closing line: "The cost of production is ₹10,900 for 1,000 units, i.e. ₹10.90 per unit."

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details show the cost of the three processes of manufacturing. The production of each process is passed on to the next till completion.

Particulars	Process A	Process B	Process C
Wages and materials	60,800	24,000	58,500
Works overheads	11,200	10,500	12,000
Production in units	72,000	75,000	96,000
Opening stock (units from preceding process on 1st July)	—	8,000	33,000
Closing stock (units from preceding process on 31st July)	—	2,000	11,000

READ THE QUESTION FIRST — WHAT TO SPOT

This problem looks like Q1 but it is not, and one phrase tells you so: “**units from preceding process**”.

- The opening and closing stock in Process B is **not half-finished work**. It is a pile of finished A-units waiting to be worked on. So it is valued at **A's rate**, not B's. Same for C — its stock is B-units, valued at B's rate.
- Because there is stock lying on both sides, the units B actually *consumed* are not the units it *received*. You must work the consumption out.
- “Production in units” is the **output** of that process. Divide by this figure to get the rate — never by the units consumed.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. **Process A first** — it has no stock, so rate = total cost ÷ production.
2. For B and C, find the units consumed: **Opening stock + Received from previous process – Closing stock**.
3. Value those consumed units at the **previous process's rate**. That is the transfer-in cost.
4. Add the process's own wages, materials and works overheads.
5. Rate for this process = that grand total ÷ **its own production in units**.
6. Now write the account. Debit side: opening stock, transfer-in, own costs. Credit side: closing stock (at the *previous* rate) and transfer-out (at *this* rate). The two sides must balance to the rupee.

WORKING NOTES

W1 Process A rate (no stock, so straight division)

Total cost = 60,800 + 11,200 = **₹72,000**

Rate = 72,000 ÷ 72,000 units = **₹1.00 per unit**

W2 Units consumed by Process B and Process C

Particulars	Process B	Process C
Opening stock of previous-process units	8,000	33,000
Add: Received from previous process	72,000	75,000
Units available	80,000	1,08,000

Particulars	Process B	Process C
Less: Closing stock of previous-process units	(2,000)	(11,000)
Units consumed in the process	78,000	97,000

Received by B = A's production 72,000. Received by C = B's production 75,000.

W3 Rate for Process B

Cost of units consumed = $78,000 \times ₹1.00 = 78,000$
Add own cost = $24,000 + 10,500 = 34,500$
Total = **₹1,12,500**
Rate = $1,12,500 \div 75,000$ units produced = **₹1.50 per unit**

W4 Rate for Process C

Cost of units consumed = $97,000 \times ₹1.50 = 1,45,500$
Add own cost = $58,500 + 12,000 = 70,500$
Total = **₹2,16,000**
Rate = $2,16,000 \div 96,000$ units produced = **₹2.25 per unit**

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Wages and materials	72,000	—	60,800	By Transfer to Process B W1: rate ₹1.00	72,000	1	72,000
To Works overheads	—	—	11,200				
Total	72,000	—	72,000	Total	72,000	—	72,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock 8,000 A-units @ ₹1.00 ← A's rate	8,000	1	8,000	By Closing stock 2,000 A-units @ ₹1.00 — still A's rate	2,000	1	2,000
To Transfer from Process A	72,000	1	72,000	By Transfer to Process C W3: 75,000 @ ₹1.50	75,000	1.50	1,12,500
To Wages and materials	—	—	24,000	(3,000 units consumed but not yielded — process shrinkage, no scrap value given)	3,000	—	—
To Works overheads	—	—	10,500				
Total	80,000	—	1,14,500	Total	80,000	—	1,14,500

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock <i>33,000 B-units @ ₹1.50 ← B's rate</i>	33,000	1.50	49,500	By Closing stock <i>11,000 B-units @ ₹1.50</i>	11,000	1.50	16,500
To Transfer from Process B	75,000	1.50	1,12,500	By Transfer to Finished Stock <i>W4: 96,000 @ ₹2.25</i>	96,000	2.25	2,16,000
To Wages and materials	—	—	58,500	<i>(1,000 units shrinkage)</i>	1,000	—	—
To Works overheads	—	—	12,000				
Total	1,08,000	—	2,32,500	Total	1,08,000	—	2,32,500



Why Process B's stock is valued at A's rate, not B's.

THE TRAP / WHERE MARKS ARE LOST

- Valuing B's opening/closing stock at B's own rate of ₹1.50. It has not been through B yet — it is worth ₹1.00. This single error breaks both sides of the account.
- Dividing by units consumed (78,000) instead of units produced (75,000). The rate must describe a unit that came out.
- Panicking when the units do not balance. Here 78,000 are consumed but only 75,000 emerge. That gap is real shrinkage. No scrap value is given, so it carries no cost — simply show it and move on. Do not invent a normal-loss working the question never asked for.

FINAL ANSWER

Cost per unit — Process A **₹ 1.00**
 Cost per unit — Process B **₹ 1.50**
 Cost per unit — Process C (final) **₹ 2.25**
 Cost transferred to Finished Stock **₹ 2,16,000**

Both B and C accounts balance exactly — 1,14,500 and 2,32,500. Always show the totals; a balancing account is self-proving.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information relating to an article which undergoes three processes for manufacture, show the cost of each process and the cost per article.

Particulars	Process A	Process B	Process C
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250

The indirect expenses amounted to ₹21,250 and the number of articles produced during the month was 240 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same shape as Q1 — one output figure (240 articles), so **no losses**.
- **The basis for splitting the indirect expenses is NOT stated.** This is deliberate. The accepted convention when no basis is given is to apportion on **direct labour**. Say so in one line in your answer — “indirect expenses apportioned on the basis of direct labour in the absence of any other basis” — and you protect the marks.
- Test whether your assumed basis is right by checking the split comes out in round figures. Here labour gives exactly 25%. That is the examiner confirming your choice.
- Two answers are wanted: **cost of each process** and **cost per article**. Give both, clearly labelled.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the chosen basis — labour: 20,000 + 50,000 + 15,000.
2. Rate = 21,250 ÷ that total. Express as a percentage if it is clean.
3. Split, then cross-check the pieces add back to 21,250.
4. Build the three process accounts in a chain, transferring the running total.
5. Divide each process total by 240 to get cost per article at that stage.
6. Final line: cost per article = Process C total ÷ 240.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: direct labour)

Total labour = 20,000 + 50,000 + 15,000 = **₹85,000**

Rate = 21,250 ÷ 85,000 = 0.25 = **25% of direct labour**

Process	Labour	× 25%	Indirect expenses
A	20,000	20,000 × 0.25	5,000
B	50,000	50,000 × 0.25	12,500
C	15,000	15,000 × 0.25	3,750
Total	85,000		21,250

Cross-check: 5,000 + 12,500 + 3,750 = 21,250 — and every figure is a round number, which confirms labour was the intended basis.

Statement of Cost of Each Process and Cost per Article

Particulars	Process A	Process B	Process C
Cost of previous process (transferred in)	—	69,000	1,62,000
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250
Indirect expenses (W1)	5,000	12,500	3,750
Total cost of the process	69,000	1,62,000	2,37,000
Cost per article (÷ 240)	287.50	675.00	987.50

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials consumed	240	—	37,500	By Transfer to Process B $69,000 \div 240 = ₹287.50$	240	287.50	69,000
To Labour	—	—	20,000				
To Direct expenses	—	—	6,500				
To Indirect expenses W1	—	—	5,000				
Total	240	—	69,000	Total	240	—	69,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	240	287.50	69,000	By Transfer to Process C $1,62,000 \div 240 = ₹675.00$	240	675	1,62,000
To Materials consumed	—	—	12,500				
To Labour	—	—	50,000				
To Direct expenses	—	—	18,000				
To Indirect expenses W1	—	—	12,500				
Total	240	—	1,62,000	Total	240	—	1,62,000

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	240	675	1,62,000	By Transfer to Finished Stock $2,37,000 \div 240 = ₹987.50$	240	987.50	2,37,000
To Materials consumed	—	—	50,000				
To Labour	—	—	15,000				
To Direct expenses	—	—	6,250				
To Indirect expenses W1	—	—	3,750				
Total	240	—	2,37,000	Total	240	—	2,37,000

WHY THIS WORKS

The statement and the three accounts say exactly the same thing. If the question says “show the cost of each process” the statement alone is enough; if it says “prepare process accounts” you must draw the accounts. When in doubt give the statement first — it is faster to write and it makes the accounts almost automatic.

FINAL ANSWER

Cost of Process A ₹ 69,000

Cost of Process B ₹ 1,62,000

Cost of Process C ₹ 2,37,000

Cost per article ₹ 987.50

Q4 By-product and residue

Chain + by-product · p.8-9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A particular brand of chemical passed through three processes during the week ended 15th January. 600 tins were produced. The following information is disclosed by each process.

Particulars	Process A	Process B	Process C
Materials	4,000	2,000	1,500
Labour expenses	3,000	2,500	2,300
Direct expenses	6,000	200	500
Cost of tins	—	—	2,030
Packing charges	—	—	325

The indirect expenses were ₹1,600. The by-product is sold for ₹240 (Process B). The residue was sold for ₹125.50 (Process C).

Prepare the process A/c with cost of production of the product per tin.

READ THE QUESTION FIRST — WHAT TO SPOT

- Again 600 tins throughout — **no process loss**.
- Two new items: a **by-product sold for 240** and a **residue sold for 125.50**. Both are **deducted** from the cost of the process in which they arise. They are not income and they do not go to Profit & Loss — they reduce the cost that the good output has to carry.
- “Cost of tins” and “packing charges” belong to Process C only. They are ordinary debits — do not treat them as anything special.
- The indirect expense basis is again not stated, so apportion on labour and say so.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Apportion the indirect expenses on labour (state the assumption).
2. Build Process A normally.
3. In Process B, debit everything, then **credit the by-product sale of 240**. The amount that transfers to C is the total *after* that deduction.
4. In Process C, debit everything including tins and packing, then **credit the residue sale of 125.50**.
5. Cost of production = final balance of Process C. Divide by 600 tins.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: labour)

Total labour = 3,000 + 2,500 + 2,300 = ₹7,800

Rate = $1,600 \div 7,800 = 0.205128$ per rupee of labour

Process	Labour	Working	Indirect expenses
A	3,000	$3,000 \times 0.205128$	615.38
B	2,500	$2,500 \times 0.205128$	512.82
C	2,300	$2,300 \times 0.205128$	471.80
Total	7,800		1,600.00

The rate does not divide neatly here, so carry two decimals and force the last figure to make the total exactly 1,600. Write “difference of ₹0.01 adjusted in Process C” if you like — examiners accept it and it shows you checked.

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	600	—	4,000	By Transfer to Process B	600	22.69	13,615.38
To Labour expenses	—	—	3,000				
To Direct expenses	—	—	6,000				
To Indirect expenses W1	—	—	615.38				
Total	600	—	13,615.38	Total	600	—	13,615.38

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	600	22.69	13,615.38	By Sale of by-product deducted from cost, NOT income	—	—	240
To Materials	—	—	2,000	By Transfer to Process C $18,828.20 - 240 = 18,588.20$	600	30.98	18,588.20
To Labour expenses	—	—	2,500				
To Direct expenses	—	—	200				
To Indirect expenses W1	—	—	512.82				
Total	600	—	18,828.20	Total	600	—	18,828.20

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	600	30.98	18,588.20	By Sale of residue deducted from cost	—	—	125.50
To Materials	—	—	1,500	By Cost of production transferred $25,715.00 - 125.50 = 25,589.50$	600	42.65	25,589.50
To Labour expenses	—	—	2,300				
To Direct expenses	—	—	500				
To Cost of tins	—	—	2,030				
To Packing charges	—	—	325				
To Indirect expenses W1	—	—	471.80				
Total	600	—	25,715	Total	600	—	25,715

THE TRAP / WHERE MARKS ARE LOST

- Adding the by-product sale to the credit of Profit & Loss instead of deducting it inside the process. In a **by-product of small value** the standard treatment is to credit the process. That is what “(Process B)” in the question is telling you.
- Deducting the residue from Process B or from the total instead of from Process C. The question names the process for each — use it.
- Rounding the per-tin figure too early. Carry the paise through and round only at the last line.

FINAL ANSWER

Total cost of production **₹ 25,589.50**

Number of tins **600**

Cost of production per tin **₹ 42.65**

Closing line: “Cost of production is ₹25,589.50 for 600 tins, i.e. ₹42.65 per tin.”

Q5 Normal loss and abnormal loss

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare a Process Account and Abnormal Loss Account from the following information.

Particulars	
Input of raw material	1,000 units @ ₹20 per unit
Direct material	₹4,200
Direct wages	₹6,000
Production overheads	₹6,000
Actual output transferred to Process II	900 units
Normal loss	5%
Value of scrap per unit	₹8

READ THE QUESTION FIRST — WHAT TO SPOT

This is the first problem with a loss, and it is the template for Q5–Q9. Four figures decide everything:

- **Input 1,000 units.** Normal loss is a percentage *of input* — so 5% of 1,000 = 50 units. Not 5% of output.
- **Expected output = 1,000 – 50 = 950 units.** Write this down. It is the figure you compare the actual against.
- **Actual output is 900**, which is *less* than expected 950. So 50 units have gone missing beyond what was allowed — that is **abnormal loss**. If actual had been more than 950 it would have been abnormal gain.
- **Scrap is worth 8 per unit.** Normal loss units are sold as scrap, so they do carry a value — and that value is removed from the cost before you divide.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits — every cost that entered the process.
2. Normal loss in **units** = given % × input units. Normal loss in **rupees** = those units × scrap rate.
3. Expected output = Input – Normal loss units.
4. Compare actual output with expected output. Shortfall = abnormal loss; excess = abnormal gain.
5. **Cost per good unit** = (Total cost – Scrap value of normal loss) ÷ (Input units – Normal loss units). Show this fraction in full.
6. Value the transfer-out AND the abnormal loss at that **same** rate.
7. Write the process account — it must balance. Then open the Abnormal Loss account: debit the process value, credit the scrap realised, and the balance goes to Profit & Loss.

WORKING NOTES

W1 Total cost debited to the process

Raw material 1,000 × ₹20 = 20,000
Direct material = 4,200
Direct wages = 6,000
Production overheads = 6,000
Total = ₹36,200

W2 The units reconciliation

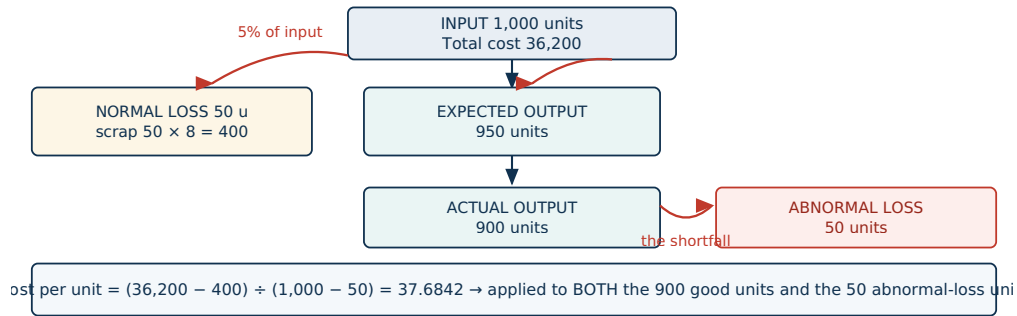
Normal loss = 5% of 1,000 = **50 units**, scrap value 50 × ₹8 = **₹400**
Expected output = 1,000 – 50 = **950 units**
Actual output = 900 units
Abnormal loss = 950 – 900 = **50 units**

W3 Cost per good unit — the formula that matters most

$$\text{₹36,200} - \text{₹400} \div 1,000 - 50 = \text{₹35,800} \div 950 = \text{₹37.6842 per unit}$$

Normal loss is taken out of the top (its scrap value) and out of the bottom (its units). That is how the cost of the normal loss gets loaded onto the good units.

Transferred out 900 × 37.6842 = **₹33,915.79**
Abnormal loss 50 × 37.6842 = **₹1,884.21**



Normal loss is expected and free. Abnormal loss is the shortfall against expectation.

Process I Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Raw material	1,000	20	20,000	By Normal loss 50 units × ₹8 scrap	50	8	400
To Direct material	—	—	4,200	By Abnormal loss W3: 50 × 37.6842	50	37.68	1,884.21
To Direct wages	—	—	6,000	By Transfer to Process II W3: 900 × 37.6842	900	37.68	33,915.79
To Production overheads	—	—	6,000				
Total	1,000	—	36,200	Total	1,000	—	36,200

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c ← from the process account	50	37.68	1,884.21	By Sale of scrap 50 units × ₹8 — abnormal loss units are also sold as scrap	50	8	400
Total	50	—	1,884.21	By Profit & Loss A/c balance — the real loss to the business	—	—	1,484.21
				Total	50	—	1,884.21

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c	50	8	400	By Cost Ledger Control A/c (cash from scrap)	50	8	400
Total	50	—	400	Total	50	—	400

THE TRAP / WHERE MARKS ARE LOST

- Dividing by 1,000 instead of 950. You would get ₹35.80 and every later figure would be wrong. **The denominator is always input minus normal loss units.**
- Forgetting to deduct the 400 scrap value from the numerator.
- Valuing the abnormal loss at the scrap rate of 8. No — abnormal loss is valued at the **good-unit rate**. It only realises 8 when sold, and the difference is the loss you take to Profit & Loss.
- Taking the whole 1,884.21 to Profit & Loss. Only the **net** figure of 1,484.21 goes there, after crediting the 400 the scrap actually fetched.

FINAL ANSWER

Cost per good unit **₹ 37.6842**

Transferred to Process II (900 units)	₹ 33,915.79
Abnormal loss (50 units, at cost)	₹ 1,884.21
Scrap realised on abnormal loss	₹ 400.00
Net abnormal loss charged to Profit & Loss	₹ 1,484.21

Note how the Process account, the Abnormal Loss account and the Normal Loss account all close cleanly. Show all three unless the question asks for fewer.

Q6 Loss, gain and neither — all in one

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The product of a company passes through 3 distinct processes. The following information is obtained from the accounts for the month ending January 31, 2024.

Particulars	Process A	Process B	Process C
Direct material	7,800	5,940	8,886
Direct wages	6,000	9,000	12,000
Production overheads	6,000	9,000	12,000

3,000 units @ ₹3 each were introduced to Process I. There was no stock of materials or work in progress. The output of each process passes directly to the next process and finally to Finished Stock A/c. The following additional data is obtained:

Process	Output	Normal loss %	Realisable value of scrap
Process 1	2,850	5%	2
Process 2	2,520	10%	4
Process 3	2,250	15%	5

Prepare Process Cost Account, Normal Loss Account and Abnormal Gain or Loss Account.

READ THE QUESTION FIRST — WHAT TO SPOT

This is **the** model problem of the module. It contains all three possibilities in one question, which is exactly why examiners like it.

- **Process 1 ends with no abnormal item at all** — actual output equals expected. Do not force an abnormal figure just because the other processes have one.
- **Process 2 has an abnormal loss** (output less than expected).
- **Process 3 has an abnormal gain** (output MORE than expected). An abnormal gain is entered on the **debit** side of the process account — the opposite side from a loss. This single fact is the most commonly dropped mark in the whole module.
- Normal loss % is always applied to **the units entering that process**, which is the previous process's output — not the original 3,000.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Work the processes strictly in order. Each one needs the previous one's output *and* its rate.
2. For each process: total the debits, compute normal loss units and scrap value, find expected output, compare with actual.
3. Rate = (Total cost – normal loss scrap) ÷ (Input units – normal loss units).

4. **Loss** → credit the process with the shortfall at that rate. **Gain** → debit the process with the excess at that rate.
5. Open one Normal Loss A/c for all three processes together. Debit each process's scrap value into it.
6. If there is an abnormal gain, the normal loss did not fully happen — so **credit Normal Loss A/c and debit Abnormal Gain A/c** with the gain units × that process's scrap rate.
7. Close Abnormal Loss and Abnormal Gain to Profit & Loss.

WORKING NOTES

W1 Process 1

Total cost = $(3,000 \times ₹3 = 9,000) + 7,800 + 6,000 + 6,000 = ₹28,800$

Normal loss = 5% of 3,000 = **150 units**; scrap $150 \times ₹2 = ₹300$

Expected output = $3,000 - 150 = 2,850$ units. Actual = 2,850 units.

Difference = NIL — no abnormal loss or gain in Process 1.

$$\text{Rate} = \frac{28,800 - 300}{3,000 - 150} = \frac{28,500}{2,850} = ₹10.00 \text{ per unit}$$

W2 Process 2

Total cost = $(2,850 \times ₹10 = 28,500) + 5,940 + 9,000 + 9,000 = ₹52,440$

Normal loss = 10% of **2,850** = **285 units**; scrap $285 \times ₹4 = ₹1,140$

Expected output = $2,850 - 285 = 2,565$ units. Actual = 2,520 units.

Abnormal LOSS = 2,565 - 2,520 = 45 units

$$\text{Rate} = \frac{52,440 - 1,140}{2,850 - 285} = \frac{51,300}{2,565} = ₹20.00 \text{ per unit}$$

Abnormal loss $45 \times 20 = ₹900$

Transfer to Process 3 $2,520 \times 20 = ₹50,400$

W3 Process 3

Total cost = $(2,520 \times ₹20 = 50,400) + 8,886 + 12,000 + 12,000 = ₹83,286$

Normal loss = 15% of **2,520** = **378 units**; scrap $378 \times ₹5 = ₹1,890$

Expected output = $2,520 - 378 = 2,142$ units. Actual = 2,250 units.

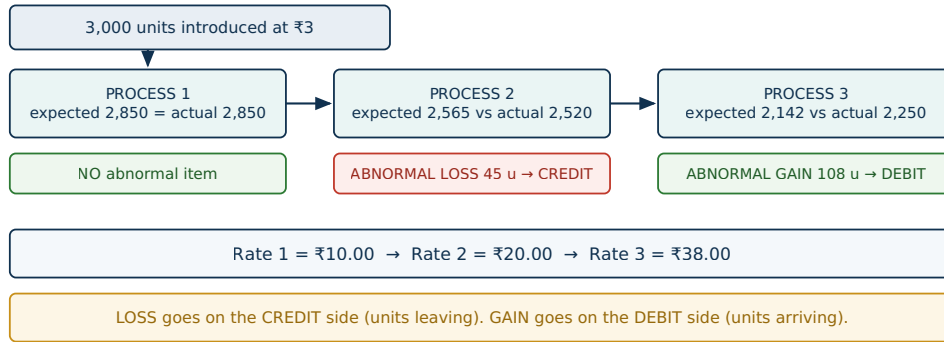
Abnormal GAIN = 2,250 - 2,142 = 108 units (actual is MORE than expected)

$$\text{Rate} = \frac{83,286 - 1,890}{2,520 - 378} = \frac{81,396}{2,142} = ₹38.00 \text{ per unit}$$

Note the rate is still built on EXPECTED output (2,142), not on the actual 2,250. The gain is then valued at this rate and added back.

Abnormal gain $108 \times 38 = ₹4,104$ (debit side)

Transfer to Finished Stock $2,250 \times 38 = ₹85,500$



One question, all three outcomes. Learn this picture and Q5 to Q9 all become the same problem.

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	3,000	3	9,000	By Normal loss $150 \text{ u} \times ₹2$	150	2	300
To Direct material	—	—	7,800	By Transfer to Process 2 $W1: 2,850 \times ₹10$	2,850	10	28,500
To Direct wages	—	—	6,000				
To Production overheads	—	—	6,000				
Total	3,000	—	28,800	Total	3,000	—	28,800

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 1	2,850	10	28,500	By Normal loss $285 \text{ u} \times ₹4$	285	4	1,140
To Direct material	—	—	5,940	By Abnormal loss $W2: 45 \times ₹20$	45	20	900
To Direct wages	—	—	9,000	By Transfer to Process 3 $W2: 2,520 \times ₹20$	2,520	20	50,400
To Production overheads	—	—	9,000				
Total	2,850	—	52,440	Total	2,850	—	52,440

Process 3 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 2	2,520	20	50,400	By Normal loss $378 \text{ u} \times ₹5$	378	5	1,890
To Direct material	—	—	8,886	By Transfer to Finished Stock $W3: 2,250 \times ₹38$	2,250	38	85,500
To Direct wages	—	—	12,000				
To Production overheads	—	—	12,000				
To Abnormal gain $W3: 108 \times ₹38$ — DEBIT side	108	38	4,104				
Total	2,628	—	87,390	Total	2,628	—	87,390

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	150	2	300	By Abnormal Gain A/c <i>108 u × ₹5 — the Process 3 normal loss that never happened</i>	108	5	540
To Process 2 A/c	285	4	1,140	By Cost Ledger Control A/c (scrap actually sold)	705	—	2,790
To Process 3 A/c	378	5	1,890				
Total	813	—	3,330	Total	813	—	3,330

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 2 A/c	45	20	900	By Cost Ledger Control A/c <i>45 u × ₹4 scrap</i>	45	4	180
Total	45	—	900	By Profit & Loss A/c balance = the real loss	—	—	720
				Total	45	—	900

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c <i>108 u × ₹5 — scrap income forgone</i>	108	5	540	By Process 3 A/c	108	38	4,104
To Profit & Loss A/c <i>balance = the real gain</i>	—	—	3,564				
Total	108	—	4,104	Total	108	—	4,104

THE TRAP / WHERE MARKS ARE LOST

- **Putting the abnormal gain on the credit side.** Think about it physically: extra units have *arrived* in the process, so they must be debited like any other input. If your Process 3 account will not balance, this is why.
- Applying the 15% normal loss to 3,000 instead of 2,520. Each percentage applies to that process's own input.
- Forgetting the entry **Abnormal Gain A/c → Normal Loss A/c 540**. Because 108 units did not perish, the company also did not earn scrap on them — so the gain must be reduced by $108 \times ₹5$. Without this the Normal Loss account will not close.
- Using the actual output 2,250 as the denominator for the rate. The denominator is always **input minus normal loss** = 2,142.

FINAL ANSWER

Cost per unit — Process 1	₹ 10.00
Cost per unit — Process 2	₹ 20.00
Cost per unit — Process 3	₹ 38.00
Abnormal loss (Process 2, 45 units)	₹ 900
Abnormal gain (Process 3, 108 units)	₹ 4,104
Transferred to Finished Stock (2,250 units)	₹ 85,500

Net loss to P&L on abnormal loss ₹ 720

Net gain to P&L on abnormal gain ₹ 3,564

Every one of the six accounts closes exactly. If yours do not, check the abnormal gain is on the debit side and that the 540 entry is present.

Q7 Abnormal gain only

Loss & gain · p.10

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Product A is obtained after it passes through three distinct processes. 2,000 kgs of material at ₹5 per kg were issued to Process 1, direct wages amounted to ₹900 and other production overheads incurred was ₹500. Normal loss is estimated at 10% of input. The wastage is sold at ₹3 per kg. The actual output is 1,850 kgs. Prepare Process 1 A/c, Normal Loss A/c and Abnormal Gain A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Only **Process 1** is asked for. The question mentions three processes to set the scene — ignore the other two. Answer exactly what is asked.
- Expected output = 2,000 – 200 = 1,800 kg. Actual is **1,850 kg — higher**. The question even names the account it wants: **Abnormal Gain A/c**. That is your confirmation before you calculate anything.
- Because there is a gain, remember the two-part consequence: debit the process with the gain at cost, and transfer the forgone scrap value to Normal Loss A/c.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits.
2. Normal loss = 10% of 2,000 = 200 kg; scrap value = 200 × 3.
3. Expected output 1,800 kg vs actual 1,850 kg → abnormal gain 50 kg.
4. Rate = (11,400 – 600) ÷ (2,000 – 200).
5. Debit the process with 50 kg at that rate; credit the transfer of 1,850 kg at that rate.
6. Normal Loss A/c: debit 200 kg at scrap rate; credit 50 kg to Abnormal Gain A/c; the rest is scrap sold.
7. Abnormal Gain A/c: debit the forgone scrap and the balance to P&L; credit the process figure.

WORKING NOTES

W1 Cost and units

Material 2,000 × ₹5 = 10,000

Direct wages = 900 | Production overheads = 500

Total cost = ₹11,400

Normal loss = 10% of 2,000 = **200 kg**; scrap 200 × ₹3 = **₹600**

Expected output = 1,800 kg | Actual output = 1,850 kg

Abnormal gain = 1,850 – 1,800 = 50 kg

$$\text{Rate} = \frac{11,400 - 600}{2,000 - 200} = \frac{10,800}{1,800} = \text{₹6.00 per kg}$$

Abnormal gain 50 × 6 = **₹300**

Transfer out 1,850 × 6 = **₹11,100**

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	5	10,000	By Normal loss 200 kg × ₹3	200	3	600
To Direct wages	—	—	900	By Transfer to Process 2 W1: 1,850 × ₹6	1,850	6	11,100
To Production overheads	—	—	500				
To Abnormal gain W1: 50 kg × ₹6	50	6	300				
Total	2,050	—	11,700	Total	2,050	—	11,700

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	200	3	600	By Abnormal Gain A/c 50 kg × ₹3 — loss that did not occur	50	3	150
Total	200	—	600	By Cost Ledger Control A/c (scrap sold)	150	3	450
				Total	200	—	600

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c scrap income forgone on 50 kg	50	3	150	By Process 1 A/c	50	6	300
To Profit & Loss A/c balance = net gain	—	—	150				
Total	50	—	300	Total	50	—	300

WHY THIS WORKS

The net gain reaching Profit & Loss is only ₹150, not ₹300. The process saved ₹300 of cost by producing 50 extra kg, but it also lost the ₹150 of scrap money those 50 kg would have fetched. Gain is always **cost saved minus scrap forgone**.

FINAL ANSWER

Cost per kg	₹ 6.00
Transferred out (1,850 kg)	₹ 11,100
Abnormal gain at cost (50 kg)	₹ 300
Scrap value forgone	₹ 150
Net abnormal gain to Profit & Loss	₹ 150

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through 2 distinct processes A and B. The normal wastage of each process is as follows: Process A – 3% of products entering the process; Process B – 5% of products entering the process.

The wastage of Process A was sold at 50 paise per unit and that of Process B at ₹1 per unit. 10,000 units were issued to Process A at ₹2 per unit. The other expenses were as follows:

Particulars	Process A	Process B
Materials	2,000	3,000
Wages	10,000	16,000
Overheads	2,100	2,375
Actual output (units)	9,500	9,100

Prepare process accounts and other ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- “3% of products entering the process” — so Process B’s 5% is on the 9,500 units it receives, not on the original 10,000.
- Process A: expected 9,700, actual 9,500 → **abnormal loss 200 units**.
- Process B: expected 9,025, actual 9,100 → **abnormal gain 75 units**.
- Two different scrap rates — 50 paise in A and ₹1 in B. Keep them apart; using one rate for both is a common slip.
- “and other ledger accounts” means: Normal Loss A/c, Abnormal Loss A/c and Abnormal Gain A/c. Draw all three.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Process A: total debits, normal loss 3% of 10,000, expected output, compare, rate, then value the loss and the transfer.
2. Carry A's output *and* A's rate into Process B as the transfer-in.
3. Process B: normal loss 5% of the units received (9,500), then the same sequence.
4. Build the three ledger accounts, remembering the Abnormal Gain → Normal Loss entry.

WORKING NOTES

W1 Process A

Total cost = $(10,000 \times ₹2 = 20,000) + 2,000 + 10,000 + 2,100 = ₹34,100$

Normal loss = 3% of 10,000 = **300 units**; scrap $300 \times ₹0.50 = ₹150$

Expected output 9,700 | Actual 9,500 → **abnormal loss 200 units**

$$\text{Rate} = \frac{34,100 - 150}{10,000 - 300} = \frac{33,950}{9,700} = ₹3.50 \text{ per unit}$$

Abnormal loss $200 \times 3.50 = ₹700$

Transfer to B $9,500 \times 3.50 = ₹33,250$

W2 Process B

Total cost = 33,250 + 3,000 + 16,000 + 2,375 = ₹54,625

Normal loss = 5% of 9,500 = 475 units; scrap 475 × ₹1 = ₹475

Expected output 9,025 | Actual 9,100 → **abnormal gain 75 units**

$$\text{Rate} = \frac{54,625 - 475}{9,500 - 475} = \frac{54,150}{9,025} = \text{₹6.00 per unit}$$

Abnormal gain 75 × 6 = ₹450 (debit side)

Transfer to Finished Stock 9,100 × 6 = ₹54,600

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	10,000	2	20,000	By Normal loss 300 u × ₹0.50	300	0.50	150
To Materials	—	—	2,000	By Abnormal loss W1: 200 × ₹3.50	200	3.50	700
To Wages	—	—	10,000	By Transfer to Process B W1: 9,500 × ₹3.50	9,500	3.50	33,250
To Overheads	—	—	2,100				
Total	10,000	—	34,100	Total	10,000	—	34,100

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	9,500	3.50	33,250	By Normal loss 475 u × ₹1	475	1	475
To Materials	—	—	3,000	By Transfer to Finished Stock W2: 9,100 × ₹6	9,100	6	54,600
To Wages	—	—	16,000				
To Overheads	—	—	2,375				
To Abnormal gain W2: 75 × ₹6 — DEBIT	75	6	450				
Total	9,575	—	55,075	Total	9,575	—	55,075

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	300	0.50	150	By Abnormal Gain A/c 75 u × ₹1 (B's scrap rate)	75	1	75
To Process B A/c	475	1	475	By Cost Ledger Control A/c (scrap sold)	700	—	550
Total	775	—	625	Total	775	—	625

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	200	3.50	700	By Cost Ledger Control A/c $200 \times ₹0.50$	200	0.50	100
Total	200	—	700	By Profit & Loss A/c	—	—	600
				Total	200	—	700

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c	75	1	75	By Process B A/c	75	6	450
To Profit & Loss A/c	—	—	375				
Total	75	—	450	Total	75	—	450

THE TRAP / WHERE MARKS ARE LOST

- Using ₹0.50 as the scrap rate in Process B. B's wastage sells for ₹1. Two processes, two rates.
- Taking 5% of 10,000 in Process B. It receives only 9,500.
- Netting the abnormal loss of Process A against the abnormal gain of Process B. They are separate accounts and separate P&L entries — never set one off against the other.

FINAL ANSWER

Cost per unit — Process A	₹ 3.50
Cost per unit — Process B	₹ 6.00
Abnormal loss (A, 200 units)	₹ 700 — net to P&L ₹ 600
Abnormal gain (B, 75 units)	₹ 450 — net to P&L ₹ 375
Transferred to Finished Stock (9,100 units)	₹ 54,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2015. Purchase of 5,400 tonnes of coconut ₹2,20,000.

Particulars	Crushing	Refining	Finishing
Cost of labour	2,750	1,100	1,650
Electric power	660	396	264
Sundry material	110	2,200	—
Repairs to machinery	308	363	154
Steam	600	495	495
Factory expenses	1,452	726	242

Cost of sacks ₹8,250. 3,200 tonnes of crude oil were produced. 2,600 tonnes of oil produced by the refining process. 2,550 tonnes of refined oil were finished for delivery. Coconut sacks sold ₹440. 1,925 tonnes of coconut residue sold ₹12,100. Loss in weight in crushing 275 tonnes. 500 tonnes of by-products obtained from refining process ₹7,425.

Show the accounts for (a) coconut crushing process, (b) refining process, (c) finishing process, to arrive at the cost per tonne of each process and the total cost per tonne of the finished oil.

READ THE QUESTION FIRST — WHAT TO SPOT

A long question, but it is only Q1 with three deductions. The one thing that makes it easy is the **units reconciliation** — do it first and the whole answer falls out.

- **Crushing:** 5,400 t in. Out: crude oil 3,200 + residue 1,925 + loss in weight 275 = 5,400 . Perfect — no abnormal item.
- **Refining:** 3,200 t in. Out: refined oil 2,600 + by-product 500 = 3,100, so loss in weight = **100 t** (you have to work this out; the question does not give it).
- **Finishing:** 2,600 t in, 2,550 t out, so loss in weight = **50 t**.
- Three items are **credited** to their process because they are saleable: sacks 440 and residue 12,100 in crushing, by-product 7,425 in refining.
- Loss in weight is pure evaporation. It has **no scrap value**, so it carries no rupee figure — only tonnes. Show the tonnes and put a dash in the amount column.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Reconcile tonnes for each stage first. Work out any loss in weight the question has not given you.
2. Debit the process with the material coming in plus all its own expenses.
3. Credit any saleable output that is *not* the main product (sacks, residue, by-product) at its sale value.
4. Credit the loss in weight in tonnes only, amount nil.
5. The balance is the cost of the main output. Divide by its tonnage for cost per tonne.
6. Carry that balance forward as the debit of the next process.

WORKING NOTES

W1 Tonnage reconciliation — do this before anything else

Stage	In (t)	Main output (t)	Other output (t)	Loss in weight (t)
Crushing	5,400	3,200 crude oil	1,925 residue	275 (given)
Refining	3,200	2,600 refined oil	500 by-product	100 (derived)
Finishing	2,600	2,550 finished oil	—	50 (derived)

Refining loss = $3,200 - 2,600 - 500 = \mathbf{100 \text{ tonnes}}$

Finishing loss = $2,600 - 2,550 = \mathbf{50 \text{ tonnes}}$

W2 Cost per tonne at each stage

$$\text{Crushing} = \frac{2,21,590}{3,200 \text{ t}} = \mathbf{₹69.25}$$

$$\text{Refining} = \frac{2,19,445}{2,600 \text{ t}} = \mathbf{₹84.40}$$

$$\text{Finishing} = \frac{2,22,250}{2,550 \text{ t}} = \mathbf{₹87.16}$$

(a) Coconut Crushing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Coconut purchased	5,400	—	2,20,000	By Sale of sacks	—	—	440
To Cost of sacks	—	—	8,250	By Sale of coconut residue	1,925	—	12,100
To Cost of labour	—	—	2,750	By Loss in weight evaporation — tonnes only, no value	275	—	—
To Electric power	—	—	660	By Crude oil to Refining W2: 2,34,130 – 440 – 12,100	3,200	69.25	2,21,590
To Sundry material	—	—	110				
To Repairs to machinery	—	—	308				
To Steam	—	—	600				
To Factory expenses	—	—	1,452				
Total	5,400	—	2,34,130	Total	5,400	—	2,34,130

(b) Refining Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Crude oil from Crushing	3,200	69.25	2,21,590	By Sale of by-products	500	—	7,425
To Cost of labour	—	—	1,100	By Loss in weight W1: derived as 100 t	100	—	—
To Electric power	—	—	396	By Refined oil to Finishing W2: 2,26,870 – 7,425	2,600	84.40	2,19,445
To Sundry material	—	—	2,200				
To Repairs to machinery	—	—	363				
To Steam	—	—	495				
To Factory expenses	—	—	726				
Total	3,200	—	2,26,870	Total	3,200	—	2,26,870

(c) Finishing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Refined oil from Refining	2,600	84.40	2,19,445	By Loss in weight W1: derived as 50 t	50	—	—
To Cost of labour	—	—	1,650	By Finished oil (cost of delivery) W2: 2,22,250 ÷ 2,550 t	2,550	87.16	2,22,250
To Electric power	—	—	264				
To Repairs to machinery	—	—	154				
To Steam	—	—	495				
To Factory expenses	—	—	242				
Total	2,600	—	2,22,250	Total	2,600	—	2,22,250

THE TRAP / WHERE MARKS ARE LOST

- Treating the sacks, residue and by-product as income in Profit & Loss. They reduce the cost of the process that produced them.
- Giving the loss in weight a rupee value. It evaporated — there is nothing to sell. Tonnes only.
- Not deriving the refining and finishing losses. If you skip them the tonnage columns will not add up and the examiner will see it immediately.
- Dividing the crushing cost by 5,400 tonnes. The cost belongs to the 3,200 tonnes of crude oil that actually came out.

FINAL ANSWER

Cost per tonne — Crushing (crude oil)	₹ 69.25
Cost per tonne — Refining (refined oil)	₹ 84.40
Cost per tonne — Finishing (finished oil)	₹ 87.16
Total cost of 2,550 tonnes delivered	₹ 2,22,250

Closing line: “The total cost per tonne of the finished oil is ₹87.16.”

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In manufacturing the main product a company processes the incidental waste into two by-products A and B. From the following data relating to the products prepare a comparative profit & loss statement showing individual cost and other details. The total cost up to separation point was ₹3,10,400.

Particulars	Main product	Product A	Product B
Sales	8,00,000	64,000	96,000
Cost after separation	80,000	12,800	14,400
Estimated net profit (% of sales)	?	20%	30%
Estimated selling expenses (% of sales)	20%	10%	15%

READ THE QUESTION FIRST — WHAT TO SPOT

- The main product's profit is a **question mark**. That tells you the whole plan: work out the two by-products completely, take their joint-cost shares away from ₹3,10,400, and whatever is left belongs to the main product.
- You are given **profit % and selling-expense % for A and B**. That is the signal for the **reverse-cost method** — start at sales and climb down.
- All percentages here are **of sales**, not of cost. Read the bracket every time.
- "Comparative profit & loss statement" means one table with a column per product and a total column. Give the total column — it is where the cross-check lives.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each by-product, start with sales. Subtract profit (% of sales). Subtract selling expenses (% of sales). What remains is that product's **total cost**.
2. Subtract its cost after separation. What remains is its **share of the joint cost**.
3. Add the by-products' shares. Subtract from the total joint cost. The balance is the **main product's share**.
4. Now build the main product's column forwards: joint share + cost after separation + selling expenses = total cost; sales – total cost = profit.
5. Express the main product's profit as a % of its sales — that is the missing '?'.
6. Cross-check: every column's total cost + profit must equal its sales.



The reverse-cost ladder. Every joint / by-product problem in Q10 to Q14 is this one picture.

WORKING NOTES

W1 Product A — climbing down from sales

Sales = 64,000

Less: Net profit 20% of 64,000 = (12,800)
 Less: Selling expenses 10% of 64,000 = (6,400)
Total cost = 44,800
 Less: Cost after separation = (12,800)
Share of joint cost = ₹32,000

W2 Product B — same ladder

Sales = 96,000
 Less: Net profit 30% of 96,000 = (28,800)
 Less: Selling expenses 15% of 96,000 = (14,400)
Total cost = 52,800
 Less: Cost after separation = (14,400)
Share of joint cost = ₹38,400

W3 Main product's share is whatever is left

Total joint cost = 3,10,400
 Less: Product A's share (W1) = (32,000)
 Less: Product B's share (W2) = (38,400)
Main product's share of joint cost = ₹2,40,000

Comparative Profit and Loss Statement

Particulars	Main product	Product A	Product B	Total
Sales	8,00,000	64,000	96,000	9,60,000
Share of joint cost (W1-W3)	2,40,000	32,000	38,400	3,10,400
Cost after separation	80,000	12,800	14,400	1,07,200
Selling expenses	1,60,000	6,400	14,400	1,80,800
Total cost	4,80,000	51,200	67,200	5,98,400
Net profit (Sales – Total cost)	3,20,000	12,800	28,800	3,61,600
Net profit as % of sales	40%	20%	30%	37.67%

Selling expenses: main 20% of 8,00,000 = 1,60,000 | A 10% of 64,000 = 6,400 | B 15% of 96,000 = 14,400.

THE TRAP / WHERE MARKS ARE LOST

- Taking the profit % on **cost** instead of on sales. Here everything is on sales.
- Forgetting to subtract selling expenses on the way down. If you skip them, A's joint share comes to 38,400 instead of 32,000 and the whole answer shifts.
- Apportioning the joint cost by sales value or by weight. The question gives you profit and selling-expense percentages precisely so you use the reverse-cost method.

FINAL ANSWER

Joint cost — Main product	₹ 2,40,000
Joint cost — Product A	₹ 32,000
Joint cost — Product B	₹ 38,400
Main product's net profit	₹ 3,20,000 (40% of sales)
Total profit of all three	₹ 3,61,600

Q11 Selling expenses have to be derived

Joint & by-product · p.11

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A factory producing an article 'X' also produces product 'Y' which is further processed into a finished product. The joint cost of manufacturing is: Materials 50,000; Labour 30,000; Overheads 20,000 — total ₹1,00,000. The subsequent costs are as follows:

Particulars	X	Y
Materials	30,000	15,000
Labour	14,000	10,000
Overheads	6,000	5,000
Total	50,000	30,000
Selling price	1,60,000	80,000
Estimated profit on selling price	25%	20%

Assume that the selling and distribution expenses are in proportion to the selling price. Show how you would apportion the joint cost of manufacturing and prepare a statement showing the cost of production of both products and the ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- This one has a hidden step. The **selling and distribution expenses are not given** — you must derive the total, then split it.
- How to derive it: total sales 2,40,000 minus total profit gives total cost. Take away the costs you already know (joint 1,00,000 + subsequent 80,000) and the gap must be the S&D expenses.
- "In proportion to the selling price" — so split in the ratio $1,60,000 : 80,000 = 2 : 1$.
- Then it is the ordinary reverse-cost ladder for each product.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Compute each product's profit from the given % of selling price. Add them.
2. Total cost = Total sales — Total profit.
3. **S&D expenses = Total cost — joint cost — subsequent costs.** This is the derived figure.
4. Split the S&D expenses in the ratio of selling prices.
5. For each product climb down: Sales — profit = total cost; then — S&D — subsequent cost = share of joint cost.
6. Check the two joint shares add back to 1,00,000.

WORKING NOTES

W1 Deriving the selling & distribution expenses

Profit on X = 25% of 1,60,000 = 40,000

Profit on Y = 20% of 80,000 = 16,000

Total profit = 56,000

Total sales = 1,60,000 + 80,000 = 2,40,000

Total cost = 2,40,000 – 56,000 = **1,84,000**
 Known costs = joint 1,00,000 + subsequent 80,000 = 1,80,000
 $\therefore$ **S&D expenses = 1,84,000 – 1,80,000 = ₹4,000**

Split 2 : 1 $\rightarrow$ X = 4,000 $\times$ 2/3 = **2,666.67** | Y = 4,000 $\times$ 1/3 = **1,333.33**

W2 Apportionment of the joint cost

Particulars	Product X	Product Y	Total
Selling price	1,60,000	80,000	2,40,000
Less: Estimated profit	(40,000)	(16,000)	(56,000)
Total cost	1,20,000	64,000	1,84,000
Less: Selling & distribution (W1)	(2,666.67)	(1,333.33)	(4,000)
Less: Subsequent cost	(50,000)	(30,000)	(80,000)
Share of joint cost	67,333.33	32,666.67	1,00,000

Cross-check: 67,333.33 + 32,666.67 = 1,00,000 exactly the joint cost given. That agreement is your proof the method is right.

Statement of Cost of Production and Profit

Particulars	Product X	Product Y	Total
Share of joint cost (W2)	67,333.33	32,666.67	1,00,000
Subsequent cost — materials	30,000	15,000	45,000
Subsequent cost — labour	14,000	10,000	24,000
Subsequent cost — overheads	6,000	5,000	11,000
Cost of production	1,17,333.33	62,666.67	1,80,000
Selling & distribution expenses	2,666.67	1,333.33	4,000
Total cost	1,20,000	64,000	1,84,000
Sales	1,60,000	80,000	2,40,000
Profit	40,000	16,000	56,000
Profit as % of selling price	25%	20%	23.33%

Product X Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: 4,000 $\times$ 2/3	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product Y Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: $4,000 \times 1/3$	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

WHY THIS WORKS

The reason the S&D expenses can be derived at all is that the question fixes the profit on *both* products. Once profit is fixed, total cost is fixed, and any cost you have not been told about must be the difference. Whenever a joint-cost question gives you every profit percentage but leaves one cost unnamed, this is the trick being tested.

FINAL ANSWER

Joint cost apportioned to X	₹ 67,333.33
Joint cost apportioned to Y	₹ 32,666.67
Selling & distribution expenses (derived)	₹ 4,000
Cost of production — X	₹ 1,17,333.33
Cost of production — Y	₹ 62,666.67
Total profit	₹ 56,000

Q12 When the printed figures do not reconcile

Joint & by-product · p.11-12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In a manufacturing concern a certain product is manufactured which also yields 2 by-products. The joint expenses of manufacture are: Materials 8,500; Labour 9,000; Overheads 7,500 — total ₹25,000. Subsequent expenses are:

Particulars	A	B	C
Materials	2,500	1,200	1,400
Labour	1,900	1,600	2,000
Overheads	1,500	900	1,050
Total	5,900	3,700	4,450

The selling prices are A — 30,000; B — 20,000 and C — 15,000. Estimated profits on selling prices are A — 40%; B — 30% and C — 25%. Show the apportionment of joint cost and prepare the product accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same ladder as Q10 and Q11, but with **no selling expenses** mentioned — so the ladder is shorter: Sales – profit – subsequent cost = share of joint cost.
- **Read the warning below before you write your answer.** The figures in this question do not reconcile, and knowing how to handle that is worth more than getting a tidy number.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each product: Sales – profit (% of sales) = total cost.
2. Total cost – subsequent cost = **notional share of joint cost**.
3. Add the three notional shares and **compare with the actual joint cost**.
4. If they agree, you are finished. **If they do not agree** (as here), apportion the *actual* joint cost in the *ratio* of the notional shares.
5. Recompute each product's real profit using its apportioned share, and state that the actual profit differs from the estimate.

WORKING NOTES

W1 Notional share of joint cost by reverse cost

Particulars	A	B	C	Total
Selling price	30,000	20,000	15,000	65,000
Less: Estimated profit (40% / 30% / 25%)	(12,000)	(6,000)	(3,750)	(21,750)
Total cost	18,000	14,000	11,250	43,250
Less: Subsequent expenses	(5,900)	(3,700)	(4,450)	(14,050)
Notional share of joint cost	12,100	10,300	6,800	29,200

W2 The figures do not reconcile — and here is how to deal with it

Notional shares add to = **29,200**

Actual joint expenses are only = **25,000**

Difference = **₹4,200** — the estimated profits are more optimistic than the actual cost structure allows.

The joint cost that actually exists is 25,000, so that is all you can apportion. Apportion it in the **ratio of the notional shares** 12,100 : 10,300 : 6,800.

Product	Notional share	Working	Apportioned joint cost
A	12,100	$25,000 \times 12,100 / 29,200$	10,359.59
B	10,300	$25,000 \times 10,300 / 29,200$	8,818.49
C	6,800	$25,000 \times 6,800 / 29,200$	5,821.92
Total	29,200		25,000.00

Statement of Cost and Actual Profit

Particulars	A	B	C	Total
Apportioned joint cost (W2)	10,359.59	8,818.49	5,821.92	25,000
Subsequent expenses	5,900	3,700	4,450	14,050

Particulars	A	B	C	Total
Total cost	16,259.59	12,518.49	10,271.92	39,050
Sales	30,000	20,000	15,000	65,000
Actual profit	13,740.41	7,481.51	4,728.08	25,950
Actual profit as % of sales	45.80%	37.41%	31.52%	39.92%
Estimated profit % (from the question)	40%	30%	25%	—

Product A Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product B Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product C Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

THE TRAP / WHERE MARKS ARE LOST

Please read this one carefully — it is about your exam, not just this sum.

- The notional shares total 29,200 but the joint expenses are 25,000. As printed, the question cannot give a tidy answer. This happens in workbooks.
- The **wrong** response is to write 12,100 / 10,300 / 6,800 as your apportionment. Those figures add to 29,200, which is ₹4,200 more joint cost than the company actually incurred.
- The **right** response is what W2 does: apportion the real 25,000 in the ratio of the notional shares, then show the actual profit percentages and note that they exceed the estimates. Write one sentence saying so. An examiner rewards a student who spots the inconsistency and handles it openly.
- **Do check this one with your professor** — if he teaches the plain notional-share answer, give that instead. Either way, understand why the two differ.

FINAL ANSWER

Apportioned joint cost — A	₹ 10,359.59
Apportioned joint cost — B	₹ 8,818.49
Apportioned joint cost — C	₹ 5,821.92
Total actual profit	₹ 25,950
Notional shares (for the ratio only)	12,100 : 10,300 : 6,800

Q13 Three joint products, full ladder

Joint & by-product · p.12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Bright Chemical Ltd. electrolyses common salt to obtain 3 joint products — caustic soda, chlorine and hydrogen. During a costing period, the expenditure relating to the inputs of the common process amounted to ₹3,50,000. After separation, expenses amounting to ₹1,60,000, ₹75,000 and ₹10,000 were incurred for caustic soda, chlorine and hydrogen respectively. The entire production was sold and ₹3,75,000, ₹2,50,000 and ₹60,000 were realised for caustic soda, chlorine and hydrogen respectively. The selling expenses were estimated at 5% of realisation from sale. The profit is 15%, 10% and 5% of realisation from sale of caustic soda, chlorine and hydrogen respectively.

Draw a columnar statement showing the apportionment of joint cost and the profitability of each product.

READ THE QUESTION FIRST — WHAT TO SPOT

- The full four-rung ladder this time: Sales — profit — selling expenses — separation expenses = share of joint cost.
- Selling expenses are **5% of sales for all three** — one rate, three amounts.
- These are **joint products**, not by-products. That changes nothing about the arithmetic; it only means all three are commercially important, so none is credited to another.
- Like Q12, the derived shares here **do not** add up to the joint cost given. The same ratio treatment applies. Do not be alarmed — be systematic.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Build the ladder in a single columnar table — one column per product, one total column.
2. Row 1 Sales. Row 2 less profit. Row 3 less selling expenses (5% of sales). Row 4 gives total cost.
3. Row 5 less separation expenses. Row 6 gives the notional joint-cost share.
4. Total row 6 and compare with the actual joint cost of 3,50,000.
5. Apportion the actual 3,50,000 in the ratio of the notional shares.
6. Rebuild each column forwards to show the actual profitability, and comment on how it compares with the estimate.

WORKING NOTES

W1 The reverse-cost ladder, all three products together

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Less: Profit (15% / 10% / 5% of sales)	(56,250)	(25,000)	(3,000)	(84,250)
Less: Selling expenses (5% of sales)	(18,750)	(12,500)	(3,000)	(34,250)
Total cost	3,00,000	2,12,500	54,000	5,66,500

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Less: Expenses after separation	(1,60,000)	(75,000)	(10,000)	(2,45,000)
Notional share of joint cost	1,40,000	1,37,500	44,000	3,21,500

W2 Reconciling with the actual joint cost

Notional shares total = **3,21,500**

Actual joint expenditure = **3,50,000**

Shortfall = **₹28,500** — the joint process cost more than the estimated profits allow for, so the actual profits will be *lower* than estimated.

Product	Notional share	Working	Apportioned joint cost
Caustic soda	1,40,000	$3,50,000 \times 1,40,000 / 3,21,500$	1,52,410.58
Chlorine	1,37,500	$3,50,000 \times 1,37,500 / 3,21,500$	1,49,688.96
Hydrogen	44,000	$3,50,000 \times 44,000 / 3,21,500$	47,900.46
Total	3,21,500		3,50,000.00

Columnar Statement of Apportionment and Profitability

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Apportioned joint cost (W2)	1,52,410.58	1,49,688.96	47,900.46	3,50,000
Expenses after separation	1,60,000	75,000	10,000	2,45,000
Selling expenses (5% of sales)	18,750	12,500	3,000	34,250
Total cost	3,31,160.58	2,37,188.96	60,900.46	6,29,250
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Actual profit / (loss)	43,839.42	12,811.04	(900.46)	55,750
Actual profit as % of sales	11.69%	5.12%	(1.50%)	8.14%
Estimated profit % (question)	15%	10%	5%	—

WHY THIS WORKS

Look at the hydrogen column: an actual **loss** of ₹900.46 against an estimated 5% profit. That is a real finding, not an error — the joint process cost ₹28,500 more than the estimates assumed, and hydrogen, having the smallest sales, cannot carry its share. Write that sentence in your answer. It shows you can read a statement instead of only filling it in.

FINAL ANSWER

Apportioned joint cost — Caustic soda **₹ 1,52,410.58**

Apportioned joint cost — Chlorine **₹ 1,49,688.96**

Apportioned joint cost — Hydrogen **₹ 47,900.46**

Total actual profit **₹ 55,750**

Notional shares (ratio basis) **1,40,000 : 1,37,500 : 44,000**

State clearly that the notional shares totalled ₹3,21,500 against actual joint cost of ₹3,50,000, and that you apportioned the actual figure in that ratio.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A by-product 'Beta' is derived in the course of manufacturing product 'Alpha'. The by-product is further processed for sale. From the following data available from the records, prepare an account showing the cost per kg of the product 'Alpha' and the by-product 'Beta'.

Particulars	Joint expenses	Separate — Alpha	Separate — Beta
Materials	10,000	6,000	500
Labour	7,000	5,000	2,000
Overheads	2,500	1,500	600
Total	19,500	12,500	3,100

The quantities produced during the period under consideration were: Alpha — 1,000 kgs and Beta — 500 kgs. The selling price was ₹120/kg on which the profit earned was 30%.

READ THE QUESTION FIRST — WHAT TO SPOT

- A **by-product** problem: Alpha is the main product, Beta is incidental. The normal treatment is to work out Beta's share of the joint cost by reverse cost and leave the balance for Alpha.
- Total cost available to share out = joint 19,500 + Alpha's own 12,500 + Beta's own 3,100 = **₹35,100**. Keep that number in your head — it is the ceiling on every cost figure in this problem.
- **Now look at the selling price.** 500 kg of Beta at 120 would be 60,000 of sales, and 1,000 kg of Alpha at 120 would be 1,20,000. Either one is far bigger than the total cost of 35,100. So the reverse-cost ladder is going to break. Read W2 before you attempt this in the exam.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Identify the main product and the by-product.
2. Apply the reverse-cost ladder to the **by-product**: Sales – profit – its own separate expenses = its share of the joint cost.
3. Joint cost less the by-product's share = the main product's share.
4. Main product cost = its joint share + its own separate expenses. Divide by its kg for cost per kg. Same for the by-product.
5. **Always sanity-check** that no share is negative and that the shares add back to the joint cost. If they do not, the data is faulty — see W2.

WORKING NOTES

W1 The reverse-cost attempt (taking ₹120/kg as Beta's price)

Beta sales = 500 kg × 120 = 60,000

Less: Profit 30% of 60,000 = (18,000)

Total cost of Beta = 42,000

Less: Beta's separate expenses = (3,100)

Beta's share of joint cost = **38,900**

But the whole joint cost is only **19,500** → Alpha's share would be **19,500 – 38,900 = (₹19,400)**, a negative cost.

W2 THE DATA AS PRINTED CANNOT BE SOLVED — AND WHAT TO DO ABOUT IT

A cost can never be negative. Whichever product you assign the ₹120/kg to, the sales value dwarfs the total cost of ₹35,100, so the ladder produces an impossible share. Check the figures against the original in your workbook — most likely the selling price, the profit percentage or the joint expenses has a typographical error.

In the exam, do this: write one line stating the inconsistency, then solve the problem on the next most defensible basis and say which basis you used. You will be given credit for the method. The standard fallback for a by-product whose selling price is unusable is to apportion the joint cost on **physical quantity**, which is shown in W3.

W3 Workable solution — joint cost apportioned on physical quantity

Quantities Alpha 1,000 kg : Beta 500 kg = **2 : 1**

Alpha's share = $19,500 \times 2/3 = \text{₹}13,000$

Beta's share = $19,500 \times 1/3 = \text{₹}6,500$

Particulars	Alpha	Beta	Total
Share of joint cost (2 : 1)	13,000	6,500	19,500
Add: Separate expenses	12,500	3,100	15,600
Total cost	25,500	9,600	35,100
Quantity produced (kg)	1,000	500	1,500
Cost per kg	25.50	19.20	—

Alpha Account (main product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 2/3$	1,000	13	13,000	By Cost of production transferred $25,500 \div 1,000 \text{ kg}$	1,000	25.50	25,500
To Separate materials	—	—	6,000				
To Separate labour	—	—	5,000				
To Separate overheads	—	—	1,500				
Total	1,000	—	25,500	Total	1,000	—	25,500

Beta Account (by-product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 1/3$	500	13	6,500	By Cost of production transferred $9,600 \div 500 \text{ kg}$	500	19.20	9,600
To Separate materials	—	—	500				
To Separate labour	—	—	2,000				
To Separate overheads	—	—	600				
Total	500	—	9,600	Total	500	—	9,600

FINAL ANSWER

Alpha — total cost	₹ 25,500
Alpha — cost per kg	₹ 25.50
Beta — total cost	₹ 9,600

Beta — cost per kg

₹ 19.20

Basis used

Physical quantity 2 : 1 (see W2 for why)

Ask your professor about this one. If he wants the reverse-cost method, the corrected selling price is needed. Show W1, W2 and W3 exactly as laid out above and you will not lose method marks either way.

Q15 Profit as % of cost, plus unrealised profit

Inter-process profit · p.12-13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through two processes X and Y. The output of Process X is charged to Process Y at a price which includes profit of **20% on actual cost** and the output of Process Y is charged to Finished Stock A/c at a price which includes **10% profit on actual cost**. The following data is provided for the month of July.

Particulars	X	Y
Material (2,500 units)	1,250	—
Labour	625	500
Overheads	1,875	750
Indirect material	—	1,250

There was no partly finished WIP. Out of the finished stock, 1,500 units had been sold for ₹7,500. Prepare the Process A/c and Finished Stock A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the **inter-process profit** topic. Each process sells to the next at a mark-up, so the transfer price is bigger than the true cost.
- **“20% on actual cost”** — profit is 20% of cost, so if cost is 3,750 the profit is 750. (Contrast Q16, where profit is a % of the transfer price — a different calculation.)
- There is **no WIP inside the processes**, which makes this the easy version. The only stock is in Finished Stock: 2,500 units produced, 1,500 sold, so **1,000 units left**.
- That closing stock of 1,000 units carries profit added by X and by Y which the company has not actually earned yet, because the goods are still on the shelf. That is the **unrealised profit**, and a reserve must be created for it.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total Process X's own costs. That is its **actual cost**.
2. Add the stated profit percentage **of that cost**. The sum is the transfer price to Y.
3. Debit Process Y with the transfer price (not the cost), add Y's own costs, total to get Y's actual cost.
4. Add Y's profit percentage of that total. The sum transfers to Finished Stock.
5. In Finished Stock, work out the per-unit transfer value, split it between units sold and units in hand, and compute the profit on sale.
6. **Unrealised profit** = total profit loaded by the processes $\times \frac{\text{closing stock units}}{\text{total units produced}}$. Create the reserve for it.

WORKING NOTES

W1 Process X

Material 1,250 + Labour 625 + Overheads 1,875

Actual cost = ₹3,750

Profit 20% of 3,750 = **₹750**

Transfer price to Process Y = ₹4,500

W2 Process Y

Transfer from X 4,500 + Indirect material 1,250 + Labour 500 + Overheads 750

Actual cost = ₹7,000

Profit 10% of 7,000 = **₹700**

Transfer to Finished Stock = ₹7,700 for 2,500 units

W3 Finished Stock

Value per unit = $7,700 \div 2,500 = \text{₹}3.08$

Cost of 1,500 units sold = $1,500 \times 3.08 = \text{₹}4,620$

Closing stock 1,000 units = $1,000 \times 3.08 = \text{₹}3,080$

Profit on sale = $7,500 - 4,620 = \text{₹}2,880$

W4 Reserve for unrealised profit — the marks everyone drops

Profit loaded by the processes = $750 (X) + 700 (Y) = \text{₹}1,450$

True cost of the 2,500 units = $7,700 - 1,450 = 6,250$ (check: $3,750 + 1,250 + 500 + 750 = 6,250$ ✓)

Closing stock is 1,000 of 2,500 units = **40%**

$$\text{Reserve} = \frac{1,000 \text{ units}}{2,500 \text{ units}} \times \text{₹}1,450 = \text{₹}580$$

Equivalently $3,080 \times 1,450 / 7,700 = 580$. Both routes give the same figure.

Process X Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,500	—	1,250	By Transfer to Process Y W1: cost 3,750 + profit 750	2,500	1.80	4,500
To Labour	—	—	625				
To Overheads	—	—	1,875				
Actual cost	—	—	3,750				
To Profit — 20% on cost W1: 20% of 3,750	—	—	750				
Total	2,500	—	4,500	Total	2,500	—	4,500

Process Y Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process X	2,500	1.80	4,500	By Transfer to Finished Stock W2: cost 7,000 + profit 700	2,500	3.08	7,700
To Indirect material	—	—	1,250				
To Labour	—	—	500				
To Overheads	—	—	750				
Actual cost	—	—	7,000				
To Profit — 10% on cost W2: 10% of 7,000	—	—	700				
Total	2,500	—	7,700	Total	2,500	—	7,700

Finished Stock Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process Y	2,500	3.08	7,700	By Sales	1,500	5	7,500
To Profit & Loss A/c (profit on sale) W3: 7,500 – 4,620	—	—	2,880	By Closing stock W3: 1,000 × ₹3.08	1,000	3.08	3,080
Total	2,500	—	10,580	Total	2,500	—	10,580

THE TRAP / WHERE MARKS ARE LOST

- **Confusing “20% on cost” with “20% on transfer price”.** On cost: profit = $0.20 \times \text{cost}$. On transfer price: profit = $0.20 \times \text{TP}$, which means profit = $0.25 \times \text{cost}$. Q16 uses the second form — compare the two side by side once and you will never mix them again.
- Forgetting the reserve for unrealised profit entirely. It is a separate, named figure and questions ask for it explicitly.
- Charging Process Y with X's cost of 3,750. Y is charged the *transfer price* of 4,500 — that is the whole point of the topic.

FINAL ANSWER

Process X — actual cost / transfer price	₹ 3,750 / ₹ 4,500
Process Y — actual cost / transfer price	₹ 7,000 / ₹ 7,700
Profit on sale of 1,500 units	₹ 2,880
Closing stock (1,000 units)	₹ 3,080
Reserve for unrealised profit	₹ 580
Total profit shown (750 + 700 + 2,880)	₹ 4,330
Real profit after the reserve	₹ 3,750

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through three processes to completion, known as A, B and C. The output of each process is charged to the next process at a price calculated to give a profit of **20% on the transfer price**. The output of Process C is charged to finished stock on a similar basis. There was no partly finished WIP in any process on December 31st, on which day the following information was obtained.

Particulars	Process A	Process B	Process C
Materials	4,000	6,000	2,000
Labour	6,000	4,000	8,000
Stock: 31st Dec	2,000	4,000	6,000

There was no stock in hand on Jan 1st and overheads were ignored. Of the goods passed into finished stock, ₹4,000 remained in hand on Dec 31st and the balance has been sold for ₹36,000. Show the Process A/c and calculate reserve for unrealised profits.

READ THE QUESTION FIRST — WHAT TO SPOT

- **“20% on the transfer price”** — not on cost. So if the transfer price is 10,000 the profit is 2,000 and the cost is 8,000. Working backwards: Transfer price = Cost ÷ 0.80.
- **Every process has closing stock this time.** That stock is deducted *before* you load the profit, because you only charge profit on what actually leaves.
- Because stock sits in B and C, and those stocks contain profit passed on from earlier processes, you must track **cost and profit in separate columns**. This is the **three-column format** — Cost | Profit | Total. Use it and this problem is straightforward; try to do it in one column and you will not be able to find the reserve.
- Process A's own stock contains **no** profit — nothing has been marked up into it yet. Only B, C and Finished Stock hold unrealised profit.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Draw three columns for every process: **Cost, Profit, Total**.
2. Enter the transfer-in split into its cost and profit parts, then the process's own costs (all in the Cost column — own costs carry no profit).
3. Total the three columns.
4. Deduct closing stock. Split it in the **same cost : profit ratio as the total debit**: stock cost part = stock $\times \frac{\text{total cost}}{\text{total debit}}$.
5. What remains is the cost of goods transferred. Transfer price = that cost ÷ 0.80; the difference is the profit for this process.
6. Repeat for the next process. In Finished Stock do the same split for the goods in hand.
7. **Reserve = the Profit column of every closing stock added together.**
8. Cross-check: total apparent profit – reserve = the realised profit from Finished Stock.

WORKING NOTES

W1 Turning “20% on transfer price” into arithmetic

$$\text{Transfer price} = \frac{\text{Cost of goods transferred}}{0.80} \quad \text{and} \quad \text{Profit} = 20\% \text{ of that price}$$

Because Cost = 80% of the transfer price. Note this is the same as 25% on cost — but always work from the price, as the question states it.

W2 Splitting each closing stock into cost and profit

Process	Total debit	of which Cost	Closing stock	Stock — cost part	Stock — profit part
A	10,000	10,000	2,000	2,000	Nil
B	20,000	18,000	4,000	$4,000 \times 18/20 = 3,600$	400
C	30,000	24,400	6,000	$6,000 \times 24,400/30,000 = 4,880$	1,120
Finished Stock	30,000	19,520	4,000	$4,000 \times 19,520/30,000 = 2,602.67$	1,397.33
Total reserve					2,917.33

Process A Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Materials	4,000	—	4,000
To Labour	6,000	—	6,000
Total debit	10,000	—	10,000
Less: Closing stock	(2,000)	—	(2,000)
Cost of goods transferred	8,000	—	8,000
Add: Profit — 20% on transfer price $8,000 \div 0.80 = 10,000$; profit 2,000	—	2,000	2,000
Transferred to Process B	8,000	2,000	10,000

Process B Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process A	8,000	2,000	10,000
To Materials	6,000	—	6,000
To Labour	4,000	—	4,000
Total debit	18,000	2,000	20,000
Less: Closing stock W2: 4,000 split 18/20	(3,600)	(400)	(4,000)
Cost of goods transferred	14,400	1,600	16,000
Add: Profit $16,000 \div 0.80 = 20,000$; profit 4,000	—	4,000	4,000
Transferred to Process C	14,400	5,600	20,000

Process C Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process B	14,400	5,600	20,000
To Materials	2,000	—	2,000
To Labour	8,000	—	8,000

Particulars	Cost ₹	Profit ₹	Total ₹
Total debit	24,400	5,600	30,000
Less: Closing stock W2: 6,000 split 24,400/30,000	(4,880)	(1,120)	(6,000)
Cost of goods transferred	19,520	4,480	24,000
Add: Profit $24,000 \div 0.80 = 30,000$; profit 6,000	—	6,000	6,000
Transferred to Finished Stock	19,520	10,480	30,000

Finished Stock Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process C	19,520	10,480	30,000
Less: Closing stock in hand W2: 4,000 split 19,520/30,000	(2,602.67)	(1,397.33)	(4,000)
Cost of sales	16,917.33	9,082.67	26,000
Sales	—	—	36,000
Profit on sale $36,000 - 26,000$	—	10,000	10,000
Total profit realised	—	19,082.67	—

Reconciliation — proof that the reserve is right

Particulars	Amount ₹
Profit shown by Process A	2,000
Profit shown by Process B	4,000
Profit shown by Process C	6,000
Profit on sale in Finished Stock	10,000
Total apparent profit	22,000.00
Less: Reserve for unrealised profit (W2)	(2,917.33)
Actual (realised) profit	19,082.67
Agrees with the Finished Stock A/c profit column ✓	19,082.67

WHY THIS WORKS

The reserve exists because the company has already recorded ₹22,000 of profit, but ₹2,917.33 of that is sitting inside goods it still owns. You cannot count profit on goods you have not sold. Removing the reserve brings the books back to the profit actually earned — and the fact that it reconciles to the paisa is your proof the whole answer is right.

FINAL ANSWER

Process A — profit / transfer price	₹ 2,000 / ₹ 10,000
Process B — profit / transfer price	₹ 4,000 / ₹ 20,000
Process C — profit / transfer price	₹ 6,000 / ₹ 30,000
Profit on sale	₹ 10,000
Total apparent profit	₹ 22,000
Reserve for unrealised profit	₹ 2,917.33
Actual realised profit	₹ 19,082.67

Reserve built up as: A nil + B 400 + C 1,120 + Finished Stock 1,397.33 = ₹2,917.33.

Equivalent Production

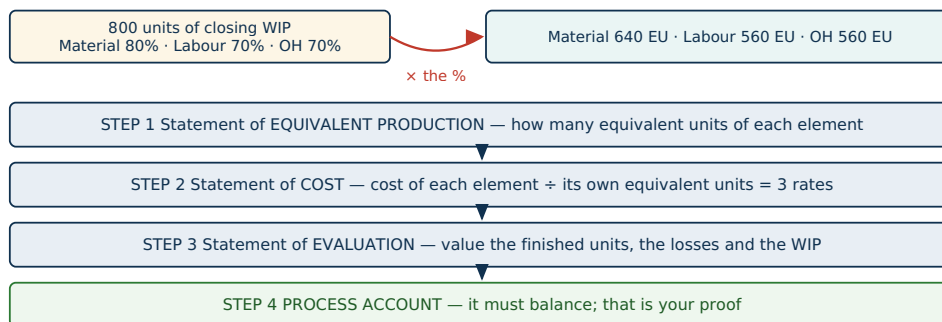
Problems 17 to 24 · workbook pages 13 to 16

THE SINGLE IDEA BEHIND ALL EIGHT PROBLEMS

Until now every unit was either finished or lost. Now some units are **half-finished** at the end of the month. You cannot divide cost by “3,000 finished units + 800 half-finished units”, because those 800 have not absorbed a full unit’s worth of cost.

So you convert them. **800 units that are 70% complete are treated as 560 complete units.** That converted figure is called an **equivalent unit**, and once you have it the problem collapses back into the ordinary “total cost ÷ units” you already know.

The one extra wrinkle: material, labour and overhead are usually at *different* stages of completion. So you do the conversion **three times** and get **three separate rates**.



Four statements, always in this order, in every one of Q17 to Q24. Never skip one.

The rules for the equivalent-production statement

What appears in the output column	Material	Labour	Overhead	Why
Units completed & transferred	100%	100%	100%	They are finished. Full cost in every element.
Normal loss	Nil	Nil	Nil	This is the rule students forget. Normal loss gets zero equivalent units — its cost is absorbed by the good units, so giving it units would double-count.
Abnormal loss	its % of completion	its %	its %	It is a real loss of real cost, so it must carry equivalent units. If the question says the scrapped units were 100% complete, use 100%.
Abnormal gain	100%	100%	100%	Always 100%, and it is deducted in the statement, not added.

What appears in the output column	Material	Labour	Overhead	Why
Closing WIP	its stated %	its stated %	its stated %	The whole reason the topic exists.
Opening WIP — FIFO only	the % still <i>to do</i>	the % to do	the % to do	Under FIFO the opening WIP was partly done last month, so only the balance belongs to this month. 40% complete means 60% to do.

FIFO VERSUS AVERAGE — THE ONLY DIFFERENCE THAT MATTERS

	FIFO method	Average method
Opening WIP units in the statement	Only the unfinished portion (the % still to do)	Not shown separately — the opening units are inside the completed figure at 100%
Opening WIP cost	Kept apart. Added back only at the evaluation stage, on top of the cost of finishing it.	Merged with this month's cost before dividing.
Which costs go into the rate	This month's costs only	Opening WIP cost + this month's costs
Use it when	The question says FIFO, or gives the opening WIP's degree of completion element by element	The question says average, or gives the opening WIP cost as one lump you cannot split

Q21 and Q22 are FIFO. Q23 is average. **Q24 asks for both** on the same data, which is the best possible way to see the difference — work it and compare the two totals.

Q17 The four statements, cleanest case

Equivalent production · p.13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In Process A on March 1st, there was no opening work-in-progress. During the month of March, 2,000 units of material was issued at a cost of ₹18,000. Labour and overheads totalled to ₹9,000 and ₹6,600 respectively. On 31st March 1,500 units were completed and transferred to the next process. Of the remaining 500 units which are incomplete, degree of completion was: Material 100%, Labour 60% and Overheads 30%.

Prepare (a) Statement of equivalent production (b) Statement of cost (c) Statement of evaluation (d) Process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **No opening WIP and no losses.** The simplest possible version — the perfect one to learn the four statements on.
- Units: 2,000 in = 1,500 transferred + 500 closing WIP. Balanced, so nothing is lost.
- **Material is 100% complete.** That is normal — material is usually issued in full at the start, so closing WIP has all its material but only part of its labour and overhead.
- Three different percentages means **three different rates**. Do not average them into one.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,500	100	1,500	100	1,500	100	1,500
Closing work-in-progress	500	100	500	60	300	30	150
Total equivalent units			2,000		1,800		1,650

(b) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	18,000	2,000	9.00
Labour	9,000	1,800	5.00
Overhead	6,600	1,650	4.00
Total cost per unit			18.00

(c) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,500 units × ₹18.00	27,000
Closing work-in-progress		
Material	500 EU × ₹9.00	4,500
Labour	300 EU × ₹5.00	1,500
Overhead	150 EU × ₹4.00	600
Value of closing WIP		6,600
Total		33,600

(d) Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	—	18,000	By Transfer to next process 1,500 × ₹18	1,500	18	27,000
To Labour	—	—	9,000	By Closing WIP statement (c)	500	—	6,600
To Overheads	—	—	6,600				
Total	2,000	—	33,600	Total	2,000	—	33,600

THE TRAP / WHERE MARKS ARE LOST

- Dividing all three costs by 2,000. Only material has 2,000 equivalent units.
- Valuing closing WIP at 500 × total rate 18 = 9,000. Wrong — you must value each element on **its own** equivalent units, giving 6,600.
- Skipping the statement of evaluation and jumping to the account. The three statements each carry marks of their own.

FINAL ANSWER

Cost per unit — Material / Labour / Overhead **₹ 9.00 / ₹ 5.00 / ₹ 4.00**

Total cost per unit **₹ 18.00**

Value of 1,500 units transferred **₹ 27,000**

Value of closing WIP (500 units) **₹ 6,600**

Process account total **₹ 33,600**

Proof: 18,000 + 9,000 + 6,600 = 33,600 = 27,000 + 6,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

During January 2,000 units were introduced into Process 1. The normal loss was estimated at 5% on input. At the end of the month 1,400 units had been produced and transferred to the next process. 460 units were incomplete and 140 units had been scrapped. It was estimated that incomplete units had reached a stage in production as follows: Material 75% complete, Labour 50% complete and overheads 50% complete.

The cost of 2,000 units introduced was ₹5,800. Direct material introduced during the process amounted to ₹1,440. Direct labour cost was ₹3,340. Production overheads incurred were ₹1,670. Units scrapped realised ₹1 each. The units scrapped have passed through the processes so were 100% complete as regards material, labour and overheads.

You are required to (a) prepare a statement of equivalent production (b) evaluate the cost of abnormal loss, finished goods and closing stock and (c) prepare the Process 1 a/c and abnormal loss a/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 2,000 = 1,400 transferred + 460 WIP + 140 scrapped. Balanced
- **Normal loss = 5% of 2,000 = 100 units.** Actually scrapped = 140. So **abnormal loss = 40 units.**
- **Normal loss gets NIL equivalent units** in the statement. Abnormal loss gets 100% in every element, because the question tells you the scrapped units were fully processed.
- The scrap money of ₹1 per unit applies to the **normal loss units only** when you reduce the material cost. The abnormal loss units also fetch ₹1 each, but that goes in the Abnormal Loss account, not in the cost statement.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,400	100	1,400	100	1,400	100	1,400
Normal loss (5% of 2,000)	100	—	Nil	—	Nil	—	Nil
Abnormal loss	40	100	40	100	40	100	40
Closing work-in-progress	460	75	345	50	230	50	230
Total equivalent units			1,785		1,670		1,670

(b1) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 5,800 + 1,440 – 100 scrap	7,140	1,785	4.00
Labour	3,340	1,670	2.00
Overhead	1,670	1,670	1.00
Total cost per unit			7.00

Material cost = cost of units introduced 5,800 + material added 1,440 = 7,240

Less: scrap value of **normal loss** 100 units × ₹1 = (100)

Net material cost = ₹7,140

(b2) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,400 × ₹7.00	9,800
Abnormal loss	40 × ₹7.00	280
Closing work-in-progress		
Material	345 EU × ₹4.00	1,380
Labour	230 EU × ₹2.00	460
Overhead	230 EU × ₹1.00	230
Value of closing WIP		2,070
Total		12,150

(c) Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	2,000	—	5,800	By Normal loss 100 units × ₹1	100	1	100
To Direct material	—	—	1,440	By Abnormal loss 40 × ₹7	40	7	280
To Direct labour	—	—	3,340	By Transfer to next process 1,400 × ₹7	1,400	7	9,800
To Production overheads	—	—	1,670	By Closing WIP statement (b2)	460	—	2,070
Total	2,000	—	12,250	Total	2,000	—	12,250

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	40	7	280	By Cash / Cost Ledger (scrap) 40 × ₹1	40	1	40
Total	40	—	280	By Profit & Loss A/c	—	—	240
				Total	40	—	280

THE TRAP / WHERE MARKS ARE LOST

- **Giving the normal loss 100 equivalent units.** It gets nil. This is the single most common error in the topic.
- Deducting the scrap of all 140 scrapped units (140) from material cost instead of only the normal loss 100.
- Forgetting that the abnormal loss must be valued at the full rate of 7.00 even though it only sells for 1.00.

FINAL ANSWER

Equivalent units — Material / Labour / Overhead	1,785 / 1,670 / 1,670
Cost per unit — Material / Labour / Overhead	₹ 4.00 / ₹ 2.00 / ₹ 1.00
Total cost per unit	₹ 7.00
Finished goods (1,400 units)	₹ 9,800
Abnormal loss (40 units)	₹ 280 — net to P&L ₹ 240
Closing stock (460 units)	₹ 2,070

Proof: 12,250 debits = 100 + 280 + 9,800 + 2,070 = 12,250

Q19 Closing WIP has to be derived

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare statement of equivalent production, statement of cost and process a/c from the following information.

Particulars	
Units introduced	7,600
Output (units)	6,000
Process cost — Materials	₹ 14,560
Process cost — Labour	₹ 21,360
Process cost — Overheads	₹ 14,240
Degree of completion for closing WIP — Materials	80%
Degree of completion for closing WIP — Labour	70%
Degree of completion for closing WIP — Overheads	70%

READ THE QUESTION FIRST — WHAT TO SPOT

- The closing WIP figure is **not given** — you must derive it: 7,600 introduced – 6,000 output = **1,600 units** of closing WIP.
- No losses are mentioned and the units balance exactly, so there is no normal or abnormal loss working at all.
- No opening WIP either, so the FIFO / average question does not arise.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	6,000	100	6,000	100	6,000	100	6,000
Closing WIP $7,600 - 6,000$	1,600	80	1,280	70	1,120	70	1,120
Total equivalent units			7,280		7,120		7,120

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	14,560	7,280	2.00
Labour	21,360	7,120	3.00
Overhead	14,240	7,120	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	$6,000 \times ₹7.00$	42,000
Closing work-in-progress		

Particulars	Working	Amount ₹
Material	1,280 EU × ₹2.00	2,560
Labour	1,120 EU × ₹3.00	3,360
Overhead	1,120 EU × ₹2.00	2,240
Value of closing WIP		8,160
Total		50,160

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	7,600	—	14,560	By Transfer to next process	6,000	7	42,000
To Labour	—	—	21,360	By Closing WIP	1,600	—	8,160
To Overheads	—	—	14,240				
Total	7,600	—	50,160	Total	7,600	—	50,160

FINAL ANSWER

Closing WIP (derived) **1,600 units**

Equivalent units — Material / Labour / Overhead **7,280 / 7,120 / 7,120**

Cost per unit **₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00**

Transferred out (6,000 units) **₹ 42,000**

Closing WIP **₹ 8,160**

Proof: 14,560 + 21,360 + 14,240 = 50,160 = 42,000 + 8,160

Q20 Normal loss, no abnormal item

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the process account from the following. Number of units introduced into the process — 4,000 units. Units completed and transferred — 3,000 units. Work-in-progress at the end of the period — 800 units (stage of completion material 80%, labour 70% and overheads 70%). Normal loss at the end of the period estimated 200 units (scrap value at ₹1 per unit). Value of materials ₹7,480, wages ₹10,680 and overheads ₹7,120.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 4,000 = 3,000 transferred + 800 WIP + 200 normal loss. Balanced exactly, so **there is no abnormal loss or gain** — do not invent one.
- Normal loss again takes **nil equivalent units**, and its scrap value of 200 × ₹1 = ₹200 comes off the material cost.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	3,000	100	3,000	100	3,000	100	3,000
Normal loss	200	—	Nil	—	Nil	—	Nil

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Closing WIP	800	80	640	70	560	70	560
Total equivalent units			3,640		3,560		3,560

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 7,480 – 200 scrap	7,280	3,640	2.00
Wages	10,680	3,560	3.00
Overhead	7,120	3,560	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	3,000 × ₹7.00	21,000
Closing work-in-progress		
Material	640 EU × ₹2.00	1,280
Wages	560 EU × ₹3.00	1,680
Overhead	560 EU × ₹2.00	1,120
Value of closing WIP		4,080
Total		25,080

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	4,000	—	7,480	By Normal loss 200 × ₹1	200	1	200
To Wages	—	—	10,680	By Transfer to next process	3,000	7	21,000
To Overheads	—	—	7,120	By Closing WIP	800	—	4,080
Total	4,000	—	25,280	Total	4,000	—	25,280

FINAL ANSWER

Equivalent units — Material / Wages / Overhead	3,640 / 3,560 / 3,560
Cost per unit	₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00
Transferred out (3,000 units)	₹ 21,000
Closing WIP (800 units)	₹ 4,080
Normal loss scrap	₹ 200
Proof: 25,280 = 200 + 21,000 + 4,080 = 25,280	

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company follows process costing and manufactures a product in one process. The work-in-process at the end of each month is valued according to **FIFO** method. At the beginning of the month of January, the inventory of WIP showed 400 units, 40% completed, valued as follows: Materials ₹3,600; Labour ₹3,400; Overheads ₹1,000 — total ₹8,000.

In the month of January, materials were purchased for ₹75,000. Wages and overheads in the month amounted to ₹79,800 and ₹21,280 respectively. **Actual issue of materials to production was ₹68,500.** Finished production taken into the stock in the month was 2,500 units. There was no loss in the process. At the end of the month, the WIP inventory was 500 units and 80% complete as regards materials and 60% complete as regards labour and overhead.

You are required to compute equivalent production and prepare process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The ₹75,000 of purchases is a deliberate trap.** Purchases are not a process cost. Only what is **issued to production** — ₹68,500 — enters the process. Use 68,500 and ignore 75,000.
- **FIFO is stated explicitly.** So the opening WIP of 400 units, already 40% done, needs only the remaining **60%** of work this month → 240 equivalent units in each element.
- Units introduced must be derived: 2,500 completed + 500 closing WIP – 400 opening WIP = **2,600 units introduced.**
- Under FIFO the **opening WIP cost of ₹8,000 is kept out of the rate.** It is added back only when you value the finished goods.

WORKING NOTES

W1 Unit flow

Opening WIP = 400 units | Completed = 2,500 units | Closing WIP = 500 units

Units introduced = 2,500 + 500 – 400 = 2,600 units

Of the 2,500 completed, 400 came from opening WIP → **2,100 were introduced and completed this month**

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do 40% done, so 60% remains	400	60	240	60	240	60	240
Introduced and completed this month	2,100	100	2,100	100	2,100	100	2,100
Closing WIP	500	80	400	60	300	60	300
Total equivalent units			2,740		2,640		2,640

Statement of Cost — this month's costs only

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material issued, not purchased	68,500	2,740	25.0000
Labour	79,800	2,640	30.2273

Element	Cost ₹	Equivalent units	Cost per unit ₹
Overhead	21,280	2,640	8.0606
Total cost per unit			63.2879

The labour and overhead rates do not come out round in this problem. Carry four decimal places through the evaluation and the process account will still balance to the rupee — which is shown below.

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening WIP completed (400 units)		
Cost brought forward	given	8,000.00
Material to finish	240 EU × ₹25.0000	6,000.00
Labour to finish	240 EU × ₹30.2273	7,254.55
Overhead to finish	240 EU × ₹8.0606	1,934.55
Total for the 400 units		23,189.10
2. Introduced and completed (2,100 units)		
At full cost	2,100 × ₹63.2879	1,32,904.55
3. Closing WIP (500 units)		
Material	400 EU × ₹25.0000	10,000.00
Labour	300 EU × ₹30.2273	9,068.18
Overhead	300 EU × ₹8.0606	2,418.18
Value of closing WIP		21,486.36
Total		1,77,580.01

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening WIP	400	—	8,000	By Finished goods transferred 23,189.10 + 1,32,904.55	2,500	—	1,56,093.65
To Materials issued	2,600	—	68,500	By Closing WIP	500	—	21,486.36
To Wages	—	—	79,800				
To Overheads	—	—	21,280				
Total	3,000	—	1,77,580	Total	3,000	—	1,77,580.01

THE TRAP / WHERE MARKS ARE LOST

- Using the ₹75,000 **purchases** figure instead of the ₹68,500 **issued** figure. Purchases go to the stores account, not the process.
- Giving the opening WIP 400 equivalent units. Under FIFO it gets only the 240 that represent work done *this* month.
- Adding the opening WIP cost of ₹8,000 into the numerator before dividing. Under FIFO it stays out of the rate and is added back at evaluation.

FINAL ANSWER

Units introduced (derived)	2,600
Equivalent units — Material / Labour / Overhead	2,740 / 2,640 / 2,640
Cost per unit — Material	₹ 25.0000
Cost per unit — Labour	₹ 30.2273
Cost per unit — Overhead	₹ 8.0606
Finished goods (2,500 units)	₹ 1,56,093.65
Closing WIP (500 units)	₹ 21,486.36

Proof: $8,000 + 68,500 + 79,800 + 21,280 = 1,77,580$ and $1,56,093.65 + 21,486.36 = 1,77,580.01$ (1 paisa rounding).

Q22 FIFO with scrap and two material inputs

Equivalent production · p.15

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare Process 2 account from the following. Opening stock 600 units at ₹1,050 (degree of completion material 80%, labour 60%, overheads 60%). Transfer from Process 1 — 11,000 units at ₹5,500. Transfer to Process 3 — 8,800 units. Direct material added in Process 2 — ₹2,410. Direct labour ₹7,155, Production overhead ₹9,540. Units scrapped — 1,200 (degree of completion material 100%, labour 70% and overheads 70%). Closing stock 1,600 units (degree of completion material 70%, labour 60%, overheads 60%). There was a normal loss in the process at 10% of production. Units scrapped would realise 50 paise per unit.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units in: opening 600 + received 11,000 = 11,600. Units out: 8,800 + 1,200 scrapped + 1,600 closing = 11,600
- “Normal loss at 10% of production” is ambiguous wording. The standard reading is 10% of the units introduced into the process from Process 1 = 10% of 11,000 = **1,100 units**. State your assumption in one line. Abnormal loss then = $1,200 - 1,100 = 100$ units.
- The opening stock’s degrees of completion are given element by element, so **use FIFO**.
- Two material figures must be added: the ₹5,500 transferred in and the ₹2,410 added here. Then take off the normal loss scrap of $1,100 \times ₹0.50 = ₹550$.

WORKING NOTES

W1 Loss analysis (assumption stated)

Normal loss = 10% of 11,000 units introduced = **1,100 units**

Units actually scrapped = 1,200 units

Abnormal loss = $1,200 - 1,100 = 100$ units

Scrap credit taken to cost statement = $1,100 \times ₹0.50 = ₹550$

W2 FIFO unit flow

Opening stock 600 units, so of the 8,800 transferred out, 600 came from opening stock

Introduced and completed = $8,800 - 600 = 8,200$ units

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening stock — work still to do Mat 20%, Lab 40%, OH 40%	600	20	120	40	240	40	240
Introduced and completed	8,200	100	8,200	100	8,200	100	8,200
Normal loss (W1)	1,100	—	Nil	—	Nil	—	Nil
Abnormal loss (W1)	100	100	100	70	70	70	70
Closing stock	1,600	70	1,120	60	960	60	960
Total equivalent units			9,540		9,470		9,470

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 5,500 + 2,410 – 550	7,360	9,540	0.7715
Labour	7,155	9,470	0.7555
Overhead	9,540	9,470	1.0074
Total cost per unit			2.5344

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening stock completed (600 units)		
Cost brought forward	given	1,050.00
Material to finish	120 EU × ₹0.7715	92.58
Labour to finish	240 EU × ₹0.7555	181.33
Overhead to finish	240 EU × ₹1.0074	241.77
Sub-total		1,565.68
2. Introduced and completed (8,200 units)		
At full cost	8,200 × ₹2.5344	20,782.28
Transferred to Process 3 (8,800 units)		22,347.96
3. Abnormal loss (100 units)		
Material / Labour / Overhead	100 × 0.7715 + 70 × 0.7555 + 70 × 1.0074	200.56
4. Closing stock (1,600 units)		
Material	1,120 EU × ₹0.7715	864.07
Labour	960 EU × ₹0.7555	725.32
Overhead	960 EU × ₹1.0074	967.10
Value of closing stock		2,556.49
Total		25,105.01

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	600	—	1,050	By Normal loss $1,100 \times ₹0.50$	1,100	0.50	550
To Transfer from Process 1	11,000	0.50	5,500	By Abnormal loss	100	—	200.56
To Direct material added	—	—	2,410	By Transfer to Process 3	8,800	—	22,347.96
To Direct labour	—	—	7,155	By Closing stock	1,600	—	2,556.49
To Production overhead	—	—	9,540				
Total	11,600	—	25,655	Total	11,600	—	25,655.01

THE TRAP / WHERE MARKS ARE LOST

About the ambiguous wording. “Normal loss at 10% of production” could be read as 10% of units introduced (1,100), 10% of total input including opening stock (1,160), or 10% of output transferred (880). I have used **10% of units introduced = 1,100**, which is the most common convention. In the exam, **write down which basis you have used** in one line. Method marks are given for a stated, consistent assumption; they are lost only for silence.

FINAL ANSWER

Normal loss / Abnormal loss	1,100 units / 100 units
Equivalent units — Material / Labour / Overhead	9,540 / 9,470 / 9,470
Cost per unit	₹ 0.7715 + ₹ 0.7555 + ₹ 1.0074 = ₹ 2.5344
Transferred to Process 3 (8,800 units)	₹ 22,347.96
Abnormal loss (100 units)	₹ 200.56
Closing stock (1,600 units)	₹ 2,556.49

Proof: debits 1,050 + 5,500 + 2,410 + 7,155 + 9,540 = 25,655 and credits 550 + 200.56 + 22,347.96 + 2,556.49 = 25,655.01

Q23 The average method

Equivalent production · p.15-16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following figures relate to a single industrial process. Opening stock 10,000 units — Material ₹2,250, Wages ₹650, Overheads ₹400, total ₹3,300. Units introduced 40,000 units — Material ₹9,250, Labour ₹4,600, Overheads ₹3,100.

During the period 30,000 units were completed and 20,000 units remained in process. The degree of completion of closing stock of WIP was: Materials 100%, Labour 25%, Overheads 25%. Make necessary computations and prepare process account by using the **average method**.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The question says average method**, so everything changes from Q21 and Q22:
- The opening stock does **not** appear as a separate line in the equivalent production statement. Its 10,000 units are simply inside the 30,000 completed, at 100%.
- The opening stock **cost is merged** with this month's cost before you divide. Material becomes 2,250 + 9,250 = 11,500, and so on.
- Units: 10,000 + 40,000 = 50,000 = 30,000 completed + 20,000 WIP no losses.

Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred <i>includes the opening 10,000 at 100%</i>	30,000	100	30,000	100	30,000	100	30,000
Closing WIP	20,000	100	20,000	25	5,000	25	5,000
Total equivalent units			50,000		35,000		35,000

Statement of Cost — opening cost merged with current cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material <i>2,250 + 9,250</i>	11,500	50,000	0.23
Wages <i>650 + 4,600</i>	5,250	35,000	0.15
Overheads <i>400 + 3,100</i>	3,500	35,000	0.10
Total cost per unit			0.48

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	30,000 × ₹0.48	14,400
Closing work-in-progress		
Material	20,000 EU × ₹0.23	4,600
Wages	5,000 EU × ₹0.15	750
Overheads	5,000 EU × ₹0.10	500
Value of closing WIP		5,850
Total		20,250

Process Account (average method)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	10,000	—	3,300	By Transfer to finished stock	30,000	0.48	14,400
To Material	40,000	—	9,250	By Closing WIP	20,000	—	5,850
To Labour	—	—	4,600				
To Overheads	—	—	3,100				
Total	50,000	—	20,250	Total	50,000	—	20,250

WHY THIS WORKS

Notice how much shorter this is than Q21. Under the average method you never separate “work done last month” from “work done this month” — you pool everything and take one average rate. That is why it is the easier method, and why examiners often specify FIFO instead.

FINAL ANSWER

Equivalent units — Material / Wages / Overheads **50,000 / 35,000 / 35,000**
 Cost per unit **₹ 0.23 + ₹ 0.15 + ₹ 0.10 = ₹ 0.48**
 Transferred out (30,000 units) **₹ 14,400**

Closing WIP (20,000 units)

₹ 5,850

Proof: $3,300 + 9,250 + 4,600 + 3,100 = 20,250 = 14,400 + 5,850$

Q24 FIFO and average on the same data

Equivalent production · p.16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the statement of equivalent production from the following under **FIFO and Average Method**.

Opening work-in-progress — 2,000 units (completed as material 80%, labour 60% and overheads 60%). Units introduced and completed 5,000 units. Units remaining as closing work-in-progress — 3,000 units (completed as material 80%, labour and overheads 60% each). There are no process losses.

READ THE QUESTION FIRST — WHAT TO SPOT

- No costs are given, so **only the two statements are wanted** — no cost statement, no evaluation, no process account. Answer exactly what is asked.
- Total units completed = opening WIP 2,000 (finished first) + 5,000 introduced and completed = **7,000 units**. Units introduced = 5,000 + 3,000 closing = **8,000 units**.
- This is the best problem in the module for **seeing** the FIFO/average difference, because the same data gives two different answers. Work both and compare the totals.

(a) Statement of Equivalent Production — FIFO method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do Mat 20%, Lab 40%, OH 40%	2,000	20	400	40	800	40	800
Introduced and completed	5,000	100	5,000	100	5,000	100	5,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			7,800		7,600		7,600

(b) Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred 2,000 opening + 5,000 = 7,000, all at 100%	7,000	100	7,000	100	7,000	100	7,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			9,400		8,800		8,800

The difference, side by side

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Material	7,800	9,400	1,600	Average counts the opening WIP's material at 100% (2,000) instead of only the 20% still to do (400). Difference = $2,000 - 400 = 1,600$.
Labour	7,600	8,800	1,200	2,000 at 100% instead of 800 → difference 1,200.

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Overhead	7,600	8,800	1,200	Same as labour.

WHY THIS WORKS

Average always gives the **larger** number of equivalent units, because it counts the whole of the opening WIP instead of just the unfinished part. Larger equivalent units with a larger cost pool (opening cost is merged in too) means the two methods usually give similar rates — but never identical ones. Whenever a question mentions opening WIP, your first job is to find out which method it wants.

FINAL ANSWER

FIFO — Material / Labour / Overhead **7,800 / 7,600 / 7,600**

Average — Material / Labour / Overhead **9,400 / 8,800 / 8,800**

Units completed **7,000**

Units introduced **8,000**

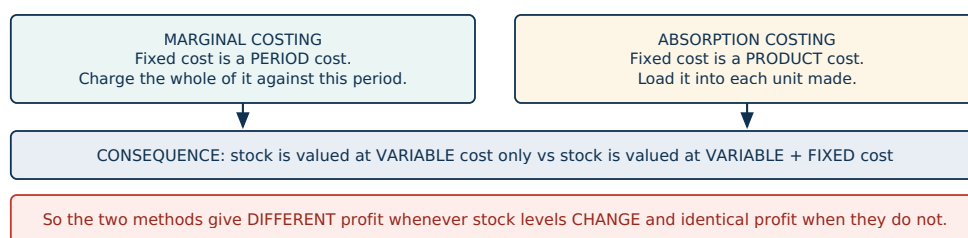
Under FIFO only the 400 / 800 / 800 of opening-WIP work belongs to this period; under average the full 2,000 units are counted at 100%.

Marginal Costing and Absorption Costing

20 problems · workbook pages 21 to 27

The one difference that creates this whole module

There is exactly **one** disagreement between marginal costing and absorption costing, and every problem in this module is a consequence of it:



Fix this picture in your head. It answers every 'why is the profit different?' question.

If ...	Then ...
Production = Sales (no change in stock)	Both methods give the same profit.
Production > Sales (stock going up)	Absorption profit is HIGHER , because some of this period's fixed cost has been parked inside the closing stock instead of being charged.
Production < Sales (stock coming down)	Absorption profit is LOWER , because last period's fixed cost comes out of opening stock and gets charged now, on top of this period's.

Absorption profit – Marginal profit = Fixed overhead in closing stock – Fixed overhead in opening stock

This is the reconciliation. Every marginal-versus-absorption problem in this module can be checked with this one line. If your two profits differ by anything else, you have made an error.

The two families of problem in Module 2

Family	What you are asked to do	Problems
A. Profit statements	Prepare a profit statement under marginal costing, under absorption costing, or both, and reconcile them. Watch stock movements and the under / over absorption of fixed overhead.	1, 15-20

Family	What you are asked to do	Problems
B. Cost-Volume-Profit	Contribution, P/V ratio, break-even point, margin of safety, required sales for a target profit. Pure formula work — the same six formulas every time.	2-14

Profit statements

Use these steps for Q1, 15, 16, 17, 18, 19 and 20

The two statement formats — learn the skeletons, fill in the numbers

MARGINAL COSTING statement	ABSORPTION COSTING statement
Sales	Sales
Less: Opening stock at VARIABLE cost per unit	Less: Opening stock at FULL cost per unit
Add: Variable cost of production	Add: Full cost of production variable + fixed absorbed
Less: Closing stock at VARIABLE cost	Less: Closing stock at FULL cost
= Variable cost of sales	= Cost of sales
CONTRIBUTION = Sales – variable cost of sales	GROSS PROFIT = Sales – cost of sales
Less: ALL fixed costs the actual amount for the period	Less / Add: Under / over absorption of fixed production overhead
	Less: Fixed selling, admin and distribution costs
= PROFIT	= PROFIT

THE SEVEN STEPS, EVERY TIME

1. **Work out the stock movement in units first.** Opening + Production – Sales = Closing. Write this line down before anything else — it tells you immediately which method will show the higher profit.
2. **Variable cost per unit** = direct material + direct labour + variable overheads. (Variable *selling* overhead is a cost of *selling*, so it never goes into stock value.)
3. **Fixed overhead absorption rate** = $\frac{\text{Budgeted fixed production overhead}}{\text{Budgeted / normal output in units}}$. Use **normal or budgeted** output, never actual.
4. **Full cost per unit** = variable production cost + that absorption rate.
5. Build the marginal statement. Charge the **whole actual fixed cost** for the period.
6. Build the absorption statement. Then compute **absorbed = actual production × rate** and compare with the **actual fixed overhead**. Absorbed more than actual = **over-absorption** (add it back); absorbed less = **under-absorption** (deduct it).
7. **Reconcile.** The difference between the two profits must equal the fixed overhead inside the change in stock. If it does not, find your error before writing the answer.

THE THREE ERRORS THAT COST THE MOST MARKS

- **Valuing stock at full cost in the marginal statement.** Under marginal costing stock carries variable cost only. This is the definition of the method.

- **Dividing fixed overhead by actual output to get the absorption rate.** The rate is fixed in advance from *budgeted* or *normal* output. That is precisely why under and over absorption exist — if you used actual output there would never be any.
- **Including variable selling overhead in the stock value.** Unsold goods have not been sold, so no selling cost has been incurred on them. It is deducted after contribution, not before.

Cost-Volume-Profit

Use these formulas for Q2 to Q14

EVERYTHING STARTS FROM ONE EQUATION

$$\text{Sales} - \text{Variable cost} = \text{CONTRIBUTION} = \text{Fixed cost} + \text{Profit}$$

Read it both ways. Left to right it tells you what contribution IS. Right to left it tells you what contribution DOES: it first pays off the fixed cost, and whatever is left over is profit. Break-even is simply the point where contribution has exactly paid the fixed cost and nothing remains.

Every one of the thirteen problems in this family is that equation rearranged. There is nothing else to learn.

The six formulas — and what each one is really asking

You want ...	Formula	In plain words
Contribution	$\text{Sales} - \text{Variable cost}$ or $\text{Fixed cost} + \text{Profit}$	What is left from the selling price after paying the costs that grow with volume.
P/V ratio	$\frac{\text{Contribution}}{\text{Sales}} \times 100$	How many paise out of every rupee of sales survives to help pay the fixed cost. It does not change when volume changes — only when price or variable cost changes.
BEP in units	$\frac{\text{Fixed cost}}{\text{Contribution per unit}}$	How many units must be sold before the fixed cost is fully paid.
BEP in rupees	$\frac{\text{Fixed cost}}{\text{P/V ratio}}$	The same point expressed as a sales value. Use this when the question gives you totals and no unit figures.
Margin of safety	$\frac{\text{Actual sales} - \text{BEP sales}}{\text{P/V ratio}}$ or $\frac{\text{Profit}}{\text{P/V ratio}}$	How far sales can fall before you start making a loss. The cushion.
Sales for a target profit	$\frac{\text{Fixed cost} + \text{Target profit}}{\text{P/V ratio}}$	Treat the target profit exactly like an extra slice of fixed cost that also has to be paid for.

WHEN THE QUESTION GIVES YOU TWO PERIODS AND NO COSTS

Q7 and Q8 give only sales and profit for two periods. You cannot see the fixed cost or the variable cost at all. There is a standard route out, and it works because **fixed cost does not change between the two periods**, so every extra rupee of contribution shows up as an extra rupee of profit.

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100$$

1. Find the P/V ratio from the change in profit over the change in sales.
2. Take any one period. Contribution = Sales × P/V ratio.

3. **Fixed cost = Contribution – Profit** for that same period.

4. Now you have everything — compute BEP and margin of safety normally.

5. **Check** your fixed cost against the *other* period. If it does not give that period's profit too, you have gone wrong.

ROUNDING THE P/V RATIO — A REAL WARNING

When the P/V ratio is a recurring decimal such as 44.44% or 54.55%, **do not round it to 44% or 54% before dividing**. In Q11 parts (v) and (vi) that rounding shifts the break-even figure by several hundred rupees. Keep the ratio as a fraction — 8,000/18,000 rather than 0.44 — and divide once at the end.

Q1 Both statements at normal volume

Statements · p.21

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The basic production data of ABC Ltd are as follows: Normal volume of production = 19,500 units per period. Sale price \$4 per unit. Variable cost \$2 per unit. Fixed cost \$1 per unit. Total fixed cost = \$19,500 (\$1 × 19,500 units, normal). Selling and distribution costs have been omitted. The opening and closing stocks consist of both finished goods and equivalent units of work-in-progress.

Present the profit and loss statement under Absorption and Marginal costing method.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Read the question again and notice what is missing.** You are given the normal volume, the three unit costs and the price — but **no actual production figure and no actual sales figure** for any period. Without those, no stock movement can be computed.
- So the only statement that can be prepared from the data as printed is the one at **normal volume**, where production = sales = 19,500 units and there is no change in stock.
- That turns out to be a useful answer rather than a wasted one, because it demonstrates the headline rule of the module: **when stock does not change, the two methods give exactly the same profit.**
- The sentence about opening and closing stocks is telling you what *would* happen if figures were supplied. The illustration at the end shows it.

(a) Marginal Costing Profit Statement — at normal volume of 19,500 units

Particulars	Per unit \$	Total \$
Sales	4.00	78,000
Less: Variable cost	(2.00)	(39,000)
Contribution	2.00	39,000
Less: Fixed cost charged in full as a period cost		(19,500)
Profit		19,500

(b) Absorption Costing Profit Statement — at normal volume of 19,500 units

Particulars	Per unit \$	Total \$
Sales	4.00	78,000
Less: Cost of sales		
Variable production cost	2.00	39,000
Fixed production overhead absorbed 19,500 × \$1	1.00	19,500
Total cost of sales	3.00	(58,500)
Profit	1.00	19,500

Absorbed overhead = 19,500 units × \$1 = \$19,500 = the actual fixed cost, so there is **no under or over absorption** at normal volume.

WHY THIS WORKS

Both methods give \$19,500. That is not a coincidence and it is the point of the problem: at normal volume the fixed overhead absorbed into production exactly equals the fixed overhead incurred, so nothing is left sitting in stock and the two profits must agree.

The illustration below shows the two cases where they diverge. Learn the direction: **stock up → absorption profit higher; stock down → absorption profit lower.**

What would happen if production and sales differed — an illustration

	Production = Sales 19,500 / 19,500	Production > Sales 19,500 / 17,500	Production < Sales 17,500 / 19,500
Closing stock (units)	Nil	2,000	Nil (2,000 opening used up)
Marginal costing profit	19,500	\$15,500	\$19,500
Absorption costing profit	19,500	\$17,500	\$17,500
Difference	Nil	+2,000	−2,000
Explanation	no stock change	2,000 units × \$1 fixed cost parked in stock	2,000 units × \$1 released from stock

THE TRAP / WHERE MARKS ARE LOST

Note on the data. The question gives no actual production or sales figures, so a full multi-period comparison cannot be prepared from it. If your professor expects a version with stock movements, he will supply the production and sales for each period — the method is exactly the one shown in Q16 and Q20. Say in one line that you have prepared the statement at normal volume, and you protect the marks.

FINAL ANSWER

Contribution per unit	\$ 2.00
Marginal costing profit	\$ 19,500
Absorption costing profit	\$ 19,500
Difference	Nil — production equals sales, so no fixed cost is held in stock

Q2 Straight marginal cost statement

CVP · p.21-22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information, find out the amount of profit earned during the year using marginal cost technique: Fixed Cost ₹5,00,000; Variable Cost ₹10 per unit; Selling Price ₹15 per unit; Output level 1,50,000 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Everything is per unit except the fixed cost, so multiply the two unit figures by the output of 1,50,000 and put the fixed cost in as a single lump.
- “Using marginal cost technique” means **present it as a statement ending in contribution then profit** — not just the final number. The layout carries marks.
- Always show the **per-unit column beside the total column**. It is free marks and it makes checking trivial.

Marginal Cost Statement

Particulars	Per unit ₹	Total ₹
Sales $1,50,000 \times ₹15$	15	22,50,000
Less: Variable cost $1,50,000 \times ₹10$	(10)	(15,00,000)
Contribution	5	7,50,000
Less: Fixed cost		(5,00,000)
Profit		2,50,000

FINAL ANSWER

Contribution per unit ₹ 5

Total contribution ₹ 7,50,000

Profit ₹ 2,50,000

Cross-check with the marginal cost equation: Contribution = Fixed cost + Profit $\rightarrow 7,50,000 = 5,00,000 + 2,50,000$

Q3 Fixed cost hidden inside the BEP

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Find the profit from the following data: Sales ₹80,000; Marginal Cost ₹60,000; Break-even point ₹60,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- “Marginal cost” is just another name for **variable cost**. So contribution = 80,000 – 60,000 = ₹20,000.
- The fixed cost is **not given** — you have to extract it from the break-even point. That is the whole trick of this problem.
- Work the BEP formula **backwards**: if $\text{BEP} = \text{Fixed cost} \div \text{P/V ratio}$, then $\text{Fixed cost} = \text{BEP} \times \text{P/V ratio}$.

WORKING NOTES

W1 Contribution and P/V ratio

Contribution = Sales – Marginal cost = 80,000 – 60,000 = **₹20,000**
P/V ratio = 20,000 ÷ 80,000 = **25%**

W2 Recovering the fixed cost from the break-even point

$$\text{BEP (in ₹)} = \frac{\text{Fixed cost}}{\text{P/V ratio}} \rightarrow \text{Fixed cost} = \text{BEP} \times \text{P/V ratio}$$

Fixed cost = 60,000 × 25% = **₹15,000**

Marginal Cost Statement

Particulars	Amount ₹
Sales	80,000
Less: Marginal (variable) cost	(60,000)
Contribution	20,000
Less: Fixed cost W2	(15,000)
Profit	5,000

FINAL ANSWER

Contribution **₹ 20,000**
P/V ratio **25%**
Fixed cost (derived) **₹ 15,000**
Profit ₹ 5,000

Check: margin of safety = 80,000 – 60,000 = 20,000; profit = MOS × P/V ratio = 20,000 × 25% = 5,000 — a second route to the same answer, worth showing.

Q4 Break-even point in units and in rupees

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following particulars, calculate the break-even point: Variable cost per unit ₹12; Fixed Expenses ₹60,000; Selling Price per unit ₹18.

READ THE QUESTION FIRST — WHAT TO SPOT

- Unit figures are given, so use the **units** form of the BEP formula: Fixed cost ÷ contribution per unit.
- Give **both** answers — units and sales value. The question says “break-even point”, and a complete answer states it both ways.

WORKING NOTES

Contribution per unit = $18 - 12 = ₹6$

$$\text{BEP (units)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}} = \frac{60,000}{6} = \mathbf{10,000 \text{ units}}$$

BEP (sales value) = $10,000 \text{ units} \times ₹18 = ₹1,80,000$

Alternative route to the same value: P/V ratio = $6 \div 18 = 33.33\%$, so BEP = $60,000 \div 0.3333 = ₹1,80,000$

FINAL ANSWER

Contribution per unit	₹ 6
BEP in units	10,000 units
BEP in sales value	₹ 1,80,000
P/V ratio	33.33%

Q5 Working the BEP formula backwards

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following particulars, find out the selling price per unit if B.E.P. is to be brought down to 9,000 units. Variable cost per unit ₹75; Fixed Expenses ₹2,70,000; Selling Price per unit ₹100.

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the BEP formula run **backwards**. You are told the answer (9,000 units) and asked for the selling price that produces it.
- First find where you currently stand: at ₹100 the contribution is ₹25 and the BEP is $2,70,000 \div 25 =$ **10,800 units**. The company wants it down to 9,000, which means each unit must contribute **more**.
- Let the new selling price be x . Then contribution per unit $= x - 75$, and $9,000 = 2,70,000 \div (x - 75)$. Solve for x .

WORKING NOTES

W1 The present position

Present contribution $= 100 - 75 = ₹25$ per unit

Present BEP $= 2,70,000 \div 25 =$ **10,800 units**

Target BEP $= 9,000$ units $\rightarrow$ the price must rise

W2 Solving for the new selling price

Required contribution per unit $= \frac{\text{Fixed cost}}{\text{Target BEP units}} = \frac{2,70,000}{9,000} =$ **₹30**

New selling price $= \text{Variable cost} + \text{Required contribution}$
 $= 75 + 30 =$ **₹105 per unit**

Proof: at ₹105 the contribution is ₹30, so BEP $= 2,70,000 \div 30 = 9,000$ units exactly as required.

FINAL ANSWER

Present contribution / BEP **₹ 25 per unit / 10,800 units**

Required contribution per unit **₹ 30**

New selling price **₹ 105 per unit**

Increase needed **₹ 5 per unit, i.e. 5%**

Q6 How a price cut moves the break-even point

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following data, calculate break-even point expressed in terms of units and also the new B.E.P., if the selling price is reduced by 10%.

Particulars	Amount
Fixed Expenses	
Depreciation	₹ 1,00,000
Salaries	₹ 1,00,000
Variable Expenses	
Materials	₹ 3 per unit
Labour	₹ 2 per unit
Selling Price	₹ 10 per unit

READ THE QUESTION FIRST — WHAT TO SPOT

- Add the two fixed items into one figure and the two variable items into one per-unit figure. Do this before you touch the formula.
- **A 10% cut in selling price does not change the fixed cost or the variable cost.** Only the contribution shrinks — and because contribution is the denominator, the break-even point rises. Expect a bigger answer for the second part.
- Notice how sharp the effect is: a 10% price cut here raises the break-even by 25%. That is the lesson of the problem, and worth one closing sentence in your answer.

WORKING NOTES

W1 Present break-even point

Fixed expenses = 1,00,000 + 1,00,000 = **₹2,00,000**

Variable cost per unit = 3 + 2 = **₹5**

Contribution per unit = 10 – 5 = **₹5**

$$\text{BEP} = \frac{2,00,000}{5} = \mathbf{40,000 \text{ units}}$$

W2 New break-even point after a 10% price cut

New selling price = 10 – 10% = **₹9**

Variable cost is unchanged at ₹5

New contribution per unit = 9 – 5 = **₹4**

$$\text{New BEP} = \frac{2,00,000}{4} = \mathbf{50,000 \text{ units}}$$

The effect of the price cut, side by side

	Present	After 10% price cut	Change
Selling price per unit	10	9	–10%
Variable cost per unit	5	5	no change
Contribution per unit	5	4	–20%
P/V ratio	50%	44.44%	–5.56 points
Break-even point (units)	40,000	50,000	+25%

WHY THIS WORKS

A 10% cut in price caused a 20% fall in contribution and a 25% rise in the break-even point. The reason is that the whole of the price cut comes out of contribution — the variable cost does not fall to share the pain. This is why price cuts are so much more dangerous than they look, and saying so in one line turns a numerical answer into an interpreted one.

FINAL ANSWER

Contribution per unit — present / new ₹ 5 / ₹ 4

Present BEP 40,000 units

New BEP after 10% price cut 50,000 units

BEP in value — present / new ₹ 4,00,000 / ₹ 4,50,000

Q7 Two periods, no costs given

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The sales turnover and profits during two periods are as under:

Period	Sales	Profit
Period I	₹ 20 lakhs	₹ 2 lakhs
Period II	₹ 30 lakhs	₹ 4 lakhs

Calculate the P/V ratio, fixed cost, break-even point and margin of safety.

READ THE QUESTION FIRST — WHAT TO SPOT

- You are given **no costs at all** — only sales and profit. This is the two-period type from Playbook B.
- The key insight: **fixed cost is the same in both periods**. So the extra ₹2 lakhs of profit in Period II came entirely from the extra ₹10 lakhs of sales. Every rupee of that increase is pure contribution.
- Once you have the P/V ratio you can work out the contribution of either period, and fixed cost falls out as contribution minus profit.
- **Always verify with the other period**. If your fixed cost does not also reproduce Period I's profit, you have made an error.

WORKING NOTES

W1 P/V ratio from the change

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100 = \frac{₹2,00,000}{₹10,00,000} \times 100 = \mathbf{20\%}$$

W2 Fixed cost

Take Period I: Contribution = Sales × P/V ratio = 20,00,000 × 20% = ₹4,00,000

Fixed cost = Contribution – Profit = 4,00,000 – 2,00,000 = **₹2,00,000**

Verification on Period II: contribution = 30,00,000 × 20% = ₹6,00,000; profit = 6,00,000 – 2,00,000 = ₹4,00,000 matches the given figure.

W3 Break-even point and margin of safety

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{2,00,000}{0.20} = \mathbf{₹10,00,000}$$

MOS Period I = 20,00,000 – 10,00,000 = **₹10,00,000** (50% of sales)

MOS Period II = 30,00,000 – 10,00,000 = **₹20,00,000** (66.67% of sales)

Marginal Cost Statement for both periods

Particulars	Period I ₹	Period II ₹
Sales	20,00,000	30,00,000
Less: Variable cost 80% of sales	(16,00,000)	(24,00,000)
Contribution 20% of sales	4,00,000	6,00,000
Less: Fixed cost W2 — same in both periods	(2,00,000)	(2,00,000)
Profit	2,00,000	4,00,000

FINAL ANSWER

P/V ratio 20%

Fixed cost ₹ 2,00,000

Break-even sales ₹ 10,00,000

Margin of safety — Period I ₹ 10,00,000 (50%)

Margin of safety — Period II ₹ 20,00,000 (66.67%)

Q8 Break-even from two years' results

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following data are obtained from the records of a company:

Particulars	First Year ₹	Second Year ₹
Sales	80,000	90,000
Profit	10,000	14,000

Calculate the break-even point.

READ THE QUESTION FIRST — WHAT TO SPOT

- Exactly the same shape as Q7. Change in profit ₹4,000 over change in sales ₹10,000 gives the P/V ratio straight away.
- The question asks only for the break-even point, but **show the fixed cost and the P/V ratio on the way** — those are the working marks.

WORKING NOTES

W1 P/V ratio

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100 = \frac{14,000 - 10,000}{90,000 - 80,000} \times 100 = \frac{4,000}{10,000} \times 100 = \mathbf{40\%}$$

W2 Fixed cost

$$\text{First year contribution} = 80,000 \times 40\% = \mathbf{₹32,000}$$

$$\text{Fixed cost} = 32,000 - 10,000 = \mathbf{₹22,000}$$

Verification: second year contribution = $90,000 \times 40\% = ₹36,000$; $36,000 - 22,000 = ₹14,000$ matches.

W3 Break-even point

$$\text{BEP} = \frac{22,000}{0.40} = \mathbf{₹55,000}$$

Marginal Cost Statement for both years

Particulars	First Year ₹	Second Year ₹
Sales	80,000	90,000
Less: Variable cost (60% of sales)	(48,000)	(54,000)
Contribution (40% of sales)	32,000	36,000
Less: Fixed cost	(22,000)	(22,000)
Profit	10,000	14,000

FINAL ANSWER

P/V ratio **40%**

Fixed cost	₹ 22,000
Break-even point	₹ 55,000
Margin of safety — first year	₹ 25,000
Margin of safety — second year	₹ 35,000

Q9 Margin of safety from profit alone

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company earned a profit of ₹30,000 during a particular year. If the marginal cost and selling price of a product are ₹8 and ₹10 per unit respectively, find out the amount of Margin of Safety.

READ THE QUESTION FIRST — WHAT TO SPOT

- You are not given sales or fixed cost, so you cannot find the BEP and subtract. You need the **other** margin-of-safety formula.
- Because profit is the part of contribution left over *after* the fixed cost is paid, the sales that generated that profit are pure margin of safety. So MOS = profit ÷ P/V ratio.
- In units it is even simpler: MOS units = profit ÷ contribution per unit.

WORKING NOTES

Contribution per unit = $10 - 8 = ₹2$

P/V ratio = $2 \div 10 = 20\%$

$$\text{MOS (in units)} = \frac{\text{Profit}}{\text{Contribution per unit}} = \frac{30,000}{2} = \mathbf{15,000 \text{ units}}$$

$$\text{MOS (in ₹)} = 15,000 \times ₹10 = \mathbf{₹1,50,000}$$

Or directly: MOS = Profit ÷ P/V ratio = $30,000 \div 0.20 = ₹1,50,000$. Both routes, same answer.

WHY THIS WORKS

Think about what margin of safety means physically. Break-even sales pay the fixed cost exactly and leave nothing. Every rupee of sales *beyond* that point contributes only profit. So if you know the profit and you know how much profit each rupee of sales produces — the P/V ratio — you can work backwards to how many rupees of sales sat above break-even. That is the margin of safety, and it is why you never needed the fixed cost.

FINAL ANSWER

Contribution per unit	₹ 2
P/V ratio	20%
Margin of safety in units	15,000 units
Margin of safety in value	₹ 1,50,000

Q10 Extracting variable cost from total cost

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following details find out (a) Profit Volume Ratio (b) BEP (c) Margin of Safety.

Particulars	₹
Sales	1,00,000
Total cost	80,000
Fixed Cost	20,000
Net profit	20,000

READ THE QUESTION FIRST — WHAT TO SPOT

- The **variable cost is not given directly** — you must strip it out: Variable cost = Total cost – Fixed cost = 80,000 – 20,000 = ₹60,000.
- Once you have that, build the marginal cost statement and the three answers drop out of it.
- The net profit of ₹20,000 is given, so use it to **check** your statement rather than to compute anything.

Marginal Cost Statement

Particulars	Amount ₹
Sales	1,00,000
Less: Variable cost 80,000 total cost – 20,000 fixed	(60,000)
Contribution	40,000
Less: Fixed cost	(20,000)
Profit agrees with the given figure ✓	20,000

WORKING NOTES

$$(a) \text{ P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{40,000}{1,00,000} \times 100 = \mathbf{40\%}$$

$$(b) \text{ BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{20,000}{0.40} = \mathbf{₹50,000}$$

$$(c) \text{ MOS} = \text{Actual sales} - \text{BEP sales} = 1,00,000 - 50,000 = \mathbf{₹50,000}$$

Check by the other route: MOS = Profit ÷ P/V ratio = 20,000 ÷ 0.40 = ₹50,000 ✓

FINAL ANSWER

Variable cost (derived)	₹ 60,000
Contribution	₹ 40,000
(a) P/V ratio	40%
(b) Break-even point	₹ 50,000

Q11 Eight situations in one statement

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information is obtained from a company for 2016: Sales ₹20,000; Variable Cost ₹10,000; Fixed Cost ₹6,000.

(a) Find P/V Ratio, Break-even point and Margin of safety at this level, and **(b)** the effect on P/V Ratio, Break-even point and Margin of safety if: (i) 20% decrease in fixed cost; (ii) 10% increase in fixed cost; (iii) 10% decrease in variable cost; (iv) 10% increase in variable cost; (v) 10% increase in selling price together with an increase of fixed overheads by ₹1,200; (vi) 10% decrease in selling price; (vii) 10% decrease in sales price accompanied by 10% decrease in variable costs.

READ THE QUESTION FIRST — WHAT TO SPOT

- Eight columns of the same four-line statement. **Do it as one wide table**, exactly as your classmate's notes do — never as eight separate little sums, or you will lose track.
- Work out which line each change touches **before** you compute anything: a fixed-cost change touches only the FC line and leaves the P/V ratio alone; a variable-cost or selling-price change moves contribution and therefore **does** change the P/V ratio.
- In (v) the selling price rises 10% so sales become 22,000, but the **variable cost stays at 10,000** — volume has not changed, only price. Same logic in (vi).
- **In (vii) both fall 10%**, so contribution falls 10% too and the **P/V ratio is unchanged at 50%**. That is the interesting result of the whole question.

Which line does each change touch?

Case	Sales	Variable cost	Fixed cost	Does the P/V ratio change?
(i) FC down 20%	—	—	↓ 4,800	No — contribution untouched
(ii) FC up 10%	—	—	↑ 6,600	No
(iii) VC down 10%	—	↓ 9,000	—	Yes — rises to 55%
(iv) VC up 10%	—	↑ 11,000	—	Yes — falls to 45%
(v) SP up 10%, FC up 1,200	↑ 22,000	—	↑ 7,200	Yes — rises to 54.55%
(vi) SP down 10%	↓ 18,000	—	—	Yes — falls to 44.44%
(vii) SP down 10% and VC down 10%	↓ 18,000	↓ 9,000	—	No — stays exactly 50%

Complete solution — all eight situations in one statement

Particulars	(a) Present	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Sales	20,000	20,000	20,000	20,000	20,000	22,000	18,000	18,000
Less: Variable cost	(10,000)	(10,000)	(10,000)	(9,000)	(11,000)	(10,000)	(10,000)	(9,000)
Contribution	10,000	10,000	10,000	11,000	9,000	12,000	8,000	9,000
Less: Fixed cost	(6,000)	(4,800)	(6,600)	(6,000)	(6,000)	(7,200)	(6,000)	(6,000)
Profit	4,000	5,200	3,400	5,000	3,000	4,800	2,000	3,000

Particulars	(a) Present	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
P/V ratio	50.00%	50.00%	50.00%	55.00%	45.00%	54.55%	44.44%	50.00%
BEP sales	12,000	9,600	13,200	10,909	13,333	13,200	13,500	12,000
Margin of safety	8,000	10,400	6,800	9,091	6,667	8,800	4,500	6,000

WORKING NOTES

How the BEP row was computed in each column

Case	Fixed cost	P/V ratio	BEP = FC ÷ P/V	MOS = Sales – BEP
(a)	6,000	50.00%	12,000	8,000
(i)	4,800	50.00%	9,600	10,400
(ii)	6,600	50.00%	13,200	6,800
(iii)	6,000	55.00%	10,909	9,091
(iv)	6,000	45.00%	13,333	6,667
(v)	7,200	$12,000/22,000 = 54.5455\%$	13,200	8,800
(vi)	6,000	$8,000/18,000 = 44.4444\%$	13,500	4,500
(vii)	6,000	$9,000/18,000 = 50.00\%$	12,000	6,000

THE TRAP / WHERE MARKS ARE LOST

Important — two figures where care is needed. In cases (v) and (vi) the P/V ratio is a recurring decimal. If you round it to 0.54 and 0.44 before dividing you get BEP figures of ₹13,333 and ₹13,636 instead of the correct ₹13,200 and ₹13,500. Your classmate's handwritten notes contain exactly this rounding, so **if your answer differs from those notes in columns (v) and (vi), you are right and the notes are rounded.**

Keep the ratio as a fraction — $12,000/22,000$ and $8,000/18,000$ — and divide only once, at the end.

WHY THIS WORKS

Look across the P/V ratio row. It does not budge for (i) and (ii) because a change in fixed cost cannot affect contribution. It moves for (iii) to (vi) because each of those changes either the price or the variable cost. And in (vii) it returns to exactly 50% because price and variable cost fell by the same percentage, so their *ratio* is unchanged — the profit falls, but the efficiency of each rupee of sales does not.

FINAL ANSWER

(a) Present — P/V / BEP / MOS	50% / ₹ 12,000 / ₹ 8,000
(i) FC down 20%	50% / ₹ 9,600 / ₹ 10,400 — profit ₹ 5,200
(ii) FC up 10%	50% / ₹ 13,200 / ₹ 6,800 — profit ₹ 3,400
(iii) VC down 10%	55% / ₹ 10,909 / ₹ 9,091 — profit ₹ 5,000
(iv) VC up 10%	45% / ₹ 13,333 / ₹ 6,667 — profit ₹ 3,000
(v) SP up 10%, FC up 1,200	54.55% / ₹ 13,200 / ₹ 8,800 — profit ₹ 4,800
(vi) SP down 10%	44.44% / ₹ 13,500 / ₹ 4,500 — profit ₹ 2,000
(vii) SP and VC both down 10%	50% / ₹ 12,000 / ₹ 6,000 — profit ₹ 3,000

The best single-sentence conclusion: a fixed-cost change moves the break-even point but never the P/V ratio; a price or variable-cost change moves both.

Q12 The four standard CVP questions

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

You are given the following data: Variable Cost ₹6,00,000; Fixed Cost ₹3,00,000; Net Profit ₹1,00,000; Sales ₹10,00,000.

Find (a) P/V Ratio (b) B.E.P. (c) Profit when sales amounted to ₹12,00,000 (d) Sales required to earn a profit of ₹2,00,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- All four parts come from the same statement, so **build the statement first** and answer from it.
- For (c) and (d) remember the fixed cost does **not** change with volume. That is what makes both parts one-line calculations.
- For (d), treat the target profit exactly like extra fixed cost that must also be covered.

Marginal Cost Statement (present position)

Particulars	Amount ₹
Sales	10,00,000
Less: Variable cost	(6,00,000)
Contribution	4,00,000
Less: Fixed cost	(3,00,000)
Profit agrees with the given ₹1,00,000 ✓	1,00,000

WORKING NOTES

$$(a) \text{ P/V ratio} = \frac{4,00,000}{10,00,000} \times 100 = 40\%$$

$$(b) \text{ BEP} = \frac{3,00,000}{0.40} = ₹7,50,000$$

(c) Profit when sales are ₹12,00,000

Contribution = 12,00,000 × 40% = 4,80,000

Less: Fixed cost = (3,00,000)

Profit = ₹1,80,000

Faster route: extra sales of ₹2,00,000 × 40% = ₹80,000 extra profit, so 1,00,000 + 80,000 = ₹1,80,000. Fixed cost does not change, so every extra rupee of contribution is extra profit.

(d) Sales required for a profit of ₹2,00,000

$$\text{Required sales} = \frac{\text{Fixed cost} + \text{Target profit}}{\text{P/V ratio}} = \frac{3,00,000 + 2,00,000}{0.40} = \frac{5,00,000}{0.40} = ₹12,50,000$$

FINAL ANSWER

(a) P/V ratio	40%
(b) Break-even point	₹ 7,50,000
(c) Profit at sales of ₹ 12,00,000	₹ 1,80,000
(d) Sales to earn ₹ 2,00,000	₹ 12,50,000
Margin of safety at present	₹ 2,50,000

Q13 Seven parts including tax

CVP · p.24

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following, find out (i) Profit Volume Ratio (ii) Break-even point (iii) Sales for 40% P/V Ratio (iv) Margin of Safety if actual sales were ₹3,00,000 (v) Net Profit if actual sales were ₹3,00,000 (vi) Required Sales to earn a profit of ₹70,000 (vii) Required sales to earn a net profit of ₹70,000 after tax, given tax rate = 60%.

Position of the Co. for the year 2025	₹
Sales	2,00,000
Variable Cost	1,50,000
Contribution	50,000
Fixed Cost	15,000
Profit	35,000

READ THE QUESTION FIRST — WHAT TO SPOT

- Seven parts, and the last two are the ones that separate students.
- **Part (iii) needs interpreting.** The P/V ratio does not depend on volume, so “sales for 40% P/V ratio” must mean: what sales value would give a 40% P/V ratio if the variable cost stays at ₹1,50,000? If contribution is to be 40% of sales, variable cost must be 60% of sales — so sales = $1,50,000 \div 0.60$.
- **Part (vii) is about tax.** ₹70,000 is wanted *after* 60% tax, so the profit before tax must be bigger. Gross it up first: pre-tax profit = $70,000 \div (1 - 0.60)$. Only then use the sales formula. Forgetting to gross up is the classic error.

WORKING NOTES

$$(i) \text{ P/V ratio} = \frac{50,000}{2,00,000} \times 100 = \mathbf{25\%}$$

$$(ii) \text{ BEP} = \frac{15,000}{0.25} = \mathbf{₹60,000}$$

(iii) Sales that would give a 40% P/V ratio

If contribution is 40% of sales, then variable cost is 60% of sales
Variable cost is unchanged at 1,50,000

$$\text{Sales} = \frac{1,50,000}{0.60} = \mathbf{₹2,50,000}$$

Check: at sales of 2,50,000 the contribution is $2,50,000 - 1,50,000 = ₹1,00,000$, which is exactly 40% of 2,50,000

(iv) and (v) — if actual sales were ₹3,00,000

$$\text{MOS} = 3,00,000 - 60,000 \text{ (BEP)} = \text{₹}2,40,000$$

$$\text{Contribution} = 3,00,000 \times 25\% = 75,000$$

$$\text{Net profit} = 75,000 - 15,000 = \text{₹}60,000$$

Or: profit = MOS $\times$ P/V ratio = 2,40,000 $\times$ 25% = ₹60,000

(vi) Sales to earn a profit of ₹70,000

$$\text{Required sales} = \frac{15,000 + 70,000}{0.25} = \frac{85,000}{0.25} = \text{₹}3,40,000$$

(vii) Sales to earn ₹70,000 after 60% tax

Step 1 — gross the profit up to a pre-tax figure:

$$\text{Profit before tax} = \frac{70,000}{1 - 0.60} = \frac{70,000}{0.40} = \text{₹}1,75,000$$

Step 2 — now use the ordinary formula on the pre-tax profit:

$$\text{Required sales} = \frac{15,000 + 1,75,000}{0.25} = \frac{1,90,000}{0.25} = \text{₹}7,60,000$$

Check: sales 7,60,000 $\times$ 25% = contribution 1,90,000; less fixed cost 15,000 = pre-tax profit 1,75,000; less tax at 60% = 1,05,000; net profit = ₹70,000

THE TRAP / WHERE MARKS ARE LOST

- **Part (vii):** dividing 70,000 by 0.25 straight away, without grossing up for tax. That gives ₹3,40,000 — the answer to part (vi), not part (vii). Tax always comes out *after* profit, so the profit you need to *earn* is always bigger than the profit you want to *keep*.
- **Part (iii):** trying to change the P/V ratio by changing volume. Volume cannot change a ratio of contribution to sales — only price or variable cost can. State your assumption that variable cost is unchanged.

FINAL ANSWER

(i) P/V ratio	25%
(ii) Break-even point	₹ 60,000
(iii) Sales for 40% P/V ratio	₹ 2,50,000
(iv) MOS at sales of ₹ 3,00,000	₹ 2,40,000
(v) Net profit at sales of ₹ 3,00,000	₹ 60,000
(vi) Sales for profit of ₹ 70,000	₹ 3,40,000
(vii) Sales for ₹ 70,000 after 60% tax	₹ 7,60,000

Q14 Merger of two plants at different capacities

CVP · p.24

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Two manufacturing companies which have the following operating details decided to merge:

Particulars	Company I	Company II
Capacity utilisation (%)	90	60
Sales (₹ in lakhs)	540	300
Variable costs (₹ in lakhs)	396	225
Fixed costs (₹ in lakhs)	80	50

Assuming that the proposal is implemented, calculate: **(a)** Break-even sales of the merged plant and the capacity utilisation at that stage; **(b)** Profitability of the merged plant at 80% capacity utilisation; **(c)** Sales turnover of the merged plant to earn a profit of ₹75 lakh; **(d)** When the merged plant is working at a capacity to earn a profit of ₹75 lakh, what percentage increase in selling price is required to sustain an increase of 5% in fixed overheads?

READ THE QUESTION FIRST — WHAT TO SPOT

This is the hardest problem in the module and it is worth learning properly, because the same structure appears in every merger question.

- **The two companies are running at different capacities** — 90% and 60%. You cannot add 540 and 300 and call it the merged sales, because those are sales at *different* levels of activity.
- **So step one is always: restate both companies at 100% capacity.** Divide sales and variable cost by the capacity percentage. Only then can you add them.
- **Fixed cost is NOT scaled up.** Fixed cost does not depend on capacity — that is what makes it fixed. Add $80 + 50 = 130$ lakh and leave it alone. This is the single most common error.
- Once you have the merged plant at 100%, everything else is ordinary CVP work.
- Part (d) is different in kind: volume stays the same and you must recover extra fixed cost purely by raising price. So the price rise, in rupees, equals the extra fixed cost; express it as a percentage of the sales at that level.

WORKING NOTES

W1 Restate both companies at 100% capacity — the essential first step

Particulars	Company I at 90%	Company I at 100%	Company II at 60%	Company II at 100%
Sales	540	$540 \div 0.90 = \mathbf{600}$	300	$300 \div 0.60 = \mathbf{500}$
Variable cost	396	$396 \div 0.90 = \mathbf{440}$	225	$225 \div 0.60 = \mathbf{375}$
Contribution	144	160	75	125
Fixed cost	80	80 NOT scaled	50	50 NOT scaled

W2 The merged plant at 100% capacity

Particulars	₹ in lakhs
Sales $600 + 500$	1,100
Less: Variable cost $440 + 375$	(815)

Particulars	₹ in lakhs
Contribution	285
Less: Fixed cost 80 + 50, unchanged	(130)
Profit at 100% capacity	155

$$\text{P/V ratio of the merged plant} = \frac{285}{1,100} \times 100 = \mathbf{25.9091\%}$$

Keep this as the fraction 285/1,100 in every later division. Rounding it to 26% will move your break-even figure by several lakh.

(a) Break-even sales and the capacity at that point

$$\text{BEP sales} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{130}{285/1,100} = \frac{130 \times 1,100}{285} = \mathbf{₹501.75 \text{ lakh}}$$

$$\text{Capacity utilisation at BEP} = \frac{501.75}{1,100} \times 100 = \mathbf{45.61\%}$$

(b) Profitability at 80% capacity utilisation

Particulars	₹ in lakhs
Sales 80% of 1,100	880.00
Less: Variable cost 80% of 815	(652.00)
Contribution 80% of 285	228.00
Less: Fixed cost	(130.00)
Profit	98.00

(c) Sales needed to earn a profit of ₹75 lakh

$$\text{Required sales} = \frac{130 + 75}{285/1,100} = \frac{205 \times 1,100}{285} = \mathbf{₹791.23 \text{ lakh}}$$

$$\text{Capacity at that level} = 791.23 \div 1,100 = \mathbf{71.93\%}$$

(d) Price increase needed to absorb a 5% rise in fixed overheads

$$\text{Increase in fixed cost} = 5\% \text{ of } 130 = \mathbf{₹6.50 \text{ lakh}}$$

Volume is unchanged, so the whole 6.50 lakh must come from a higher selling price.

$$\text{Sales at that level (from part c)} = \mathbf{₹791.23 \text{ lakh}}$$

$$\text{Required increase in selling price} = \frac{6.50}{791.23} \times 100 = \mathbf{0.82\%}$$

Check: new sales = 791.23 × 1.0082 = ₹797.73 lakh. Contribution rises by the full 6.50 (variable cost is unchanged because volume is unchanged) to 211.50; less new fixed cost of 136.50 = ₹75 lakh profit

THE TRAP / WHERE MARKS ARE LOST

- **Scaling up the fixed cost along with sales.** Fixed cost stays at 130 lakh at every capacity. If you scale it you will get every one of the four answers wrong.
- **Adding $540 + 300 = 840$ and treating that as merged sales.** Those are sales at 90% and 60% respectively. Restate to 100% first — $600 + 500 = 1,100$.
- **Rounding the P/V ratio to 26%.** It is 25.9091%. Use the fraction $285/1,100$ throughout.
- **In part (d), also raising the variable cost.** Volume has not changed, so variable cost has not changed. Only the price moves.

FINAL ANSWER

Merged plant at 100% — sales / contribution / fixed cost	₹ 1,100 L / ₹ 285 L / ₹ 130 L
P/V ratio of the merged plant	25.9091%
(a) Break-even sales	₹ 501.75 lakh
(a) Capacity utilisation at BEP	45.61%
(b) Profit at 80% capacity	₹ 98 lakh
(c) Sales for a profit of ₹ 75 lakh	₹ 791.23 lakh (71.93% capacity)
(d) Price increase for 5% higher fixed cost	0.82%

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Rain Until September Co makes a product, the Splash, which has a variable production cost of \$6 per unit and a sales price of \$10 per unit. At the beginning of September 20X0 there were no opening inventories and production during the month was 20,000 units. Fixed costs for the month were \$45,000 (production, administration, sales and distribution). There were no variable marketing costs.

Calculate the contribution and profit for September 20X0, using marginal costing principles, if sales were (a) 10,000 Splashes (b) 15,000 Splashes (c) 20,000 Splashes.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Marginal costing only** — the question says so. That makes it simple: fixed cost of \$45,000 is charged in full in every one of the three cases, no matter how much was sold.
- Contribution per unit = $10 - 6 = \$4$. Multiply by **units SOLD**, not units produced. Contribution comes from selling, not from making.
- Production is 20,000 in all three cases, so the closing inventory differs: 10,000, 5,000 and nil units. Under marginal costing it is valued at the variable cost of \$6.
- Notice case (a) makes a **loss**. Do not assume every answer is a profit — write the bracket.

Marginal Costing Profit Statement — September 20X0

Particulars	(a) Sales 10,000	(b) Sales 15,000	(c) Sales 20,000
Sales $\text{units sold} \times \10	1,00,000	1,50,000	2,00,000
Less: Variable cost of sales $\text{units sold} \times \6	(60,000)	(90,000)	(1,20,000)
CONTRIBUTION	40,000	60,000	80,000
Less: Fixed costs $\text{charged in full every time}$	(45,000)	(45,000)	(45,000)
PROFIT / (LOSS)	(5,000)	15,000	35,000
Memo: closing inventory in units	10,000	5,000	Nil
Memo: closing inventory value at \$6	60,000	30,000	Nil

WORKING NOTES

$$\text{Contribution per unit} = 10 - 6 = \$4$$

$$\text{Break-even point} = \frac{45,000}{4} = 11,250 \text{ units}$$

That single figure explains all three answers at a glance: 10,000 units is below break-even so there is a loss; 15,000 and 20,000 are above it so there is a profit.

WHY THIS WORKS

Because the fixed cost is charged in full every time, profit moves in exact step with contribution: each extra 5,000 units sold adds $5,000 \times \$4 = \$20,000$ of profit. That straight-line relationship is the whole practical value of marginal costing — a manager can read the effect of a volume change immediately, which absorption costing does not allow.

FINAL ANSWER

Contribution per unit **\$ 4**

Break-even point	11,250 units
(a) 10,000 units	Contribution \$ 40,000 — Loss \$ 5,000
(b) 15,000 units	Contribution \$ 60,000 — Profit \$ 15,000
(c) 20,000 units	Contribution \$ 80,000 — Profit \$ 35,000

Q16 Both methods, two periods, full reconciliation

Statements · p.25

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company makes and sells a single product. At the beginning of period 1 there is no opening stock of the product, for which the variable production cost is ₹4 and the sale price is ₹6 per unit. Fixed costs are ₹2,000 per period, of which ₹1,500 are fixed production costs.

Particulars	Period 1	Period 2
Sales	1,200 units	1,800 units
Production	1,500 units	1,500 units

What would be the profit in each period using (a) Absorption costing (assume normal output is 1,500 units per period) and (b) Marginal costing?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Do the stock movement first.** Period 1: $0 + 1,500 - 1,200 = 300$ units closing. Period 2: $300 + 1,500 - 1,800 = \text{nil closing}$.
- So stock **rises** in period 1 (absorption profit will be higher) and **falls back to zero** in period 2 (absorption profit will be lower). Predict the direction before you calculate — then your answer confirms it.
- Only ₹1,500 of the ₹2,000 fixed cost is a **production** cost, so only that part is absorbed into units. The other ₹500 is a non-production fixed cost and is deducted below gross profit in the absorption statement.
- Production equals normal output (1,500) in both periods, so there is **no under or over absorption** anywhere in this problem. That is deliberate — it isolates the stock effect.

WORKING NOTES

W1 Stock movement and the absorption rate

	Period 1	Period 2
Opening stock (units)	Nil	300
Add: Production	1,500	1,500
Less: Sales	(1,200)	(1,800)
Closing stock (units)	300	Nil

$$\text{Fixed production overhead rate} = \frac{\text{₹1,500}}{1,500 \text{ units normal output}} = \text{₹1 per unit}$$

Full production cost per unit = variable 4 + fixed 1 = **₹5**

Marginal (variable) cost per unit = **₹4**

(a) Absorption Costing Profit Statement

Particulars	Period 1 ₹	Period 2 ₹
Sales $\text{units} \times ₹6$	7,200	10,800
Opening stock $\text{at } ₹5 \text{ full cost}$	Nil	1,500
Add: Cost of production $1,500 \times ₹5$	7,500	7,500
Less: Closing stock $\text{at } ₹5$	(1,500)	Nil
Cost of sales	(6,000)	(9,000)
Gross profit	1,200	1,800
Less: Non-production fixed costs $2,000 - 1,500$	(500)	(500)
PROFIT	700	1,300

No under or over absorption: production of 1,500 units equals normal output, so absorbed overhead of $1,500 \times ₹1 = ₹1,500$ exactly equals the actual fixed production cost.

(b) Marginal Costing Profit Statement

Particulars	Period 1 ₹	Period 2 ₹
Sales	7,200	10,800
Opening stock $\text{at } ₹4 \text{ variable cost}$	Nil	1,200
Add: Variable cost of production $1,500 \times ₹4$	6,000	6,000
Less: Closing stock $\text{at } ₹4$	(1,200)	Nil
Variable cost of sales	(4,800)	(7,200)
CONTRIBUTION	2,400	3,600
Less: Fixed costs $\text{all } ₹2,000, \text{ in full}$	(2,000)	(2,000)
PROFIT	400	1,600

Reconciliation — the proof

Particulars	Period 1 ₹	Period 2 ₹	Total ₹
Marginal costing profit	400	1,600	2,000
Add: Fixed overhead in closing stock $300 \times ₹1 / \text{nil}$	300	Nil	—
Less: Fixed overhead in opening stock $\text{nil} / 300 \times ₹1$	Nil	(300)	—
Absorption costing profit	700	1,300	2,000

Over the two periods together both methods give ₹2,000. The methods differ only in **timing**, never in total profit over the life of the stock.

WHY THIS WORKS

Follow the ₹300 through. In period 1, 300 unsold units carry ₹1 of fixed cost each into the balance sheet instead of the profit and loss account — so absorption profit is ₹300 higher. In period 2 those same units are sold, that ₹300 comes back out of stock and is charged — so absorption profit is ₹300 lower. Over both periods it cancels exactly.

That is the whole marginal-versus-absorption argument in one number: absorption costing lets profit be moved between periods by making stock. Marginal costing does not.

FINAL ANSWER

Closing stock — Period 1 / Period 2	300 units / Nil
Fixed production overhead rate	₹ 1 per unit
Absorption profit — Period 1 / Period 2	₹ 700 / ₹ 1,300
Marginal profit — Period 1 / Period 2	₹ 400 / ₹ 1,600
Total over both periods (either method)	₹ 2,000

Q17 Two products, marginal costing

Statements · p.25-26

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Mill Stream makes two products, the Mill and the Stream. Information relating to each for April 20X1 is as follows.

Particulars	Mill	Stream
Opening inventory	Nil	Nil
Production (units)	15,000	6,000
Sales (units)	10,000	5,000
Sales price per unit	\$20	\$30
Unit costs		
Direct materials	8	14
Direct labour	4	2
Variable production overhead	2	1
Variable sales overhead	2	3
Fixed costs for the month		
Production costs	\$40,000	
Administration costs	\$15,000	
Sales and distribution costs	\$25,000	

(a) Using marginal costing principles, calculate the profit in April 20X1. **(b)** Calculate the profit if sales had been 15,000 units of Mill and 6,000 units of Stream.

READ THE QUESTION FIRST — WHAT TO SPOT

- Marginal costing, so **total the fixed costs into one figure**: $40,000 + 15,000 + 25,000 = \$80,000$. It is charged in full in both parts.
- **Variable sales overhead is included in the contribution calculation** because it varies with each unit sold — but it must **never** go into the value of unsold stock. Here it does not matter for stock because we only need the profit, but note the principle.
- Contribution per unit must be found for each product separately, then multiplied by **that product's** sales units.
- In part (b), production has not changed — only sales. Since fixed cost is unchanged too, the extra profit is simply the extra contribution.

WORKING NOTES

W1 Contribution per unit for each product

Particulars	Mill \$	Stream \$
Selling price	20	30
Less: Direct materials	(8)	(14)
Less: Direct labour	(4)	(2)
Less: Variable production overhead	(2)	(1)
Less: Variable sales overhead	(2)	(3)
Contribution per unit	4	10
<i>Memo: variable production cost per unit for stock valuation</i>	14	17

W2 Total fixed costs for the month

Production 40,000 + Administration 15,000 + Sales and distribution 25,000 = **\$80,000**

(a) Marginal Costing Profit Statement — actual sales

Particulars	Mill \$	Stream \$	Total \$
Sales units	10,000	5,000	—
Contribution per unit W1	4	10	—
Total contribution	40,000	50,000	90,000
Less: Fixed costs W2			(80,000)
PROFIT			10,000

(b) Marginal Costing Profit Statement — if all production had been sold

Particulars	Mill \$	Stream \$	Total \$
Sales units	15,000	6,000	—
Contribution per unit	4	10	—
Total contribution	60,000	60,000	1,20,000
Less: Fixed costs			(80,000)
PROFIT			40,000

WORKING NOTES

Extra units sold = 5,000 Mill + 1,000 Stream

Extra contribution = $(5,000 \times 4) + (1,000 \times 10) = 20,000 + 10,000 = \mathbf{\$30,000}$

Profit rises from \$10,000 to \$40,000 — an increase of exactly \$30,000 ✓

THE TRAP / WHERE MARKS ARE LOST

- Multiplying contribution per unit by **production** units (15,000 and 6,000) in part (a). Only 10,000 and 5,000 were sold. Contribution comes from sales.
- Leaving the variable **sales** overhead out of the contribution calculation. It is a variable cost and must be deducted — Mill's contribution is \$4, not \$6.

- Splitting the fixed costs between the two products. There is no basis given and no need — under marginal costing you deduct total fixed cost from total contribution.

FINAL ANSWER

Contribution per unit — Mill / Stream	\$ 4 / \$ 10
Total fixed costs	\$ 80,000
(a) Profit at actual sales	Contribution \$ 90,000 – \$ 80,000 = \$ 10,000
(b) Profit if all output sold	Contribution \$ 1,20,000 – \$ 80,000 = \$ 40,000

Q18 Annual budget, quarterly statement

Statements · p.26

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Big Woof Co manufactures a single product, the Bark: Selling price \$180.00; Direct materials \$40.00; Direct labour \$16.00; Variable overheads \$10.00 per unit.

Annual fixed production overheads are budgeted to be \$1.6 million and Big Woof expects to produce 1,280,000 units of the Bark each year. Overheads are absorbed on a per unit basis. Actual overheads are \$1.6 million for the year. Budgeted fixed selling costs are \$320,000 per quarter. Actual sales and production for the first quarter of 20X8: **Sales 240,000 units; Production 280,000 units.** There is no opening inventory at the beginning of January.

Prepare statements of profit or loss for the quarter using (a) Marginal costing (b) Absorption costing.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Careful with the time periods.** The fixed production overhead of \$1.6 million is an **annual** figure, but you are preparing a **quarterly** statement. Divide by four: \$400,000 for the quarter. The selling cost of \$320,000 is already quarterly — do not divide it.
- The **absorption rate is built from the annual budget**: $1,600,000 \div 1,280,000 \text{ units} = \1.25 per unit . Do not use the quarterly figures to build the rate.
- Production 280,000 exceeds sales 240,000, so closing inventory is **40,000 units** and stock is rising — therefore **absorption profit will be higher**. Predict it, then prove it.
- Absorbed overhead = $280,000 \times 1.25 = \$350,000$ against an actual quarterly charge of \$400,000, so there is an **under-absorption of \$50,000** to be deducted.

WORKING NOTES

W1 Unit costs and the absorption rate

Variable cost per unit = $40 + 16 + 10 = \$66$

Contribution per unit = $180 - 66 = \$114$

$$\text{Fixed production overhead absorption rate} = \frac{\$1,600,000 \text{ per year}}{1,280,000 \text{ units per year}} = \$1.25 \text{ per unit}$$

Full production cost per unit = $66 + 1.25 = \$67.25$

W2 Quarterly fixed costs and inventory

Fixed production overhead for the quarter = $1,600,000 \div 4 = \$400,000$

Fixed selling cost for the quarter = **\$320,000** (already quarterly)

Closing inventory = $0 + 280,000 - 240,000 = 40,000$ units

W3 Under / over absorption

Overhead absorbed = $280,000 \text{ units} \times \$1.25 = 350,000$

Actual overhead for the quarter = 400,000

UNDER-absorbed = \$50,000 — deducted in the absorption statement

(a) Marginal Costing Statement of Profit or Loss — quarter to March 20X8

Particulars	Amount \$
Sales $240,000 \times \$180$	43,200,000
Less: Variable cost of sales $240,000 \times \$66$	(15,840,000)
CONTRIBUTION $240,000 \times \$114$	27,360,000
Less: Fixed production overhead W2	(400,000)
Less: Fixed selling costs	(320,000)
PROFIT	26,640,000

(b) Absorption Costing Statement of Profit or Loss — quarter to March 20X8

Particulars	Amount \$
Sales	43,200,000
Cost of production $280,000 \times \$67.25$	18,830,000
Less: Closing inventory $40,000 \times \$67.25$	(2,690,000)
Cost of sales $240,000 \times \$67.25$	(16,140,000)
Gross profit	27,060,000
Less: Under-absorbed fixed production overhead W3	(50,000)
Less: Fixed selling costs	(320,000)
PROFIT	26,690,000

Reconciliation

Particulars	Amount \$
Marginal costing profit	26,640,000
Add: Fixed production overhead carried in closing inventory $40,000 \times \$1.25$	50,000
Absorption costing profit	26,690,000

THE TRAP / WHERE MARKS ARE LOST

- **Charging the whole \$1.6 million in a three-month statement.** It is an annual cost — only a quarter of it belongs to this quarter.
- **Dividing the \$320,000 selling cost by four.** The question says “per quarter”. Read the units on every figure.
- **Building the absorption rate from 280,000 actual units instead of 1,280,000 budgeted.** That would give \$1.43 and destroy the whole answer — and you would also find no under-absorption, which should ring a warning bell.

FINAL ANSWER

Contribution per unit	\$ 114
Absorption rate / full cost per unit	\$ 1.25 / \$ 67.25
Closing inventory	40,000 units = \$ 2,690,000
Under-absorbed overhead	\$ 50,000
(a) Marginal costing profit	\$ 26,640,000
(b) Absorption costing profit	\$ 26,690,000
Difference	\$ 50,000 = 40,000 units × \$ 1.25 ✓

Q19 Opening and closing stock at different rates

Statements · p.26-27

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Eagle makes one product. Budgeted production and sales for the year just ended were 32,000 units. Actual sales for the year were 34,000 units at a selling price of \$34 per unit, and actual production was 36,000 units. Opening inventory was 5,000 units valued at the unit cost for the preceding year. Using an absorption costing system based on budgeted levels of output, unit costs for the year and the preceding year were as follows:

Particulars	Year \$	Preceding year \$
Prime cost	24	22
Variable production overhead	3.5	1
Fixed production overhead	1.5	3
Selling and administrative overhead (fixed)	1	1
Total cost	30	27
Profit	4	3
Selling price	34	30

Fixed production overhead for the year just ended was \$48,000 (budget and actual). Actual variable costs per unit were the same as budgeted. Prepare statements of profit or loss for the year just ended on the basis that (i) an absorption costing system is used and (ii) a marginal costing system is used.

READ THE QUESTION FIRST — WHAT TO SPOT

The hardest statement problem in the module, because **opening and closing stock are valued at different rates** — last year's and this year's.

- **Opening inventory of 5,000 units is at the PRECEDING year's cost.** For absorption that is $22 + 1 + 3 = \$26$ per unit. For marginal it is $22 + 1 = \$23$ per unit. Note the fixed element was \$3 last year, not \$1.50.
- Closing inventory = $5,000 + 36,000 - 34,000 = 7,000$ units, valued at **this** year's cost: absorption $24 + 3.5 + 1.5 = \$29$; marginal $24 + 3.5 = \$27.50$.
- **Check the fixed overhead rate:** $\$48,000 \div 32,000$ budgeted units = \$1.50 which matches the table. That confirms the rate is built on budgeted output.
- Actual production 36,000 exceeds budget 32,000, so overhead is **OVER-absorbed**: $36,000 \times 1.50 = \$54,000$ absorbed against \$48,000 actual, a credit of \$6,000.
- Selling and administrative overhead is **fixed**, so it is $\$1 \times 32,000$ budgeted units = \$32,000 for the year — not $\$1 \times 34,000$.

WORKING NOTES

W1 Inventory in units

Opening 5,000 + Production 36,000 – Sales 34,000 = **Closing 7,000 units**

W2 Unit values for stock — two different years, two different rates

	Absorption \$	Marginal \$
Opening stock — preceding year rates		
Prime cost	22	22
Variable production overhead	1	1
Fixed production overhead	3	—
Per unit	26	23
Closing stock — this year's rates		
Prime cost	24	24
Variable production overhead	3.5	3.5
Fixed production overhead	1.5	—
Per unit	29	27.50

Opening stock value — absorption = 5,000 × 26 = **\$130,000** | marginal = 5,000 × 23 = **\$115,000**

Closing stock value — absorption = 7,000 × 29 = **\$203,000** | marginal = 7,000 × 27.50 = **\$192,500**

W3 Over-absorption of fixed production overhead

Absorbed = 36,000 units × \$1.50 = 54,000

Actual = 48,000

OVER-absorbed = \$6,000 — added back

(i) Absorption Costing Statement of Profit or Loss

Particulars	Amount \$
Sales 34,000 × \$34	1,156,000
Opening inventory W2: 5,000 × \$26	130,000
Add: Cost of production 36,000 × \$29	1,044,000
Less: Closing inventory W2: 7,000 × \$29	(203,000)
Cost of sales	(971,000)
Gross profit	185,000
Add: Over-absorbed fixed production overhead W3	6,000
Less: Selling and administrative overhead fixed, \$1 × 32,000 budget	(32,000)
PROFIT	159,000

(ii) Marginal Costing Statement of Profit or Loss

Particulars	Amount \$
Sales	1,156,000
Opening inventory W2: $5,000 \times \$23$	115,000
Add: Variable cost of production $36,000 \times \$27.50$	990,000
Less: Closing inventory W2: $7,000 \times \$27.50$	(192,500)
Variable cost of sales	(912,500)
CONTRIBUTION	243,500
Less: Fixed production overhead actual	(48,000)
Less: Selling and administrative overhead	(32,000)
PROFIT	163,500

Reconciliation

Particulars	Amount \$
Marginal costing profit	163,500
Add: Fixed overhead in closing inventory $7,000 \times \$1.50$	10,500
Less: Fixed overhead in opening inventory $5,000 \times \$3.00$	(15,000)
Absorption costing profit	159,000

Here the absorption profit is **lower** even though inventory rose in units — because the fixed overhead rate *halved* from \$3.00 to \$1.50. Always reconcile on the **rupee value** of fixed overhead in stock, never on units.

WHY THIS WORKS

This problem exists to break a lazy rule. Students learn “stock up → absorption profit higher” and apply it blindly. Here inventory rose from 5,000 to 7,000 units, yet absorption profit is \$4,500 **lower**.

The reason is that the fixed overhead rate fell from \$3.00 to \$1.50. The 5,000 opening units released \$15,000 of last year’s fixed cost into this year’s charge, while the 7,000 closing units only carried \$10,500 forward. The rule is really about the **value** of fixed overhead held in stock, not the number of units.

FINAL ANSWER

Closing inventory	7,000 units
Stock value — absorption: opening / closing	\$ 130,000 / \$ 203,000
Stock value — marginal: opening / closing	\$ 115,000 / \$ 192,500
Over-absorbed overhead	\$ 6,000
(i) Absorption costing profit	\$ 159,000
(ii) Marginal costing profit	\$ 163,500
Difference	\$ 4,500 = 10,500 – 15,000 ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Stoney is drafting a budget on the basis of the following data: Direct material \$10 per unit; Direct labour \$5 per unit; Variable production expenses \$8 per unit; Fixed production costs \$27,000 per month; Normal output 9,000 units per month (90% capacity); Sales price \$30 per unit.

In order to build up inventory in anticipation of an increase in demand expected later in the year, production is to exceed sales in the first three months as follows:

	Month 1	Month 2	Month 3
Production	6,500	9,000	10,000
Sales	5,000	8,500	9,500

Prepare two statements of profit or loss, each in comparative columnar form, covering each of the three months: (i) on a marginal costing basis and (ii) on a full absorption costing basis.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Track the inventory across all three months first** — the closing stock of one month is the opening stock of the next. Get this table right and the rest is mechanical.
- Absorption rate = $27,000 \div 9,000$ **normal output** = \$3 per unit. Note it is built on **normal** output, not on each month's actual production.
- That is why **each month has a different under/over absorption**: month 1 produces only 6,500 (under-absorbed), month 2 produces exactly 9,000 (nil), month 3 produces 10,000 (over-absorbed). This is the point of the problem.
- Stock rises every month, so **absorption profit is higher in every month**.

WORKING NOTES

W1 Inventory movement in units — carry it across the months

	Month 1	Month 2	Month 3
Opening stock	Nil	1,500	2,000
Add: Production	6,500	9,000	10,000
Less: Sales	(5,000)	(8,500)	(9,500)
Closing stock	1,500	2,000	2,500

W2 Unit costs

Variable production cost per unit = $10 + 5 + 8 = \$23$

Contribution per unit = $30 - 23 = \$7$

$$\text{Fixed overhead absorption rate} = \frac{\$27,000}{9,000 \text{ units normal output}} = \$3 \text{ per unit}$$

Full production cost per unit = $23 + 3 = \$26$

W3 Under / over absorption, month by month

	Month 1	Month 2	Month 3
Production (units)	6,500	9,000	10,000
Overhead absorbed at \$3	19,500	27,000	30,000
Actual fixed overhead	27,000	27,000	27,000
Under (–) / Over (+) absorbed	(7,500)	Nil	+3,000

(i) Marginal Costing Statements of Profit or Loss

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Sales $\text{units sold} \times \30	150,000	255,000	285,000
Opening stock $\text{at } \$23$	Nil	34,500	46,000
Add: Variable cost of production $\times \$23$	149,500	207,000	230,000
Less: Closing stock $\text{at } \$23$	(34,500)	(46,000)	(57,500)
Variable cost of sales	(115,000)	(195,500)	(218,500)
CONTRIBUTION $\text{units sold} \times \7	35,000	59,500	66,500
Less: Fixed production costs $\text{in full every month}$	(27,000)	(27,000)	(27,000)
PROFIT	8,000	32,500	39,500

(ii) Absorption Costing Statements of Profit or Loss

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Sales	150,000	255,000	285,000
Opening stock $\text{at } \$26$	Nil	39,000	52,000
Add: Cost of production $\times \$26$	169,000	234,000	260,000
Less: Closing stock $\text{at } \$26$	(39,000)	(52,000)	(65,000)
Cost of sales $\text{units sold} \times \26	(130,000)	(221,000)	(247,000)
Gross profit	20,000	34,000	38,000
Under (–) / Over (+) absorbed overhead W3	(7,500)	Nil	3,000
PROFIT	12,500	34,000	41,000

Reconciliation, month by month

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Marginal costing profit	8,000	32,500	39,500
Add: Fixed overhead in closing stock $\text{closing units} \times \3	4,500	6,000	7,500
Less: Fixed overhead in opening stock $\text{opening units} \times \3	Nil	(4,500)	(6,000)
Absorption costing profit	12,500	34,000	41,000
<i>Memo:</i> increase in stock (units)	1,500	500	500
<i>Memo:</i> difference in profit $= \text{increase} \times \3	4,500	1,500	1,500

WHY THIS WORKS

Month 2 is the instructive one. Production of 9,000 exactly equals normal output, so there is **no under or over absorption at all** — yet the two profits still differ by \$1,500, because stock still rose by 500 units. That separates the two effects cleanly:

- **Under/over absorption** arises when actual production differs from *normal output*.
- **The marginal-versus-absorption profit gap** arises when production differs from *sales*.

They are different things and a good answer does not confuse them.

FINAL ANSWER

Contribution per unit / full cost per unit	\$ 7 / \$ 26
Absorption rate	\$ 3 per unit (27,000 ÷ 9,000 normal)
Closing stock — months 1 / 2 / 3	1,500 / 2,000 / 2,500 units
(i) Marginal profit — M1 / M2 / M3	\$ 8,000 / \$ 32,500 / \$ 39,500
(ii) Absorption profit — M1 / M2 / M3	\$ 12,500 / \$ 34,000 / \$ 41,000
Under / over absorption — M1 / M2 / M3	(\$ 7,500) / Nil / +\$ 3,000

Cost-Volume-Profit Analysis

18 problems · workbook pages 31 to 37

What Module 3 is really about

Module 2 taught you the formulas. Module 3 asks you to **make a decision** with them. Almost every problem here is a manager saying “should I do this or not?”, and your job is to answer with a number and a recommendation.

THE SINGLE RULE THAT DECIDES EVERY PROBLEM IN THIS MODULE

Ignore fixed cost. Compare CONTRIBUTION.

Fixed cost is going to be incurred whether you accept the order or not, so it cannot help you choose. Only the amounts that CHANGE between the alternatives matter. In this module that is almost always contribution.

There are only two exceptions, and both are flagged where they arise:

- **When fixed cost itself changes** because of the decision — new machinery, an extra sales office, a shut-down saving. Then the change in fixed cost is relevant and must be brought in.
- **When capacity runs out.** Then you are not choosing whether to produce, but WHAT to produce, and the answer is contribution per unit of the scarce resource — not contribution per unit of product.

The four decision types — recognise which one you are in

Type	The question sounds like ...	What you compare	Problems
1. Special order & pricing	“An export order at a reduced price ... should we accept?” “What minimum price?”	Extra contribution from the order against any extra fixed cost it causes. Spare capacity is the key condition.	1, 2, 4, 5, 6, 11
2. Make or buy	“It costs us ₹6.25 to make; the market price is ₹4.85.”	The variable cost of making against the buying price. Fixed cost continues either way, so exclude it.	3
3. Key factor / product mix	“Material is restricted to 18,400 kg” “the limiting factor is labour”	Contribution per unit of the limiting factor , then rank and allocate.	7, 8, 9, 10
4. Shut-down	“Management plans to shut down the plant ... should it?”	Loss if you continue against loss if you shut. Shut only if the shut-down loss is smaller.	12, 13, 14

Type	The question sounds like ...	What you compare	Problems
5. Charts	"Draw the break-even chart" "draw the profit-volume graph"	Nothing — but you must plot correctly and read the BEP off the graph, then verify by calculation.	15-18

HOW TO ANSWER ANY DECISION QUESTION AND NOT LOSE MARKS

1. **Separate fixed from variable** in whatever the question gives you. Nothing can start until this is done.
2. **Find the contribution per unit** — selling price minus *all* variable costs including variable selling expenses.
3. **Check the capacity.** Is there spare capacity for the new order, or must something be given up? This single check changes the whole answer.
4. **Build a comparative statement** with one column per alternative, including the present position as a column. Never answer a decision question with loose calculations.
5. **Write the recommendation in words**, with the amount. "Accept the order, as it increases profit by ₹24,000." The sentence carries marks of its own.
6. **Mention the non-financial point** if there is an obvious one — effect on the existing market, reliability of the supplier, the workforce. One line is enough and it is often the difference between a good answer and a full-mark one.

Q1 Minimum price for a special order

Special order · p.31

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A manufacturer makes an average profit of ₹2.5 on a selling price of ₹14.5. He produces and sells 60,000 units at 60% capacity. His cost of sales per unit is: Direct Materials ₹4; Direct Wages ₹1; Factory overheads (Variable) ₹3; Selling overheads (Variable) 25 paise. Total fixed cost ₹2,25,000.

During the current year he anticipates that (a) fixed cost will increase by 10% and (b) material and labour cost will increase by 5% each. Under these circumstances he obtains an offer for a further 20% utilisation of his capacity. What minimum price would you recommend to accept the offer so as to ensure an overall profit of ₹1,60,000?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Verify the given data first.** Variable cost = $4 + 1 + 3 + 0.25 = ₹8.25$; fixed per unit = $2,25,000 \div 60,000 = ₹3.75$; total ₹12.00; profit = $14.50 - 12.00 = ₹2.50$. That tie-up proves you have read the question correctly.
- **60,000 units is 60% capacity**, so 100% = 1,00,000 units and the further 20% = **20,000 units**.
- The two cost increases apply to **everything**, existing units as well as the new order.
- This is a **working-backwards** problem: the profit is given and the price is unknown. So build the required total contribution first, take away what the existing 60,000 units provide, and whatever is left must come from the 20,000 new units.

WORKING NOTES

W1 Revised variable cost per unit

Element	Present ₹	Change	Revised ₹
Direct materials	4.00	+5%	4.20
Direct wages	1.00	+5%	1.05
Factory overheads (variable)	3.00	no change	3.00
Selling overheads (variable)	0.25	no change	0.25
Variable cost per unit	8.25		8.50

W2 Revised fixed cost and capacity

Revised fixed cost = 2,25,000 + 10% = **₹2,47,500**

60,000 units = 60% capacity → 100% = 1,00,000 units

Further 20% of capacity = **20,000 units**

W3 Working backwards to the minimum price

Required profit = 1,60,000

Add: Revised fixed cost = 2,47,500

Total contribution required = ₹4,07,500

Less: Contribution from existing 60,000 units = 60,000 × (14.50 – 8.50) = 60,000 × 6 = (3,60,000)

Contribution needed from the new 20,000 units = ₹47,500

$$\text{Contribution per unit on the offer} = \frac{47,500}{20,000} = ₹2.375$$

Minimum price = Variable cost + Required contribution = 8.50 + 2.375 = ₹10.875 per unit

Round up to ₹10.88 in practice — rounding down would leave the profit target just short.

Proof — profit statement at the recommended price

Particulars	Amount ₹
Sales — existing 60,000 × ₹14.50	8,70,000
Sales — offer 20,000 × ₹10.875	2,17,500
Total sales	10,87,500
Less: Variable cost 80,000 × ₹8.50	(6,80,000)
Contribution	4,07,500
Less: Fixed cost (revised)	(2,47,500)
Profit exactly the target ✓	1,60,000

THE TRAP / WHERE MARKS ARE LOST

- Applying the 5% increase only to the new order. The cost rise affects the whole year's production.
- Increasing the factory and selling overheads by 5% too. The question says **material and labour** only.
- Adding the old fixed cost per unit of ₹3.75 to the minimum price. The fixed cost is already covered by the existing 60,000 units, which is why the offer can be accepted at ₹10.875 — far below the normal ₹14.50 — and still increase profit.

FINAL ANSWER

Revised variable cost per unit	₹ 8.50
Revised fixed cost	₹ 2,47,500
Units in the offer (20% of capacity)	20,000
Total contribution required	₹ 4,07,500
Contribution needed per offer unit	₹ 2.375
Minimum price recommended	₹ 10.875 (say ₹ 10.88) per unit

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

ABC Company produces a single product sold at ₹75 per unit. Present production and sale is 40,000 units per month, representing 50% of capacity. Variable cost per unit ₹50; fixed costs per month ₹10 lakhs. Three proposals are on hand:

- I. Accept an export supply order for 30,000 units per month at a reduced price of ₹60 per unit, incurring an additional variable cost of ₹5 per unit towards export packing, duties etc.
- II. Increase domestic market sales by selling to a domestic chain store 30,000 units at ₹55 per unit, retaining the existing sales at the existing price.
- III. Reduce the selling price for increased domestic sale as advised by the sales department:

Reduce selling price per unit (₹)	Increase in sales expected (units)
5	10,000
8	30,000
11	35,000

Prepare a table to present the results of the above proposals and give your comments and advice.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Capacity check first.** 40,000 units = 50%, so 100% = 80,000 units per month and there are **40,000 units of spare capacity**. Every proposal must be tested against this ceiling.
- **The present position is break-even.** Contribution $40,000 \times 25 = ₹10,00,000$, which exactly equals the fixed cost. Profit is nil. Say this in your answer — it explains why the management is looking at proposals at all.
- **Proposals I and II add sales without touching the existing price.** So compute only the extra contribution and add it to the present profit of nil.
- **Proposal III cuts the price on ALL units**, existing and new. That is the crucial difference — you must recompute the whole contribution, not just the increment. Most students get this wrong.

WORKING NOTES

W1 Present position and capacity

Contribution per unit = $75 - 50 = ₹25$

Present contribution = $40,000 \times 25 = ₹10,00,000$

Less fixed cost = (10,00,000) → **present profit = NIL (break-even)**

40,000 units = 50% capacity → 100% = 80,000 units; **spare capacity 40,000 units**

W2 Proposal III — the whole price is cut, so recompute everything

Price cut ₹	New selling price	Total units	Contribution per unit	Total contribution	Within capacity?
5	70	50,000	20	10,00,000	Yes
8	67	70,000	17	11,90,000	Yes
11	64	75,000	14	10,50,000	Yes

Total units = 40,000 existing + the stated increase. Contribution per unit = new price – ₹50 variable cost.

Comparative Statement of the Present Position and All Proposals

Particulars	Present	Proposal I (export)	Proposal II (chain store)	III — cut ₹5	III — cut ₹8	III — cut ₹11
Units sold	40,000	70,000	70,000	50,000	70,000	75,000
Selling price on existing units	75	75	75	70	67	64
Selling price on additional units	—	60	55	70	67	64
Contribution — existing 40,000 units	10,00,000	10,00,000	10,00,000	8,00,000	6,80,000	5,60,000
Contribution — additional units	—	1,50,000	1,50,000	2,00,000	5,10,000	4,90,000
Total contribution	10,00,000	11,50,000	11,50,000	10,00,000	11,90,000	10,50,000
Less: Fixed cost	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)
PROFIT	Nil	1,50,000	1,50,000	Nil	1,90,000	50,000

WHY THIS WORKS

Advice: accept Proposal III with a price reduction of ₹8, giving a profit of ₹1,90,000 per month. It is the best of the six columns.

Two things are worth commenting on, and both earn marks:

- **Proposals I and II give identical profit** of ₹1,50,000. The export order sells at ₹60 but carries ₹5 of extra cost; the chain store sells at ₹55 with no extra cost. Both leave a contribution of ₹5 per unit. On financial grounds there is nothing to choose — so decide on non-financial grounds. The export order does not disturb the home market, whereas selling to a domestic chain at ₹55 risks existing customers demanding the same price. **On that basis Proposal I is safer than Proposal II.**
- **Cutting the price by ₹11 is worse than cutting it by ₹8**, even though it sells 5,000 more units. The extra volume does not make up for the contribution given away on all 75,000 units. This is the danger of chasing volume through price cuts.

THE TRAP / WHERE MARKS ARE LOST

- Treating Proposal III as an *increment* and adding only the new units' contribution. The price cut applies to the existing 40,000 units too, which is why the ₹5 cut leaves profit at nil rather than improving it.
- Forgetting to check capacity. All the options here fit inside 80,000 units, but had any exceeded it you would have had to turn away domestic sales — a completely different calculation.

FINAL ANSWER

Contribution per unit (present)	₹ 25
Present profit	Nil — the company is at break-even
Proposal I — export at ₹60	Profit ₹ 1,50,000
Proposal II — chain store at ₹55	Profit ₹ 1,50,000
Proposal III — cut ₹5 / ₹8 / ₹11	Nil / ₹ 1,90,000 / ₹ 50,000
Recommendation	Proposal III with an ₹ 8 cut — profit ₹ 1,90,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A radio manufacturing company finds that while it costs ₹6.25 to make each component X 2730, the same is available in the market at ₹4.85 each, with an assurance of continued supply. The breakdown of cost is: Materials ₹2.75 each; Labour ₹1.75 each; Other variables ₹0.50 each; Depreciation and other fixed costs ₹1.25 each — total ₹6.25 each.

(a) Should you make or buy? **(b)** What would be your decision if the supplier offered the component at ₹5.85 each?

READ THE QUESTION FIRST — WHAT TO SPOT

- The whole problem turns on one question: **which costs disappear if you stop making?**
- Materials, labour and other variables disappear — they are only incurred when you make. **Depreciation and other fixed costs do not.** The machine still depreciates whether you use it or not.
- So the relevant cost of making is $2.75 + 1.75 + 0.50 = \text{₹}5.00$, not ₹6.25. Comparing 6.25 with the market price is the trap the question is built around.

Relevant cost comparison

Particulars	Per component ₹	Relevant?
Materials	2.75	Yes — avoided if bought
Labour	1.75	Yes — avoided if bought
Other variables	0.50	Yes — avoided if bought
Relevant (marginal) cost of making	5.00	
Depreciation and other fixed costs	1.25	No — incurred anyway, so it is irrelevant to the decision
Total cost as shown in the books	6.25	— misleading for this decision

The decision under each supplier price

	(a) Supplier at ₹4.85	(b) Supplier at ₹5.85
Relevant cost of making	5.00	5.00
Cost of buying	4.85	5.85
Saving by buying / (by making)	0.15 by buying	(0.85) — make instead
Decision	BUY	MAKE

WHY THIS WORKS

(a) Buy. At ₹4.85 the supplier is cheaper than the ₹5.00 it costs in avoidable terms, so buying saves ₹0.15 per component.

(b) Make. At ₹5.85 the supplier is dearer than ₹5.00, so making saves ₹0.85 per component. Note that the company should *still make* even though the book cost of ₹6.25 looks higher than ₹5.85 — because ₹1.25 of that book cost will be incurred whether the component is made or not.

Non-financial points worth one line: the question mentions “an assurance of continued supply”, which removes the usual worry about depending on an outside supplier. Against that, buying leaves the company’s own capacity idle — if that freed capacity can be used for something else, its contribution should be added to the case for buying.

FINAL ANSWER

Relevant (avoidable) cost of making	₹ 5.00 per component
Irrelevant fixed cost excluded	₹ 1.25 per component
(a) At ₹ 4.85	BUY — saves ₹ 0.15 per component
(b) At ₹ 5.85	MAKE — saves ₹ 0.85 per component

Q4 Foreign order using idle capacity

Special order · p.32

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Indo-British Company has capacity to produce 5,000 articles but actually produces only 2,000 articles for the home market at the following costs: Materials ₹40,000; Wages ₹36,000; Factory overheads — Fixed ₹12,000, Variable ₹20,000; Administration overhead — Fixed ₹18,000; Selling and Distribution overheads — Fixed ₹10,000, Variable ₹16,000. Total cost ₹1,52,000.

The home market can consume only 2,000 articles at a selling price of ₹80 per article. An additional order for the supply of 3,000 articles is received from a foreign country at ₹65 per article. Should this order be accepted or not, if execution of this order entails an additional packing cost of ₹3,000?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Split the costs.** Variable: materials 40,000 + wages 36,000 + factory variable 20,000 + selling variable 16,000 = ₹1,12,000 for 2,000 units, i.e. **₹56 per unit**. Fixed: 12,000 + 18,000 + 10,000 = **₹40,000**.
- **Capacity is exactly right.** 2,000 + 3,000 = 5,000 = full capacity. So no domestic sales have to be sacrificed — the order uses idle capacity only.
- Because fixed cost is already fully covered by the home market, the only test is whether the ₹65 price beats the ₹56 variable cost plus the ₹3,000 packing cost.
- The packing cost of ₹3,000 is a **lump sum specific to this order**, so it is relevant even though it is not a per-unit cost.

WORKING NOTES**W1 Splitting the present cost**

Element	Total ₹	Nature	Per unit ₹
Materials	40,000	Variable	20
Wages	36,000	Variable	18
Factory overheads — variable	20,000	Variable	10
Selling and distribution — variable	16,000	Variable	8
Total variable cost (2,000 units)	1,12,000		56
Factory overheads — fixed	12,000	Fixed	—
Administration overhead — fixed	18,000	Fixed	—
Selling and distribution — fixed	10,000	Fixed	—
Total fixed cost	40,000		—
Total cost	1,52,000		76

Comparative Statement of Profitability

Particulars	Present 2,000 units ₹	Foreign order 3,000 units ₹	Total 5,000 units ₹
Sales $2,000 \times 80$ $3,000 \times 65$	1,60,000	1,95,000	3,55,000
Less: Variable cost at ₹56 per unit	(1,12,000)	(1,68,000)	(2,80,000)
Less: Additional packing cost	—	(3,000)	(3,000)
Contribution	48,000	24,000	72,000
Less: Fixed cost <i>unchanged — already borne by the home market</i>	(40,000)	—	(40,000)
PROFIT	8,000	24,000	32,000

WHY THIS WORKS

Accept the order. It adds ₹24,000 of contribution and, because the fixed cost of ₹40,000 is already fully recovered from the home market, the whole of that ₹24,000 drops straight into profit. Profit rises from ₹8,000 to ₹32,000 — a fourfold increase.

Notice that the order price of ₹65 is **below the full cost of ₹76 per unit**. A student who compares 65 with 76 will reject the order and lose the entire question. The correct comparison is 65 against the ₹56 variable cost plus the ₹1 per unit of packing.

One non-financial line: the order is from a foreign country, so it does not undercut the home market where the price stays at ₹80. That is what makes accepting it safe.

FINAL ANSWER

Variable cost per unit	₹ 56
Total fixed cost	₹ 40,000
Contribution from the order	₹ 1,95,000 – ₹ 1,71,000 = ₹ 24,000
Present profit / profit after acceptance	₹ 8,000 / ₹ 32,000
Decision	ACCEPT — profit increases by ₹ 24,000

Q5 Export order needing more capacity than exists

Special order · p.32-33

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company currently operating at 80% capacity has the following particulars: Sales ₹32,00,000; Direct materials ₹10,00,000; Direct labour ₹4,00,000; Variable overheads ₹2,00,000; Fixed overheads ₹13,00,000.

An export order has been received that would utilise half the capacity of the factory. This order cannot be split — it has to be taken in full and executed at 10% below the normal domestic prices, or rejected totally. The alternatives available are:

- (1) Reject the order and continue with domestic sales only (as at present); or
- (2) Accept the order, split capacity between overseas and domestic sales and turn away excess domestic demand; or
- (3) Increase capacity to accept the export order and maintain present domestic sales by (a) buying equipment that will increase capacity, resulting in an increase of ₹1,00,000 in fixed cost, or (b) increasing the normal wage rate by 1½ times to increase output.

Prepare a comparative statement of profitability and suggest the best alternative.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Scale everything to a per-1% basis first.** The figures are at 80% capacity, so divide by 80 to get the value of 1% and then build any level you need. Sales are ₹40,000 per 1%; variable cost is ₹20,000 per 1%.
- **The export order needs 50% of capacity** and cannot be split. Present domestic use is 80%. So $80 + 50 = 130\%$ — more than the factory has. That conflict is the whole problem.
- Alternative 2 keeps total capacity at 80%: 50% export + only 30% domestic. You **give up** 50 percentage points of domestic sales.
- Alternative 3 runs at 130% by either buying equipment (extra fixed cost) or paying $1\frac{1}{2}$ times the wage rate on the extra output.
- **Fixed overhead of ₹13,00,000 stays the same in alternatives 1, 2 and 3(b)** — only 3(a) changes it.

WORKING NOTES

W1 Value of 1% of capacity

Element	At 80% capacity ₹	Per 1% ₹	At 50% ₹	At 30% ₹
Sales	32,00,000	40,000	20,00,000	12,00,000
Direct materials	10,00,000	12,500	6,25,000	3,75,000
Direct labour	4,00,000	5,000	2,50,000	1,50,000
Variable overheads	2,00,000	2,500	1,25,000	75,000
Total variable cost	16,00,000	20,000	10,00,000	6,00,000
Contribution	16,00,000	20,000	10,00,000	6,00,000

W2 The export order at 50% capacity, priced 10% below normal

Sales at normal price = ₹20,00,000
 Less 10% = **export sales value ₹18,00,000**
 Variable cost at 50% = ₹10,00,000
Export contribution = ₹8,00,000

W3 Alternative 3(b) — wages at $1\frac{1}{2}$ times on the extra 50%

Direct materials at 50% = 6,25,000
 Direct labour at 50% $\times 1.5 = 2,50,000 \times 1.5 = 3,75,000$
 Variable overheads at 50% = 1,25,000
Variable cost of the export order = ₹11,25,000
 Export contribution = $18,00,000 - 11,25,000 = \text{₹}6,75,000$

Comparative Statement of Profitability

Particulars	Alt 1 Reject	Alt 2 Split 50+30	Alt 3(a) Buy equipment	Alt 3(b) Overtime wages
Capacity used	80% domestic	50% export 30% domestic	80% domestic 50% export	80% domestic 50% export
Domestic sales	32,00,000	12,00,000	32,00,000	32,00,000
Export sales W2	—	18,00,000	18,00,000	18,00,000
Total sales	32,00,000	30,00,000	50,00,000	50,00,000

Particulars	Alt 1 Reject	Alt 2 Split 50+30	Alt 3(a) Buy equipment	Alt 3(b) Overtime wages
Less: Variable cost — domestic	(16,00,000)	(6,00,000)	(16,00,000)	(16,00,000)
Less: Variable cost — export	—	(10,00,000)	(10,00,000)	(11,25,000)
Total contribution	16,00,000	14,00,000	24,00,000	22,75,000
Less: Fixed overheads	(13,00,000)	(13,00,000)	(13,00,000)	(13,00,000)
Less: Additional fixed cost for equipment	—	—	(1,00,000)	—
PROFIT	3,00,000	1,00,000	10,00,000	9,75,000
Ranking	3rd	4th — worst	1st — BEST	2nd

WHY THIS WORKS

Recommendation: Alternative 3(a) — buy the equipment and accept the export order while keeping domestic sales at 80%. Profit ₹10,00,000.

- **Alternative 2 is the worst of all**, at ₹1,00,000 — worse even than rejecting the order. Giving up 50 percentage points of full-price domestic business to take on 50 points of discounted export business destroys contribution. This is the trap the question is built around: an export order at a discount is only worth having if it uses **spare** capacity.
- 3(a) beats 3(b) by ₹25,000 because the extra fixed cost of ₹1,00,000 is cheaper than the ₹1,25,000 of extra wages. But note that 3(a) commits the company permanently while overtime can be stopped — worth one sentence if the order is a one-off.

FINAL ANSWER

Value of 1% capacity — sales / variable cost	₹ 40,000 / ₹ 20,000
Export contribution at normal wages	₹ 8,00,000
Alt 1 — reject	Profit ₹ 3,00,000
Alt 2 — split capacity	Profit ₹ 1,00,000 — worst
Alt 3(a) — buy equipment	Profit ₹ 10,00,000 — BEST
Alt 3(b) — overtime at 1½ times	Profit ₹ 9,75,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prestige Company Private Limited, manufacturing pressure cookers, has drawn up the following budget for the year 20X5-X6: Raw materials ₹20,00,000; Labour, stores, power and other variable costs ₹6,00,000; Manufacturing overheads (Fixed) ₹7,00,000; Variable distribution costs ₹4,00,000; General overheads including selling (Fixed) ₹3,00,000 — total ₹40,00,000. Income from sales ₹50,00,000; Budgeted profit ₹10,00,000.

The general manager suggests reducing selling price by 5% and expects to achieve an additional volume of 50%. There is sufficient manufacturing capacity. A more intensive manufacturing programme will involve additional costs of ₹50,000 for production planning. It will also be necessary to open an additional sales office at a cost of ₹1,00,000 per annum.

The sales manager suggests increasing selling price by 10%, which is estimated to reduce sales volume by 10%. At the same time, savings in manufacturing overheads and general overheads of ₹50,000 and ₹1,00,000 per annum respectively are expected on this reduced volume.

Which of these two proposals would you accept and why?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Split the budget into variable and fixed first.** Variable = raw materials 20,00,000 + labour and other variable 6,00,000 + variable distribution 4,00,000 = **₹30,00,000**. Fixed = 7,00,000 + 3,00,000 = **₹10,00,000**.
- **Handle volume and price separately.** Variable cost changes with **volume only**; sales value changes with **both** volume and price. Doing them in one step is where errors creep in.
- GM: volume $\times 1.50$ and price $\times 0.95$, so sales = $50,00,000 \times 1.50 \times 0.95$. Variable cost = $30,00,000 \times 1.50$ (the price cut does not change costs).
- Sales manager: volume $\times 0.90$ and price $\times 1.10$, so sales = $50,00,000 \times 0.90 \times 1.10$. Variable cost = $30,00,000 \times 0.90$.
- Then adjust fixed cost: GM **adds** 1,50,000; sales manager **saves** 1,50,000.

WORKING NOTES

W1 Splitting the budget

Element	Amount ₹	Nature
Raw materials	20,00,000	Variable
Labour, stores, power and other variable costs	6,00,000	Variable
Variable distribution costs	4,00,000	Variable
Total variable cost	30,00,000	
Manufacturing overheads	7,00,000	Fixed
General overheads including selling	3,00,000	Fixed
Total fixed cost	10,00,000	

Budgeted contribution = $50,00,000 - 30,00,000 = ₹20,00,000 \rightarrow$ P/V ratio **40%**

Budgeted profit = $20,00,000 - 10,00,000 = ₹10,00,000 \checkmark$ agrees with the question

W2 Building each proposal

General manager: sales = $50,00,000 \times 1.50 \times 0.95 = ₹71,25,000$

variable cost = $30,00,000 \times 1.50 = ₹45,00,000$

fixed cost = $10,00,000 + 50,000 + 1,00,000 = ₹11,50,000$

Sales manager: sales = $50,00,000 \times 0.90 \times 1.10 = ₹49,50,000$

variable cost = $30,00,000 \times 0.90 = ₹27,00,000$

fixed cost = $10,00,000 - 50,000 - 1,00,000 = ₹8,50,000$

Comparative Statement of Profitability

Particulars	Original budget ₹	General manager ₹	Sales manager ₹
Volume index	100%	150%	90%
Price index	100%	95%	110%
Sales	50,00,000	71,25,000	49,50,000
Less: Variable cost	(30,00,000)	(45,00,000)	(27,00,000)
Contribution	20,00,000	26,25,000	22,50,000
P/V ratio	40.00%	36.84%	45.45%
Less: Fixed cost	(10,00,000)	(11,50,000)	(8,50,000)
PROFIT	10,00,000	14,75,000	14,00,000
Increase over budget	—	+4,75,000	+4,00,000

WHY THIS WORKS

Accept the general manager's proposal. It gives ₹14,75,000 against the sales manager's ₹14,00,000 — ₹75,000 better, and ₹4,75,000 above the original budget.

But the reasoning matters more than the ₹75,000, and a full-mark answer says so:

- The GM wins on **volume**: his P/V ratio actually *falls* from 40% to 36.84% because he cut the price, yet 50% more volume more than makes up for it.
- The sales manager wins on **margin**: his P/V ratio *rises* to 45.45% and he even saves fixed cost — but on 10% less volume.
- **Risk cuts the other way.** The GM's plan depends on actually selling 50% more, which is a large assumption, and it commits ₹1,50,000 of new fixed cost that cannot easily be reversed. The sales manager's plan needs no extra spending and earns nearly as much. If you doubt the 50% volume increase, the sales manager's proposal is the safer choice — say so.

FINAL ANSWER

Variable cost / fixed cost in the budget	₹ 30,00,000 / ₹ 10,00,000
Budgeted P/V ratio	40%
General manager — profit	₹ 14,75,000 (P/V 36.84%)
Sales manager — profit	₹ 14,00,000 (P/V 45.45%)
Recommendation	General manager's proposal — ₹ 75,000 better

Key Factor and Product Mix

Use these steps for Q7, Q8, Q9 and Q10

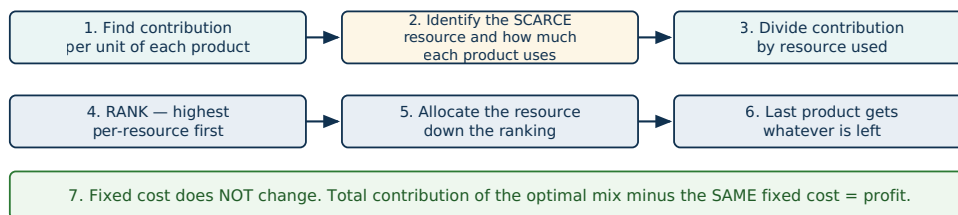
THE IDEA, AND WHY THE OBVIOUS ANSWER IS WRONG

When something runs short — material, labour hours, machine hours — you cannot make everything. You must choose. And the instinct is to make the product with the **highest contribution per unit**.

That instinct is wrong. If product A earns ₹30 of contribution but eats 15 labour hours, while product B earns only ₹16 but needs just 2 hours, then in the 15 hours that one A takes you could have made seven and a half B's and earned ₹120. B is four times better, even though its contribution per unit is half.

Rank by $\frac{\text{Contribution per unit}}{\text{Units of the scarce resource per unit}}$

This is called contribution per unit of key factor, or contribution per limiting factor. It is the entire topic in one line.



The six steps. Step 3 is the one that decides the answer; step 7 is the one students forget.

THREE THINGS TO BE CAREFUL ABOUT

- **Work out the resource per unit from the money figure.** If materials cost ₹80 per unit at ₹20 per kg, then the product uses 4 kg. Questions almost never give you the quantity directly.
- **Fixed cost stays the same in the optimal mix.** It was fixed before the shortage and it is fixed after. Do not recompute it per unit — carry the same total across.
- **Check whether demand also limits you.** You cannot make more of a product than can be sold, even if it ranks first. In Q8 and Q10 the budgeted quantity is the demand ceiling.

Q7 Three sales mixtures compared

Product mix · p.33-34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Present the following information to show to the management (a) the marginal product cost and the contribution per unit and (b) the total contribution and profits resulting from each of the following sales mixtures.

Particulars	Product A	Product B
Direct materials per unit	₹ 10	₹ 9
Direct wages per unit	₹ 3	₹ 2
Sales price per unit	₹ 20	₹ 15

Fixed expenses ₹800. Variable expenses are allocated to products as 100% of direct wages.

Sales mixtures: **(i)** 1,000 units of A and 2,000 units of B; **(ii)** 1,500 units of A and 1,500 units of B; **(iii)** 2,000 units of A and 1,000 units of B.

READ THE QUESTION FIRST — WHAT TO SPOT

- “Variable expenses are 100% of direct wages” means variable overhead = the same amount as wages: A gets ₹3, B gets ₹2. Add it as a third variable element.
- There is **no scarce resource here** — the total is 3,000 units in every mixture. So this is not a ranking problem; you simply evaluate three given mixes and pick the best.
- Because total units are the same in all three, the best mix is automatically the one with more of the **higher-contribution** product. A earns ₹4 and B earns ₹2, so expect mixture (iii) to win.

(a) Marginal cost and contribution per unit

Particulars	Product A ₹	Product B ₹
Selling price	20	15
Less: Direct materials	(10)	(9)
Less: Direct wages	(3)	(2)
Less: Variable expenses 100% of direct wages	(3)	(2)
Marginal (variable) cost per unit	16	13
Contribution per unit	4	2
P/V ratio	20%	13.33%

(b) Total contribution and profit under each sales mixture

Particulars	(i) 1,000 A + 2,000 B	(ii) 1,500 A + 1,500 B	(iii) 2,000 A + 1,000 B
Contribution from A units × ₹4	4,000	6,000	8,000
Contribution from B units × ₹2	4,000	3,000	2,000
Total contribution	8,000	9,000	10,000
Less: Fixed expenses	(800)	(800)	(800)
PROFIT	7,200	8,200	9,200
Ranking	3rd	2nd	1st — BEST

WHY THIS WORKS

Mixture (iii) is best, giving a profit of ₹9,200.

The reason is simple once you see it: all three mixes sell exactly 3,000 units, and A contributes ₹4 against B's ₹2. So every unit of B swapped for a unit of A adds ₹2. Moving from mixture (i) to (iii) swaps 1,000 units of B for 1,000 units of A and gains exactly ₹2,000 — which is the difference between ₹7,200 and ₹9,200.

Recommendation: sell as much of A as the market will absorb. The only reason to keep B in the mix at all is if demand for A is limited to 2,000 units.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 16 / ₹ 13
Contribution per unit — A / B	₹ 4 / ₹ 2
(i) Contribution / profit	₹ 8,000 / ₹ 7,200
(ii) Contribution / profit	₹ 9,000 / ₹ 8,200
(iii) Contribution / profit	₹ 10,000 / ₹ 9,200 — BEST

Q8 Material shortage — optimal mix

Key factor · p.34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company manufactures three products. The budgeted quantity, selling prices and unit costs are as under:

Particulars	A	B	C
Raw materials (@ ₹20 per kg)	80	40	20
Direct wages (@ ₹5 per hour)	5	15	10
Variable overheads	10	30	20
Fixed overheads	9	22	18
Budgeted production (in units)	6,400	3,200	2,400
Selling price per unit (₹)	140	120	90

(i) Present a statement of budgeted profit. **(ii)** Set the optimal product mix and determine the profit if the supply of raw materials is restricted to 18,400 kgs.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Convert the material cost into kilograms.** At ₹20 per kg: A uses $80 \div 20 = 4$ kg, B uses $40 \div 20 = 2$ kg, C uses $20 \div 20 = 1$ kg. This is the step the whole of part (ii) depends on.
- **Fixed overhead per unit is given**, so total fixed cost = $(6,400 \times 9) + (3,200 \times 22) + (2,400 \times 18) = ₹1,71,200$. That total **does not change** in part (ii).
- **Check the shortage.** Budget needs $(6,400 \times 4) + (3,200 \times 2) + (2,400 \times 1) = 25,600 + 6,400 + 2,400 = 34,400$ kg, but only 18,400 kg is available — a severe shortage of 16,000 kg.
- So rank by **contribution per kg**, and remember the budgeted production is the **demand ceiling** — you cannot make more of C than 2,400 units even though it ranks first.

(i) Statement of Budgeted Profit

Particulars	A	B	C	Total ₹
Selling price per unit	140	120	90	—
Less: Raw materials	(80)	(40)	(20)	—
Less: Direct wages	(5)	(15)	(10)	—
Less: Variable overheads	(10)	(30)	(20)	—
Variable cost per unit	95	85	50	—
Contribution per unit	45	35	40	—
Budgeted units	6,400	3,200	2,400	—
Total contribution	2,88,000	1,12,000	96,000	4,96,000
Fixed overheads units × rate	(57,600)	(70,400)	(43,200)	(1,71,200)
BUDGETED PROFIT				3,24,800

(ii) Step 1 — Ranking by contribution per kg of raw material

Particulars	A	B	C
Contribution per unit ₹	45	35	40
Raw material per unit cost ÷ ₹20	4 kg	2 kg	1 kg
Contribution per kg	11.25	17.50	40.00
Rank	3rd	2nd	1st

(ii) Step 2 — Allocating the 18,400 kg down the ranking

Rank	Product	Units made	kg per unit	kg used	kg remaining
1st	C — full demand 2,400 units	2,400	1	2,400	16,000
2nd	B — full demand 3,200 units	3,200	2	6,400	9,600
3rd	A — only what is left	2,400	4	9,600	Nil
	Total			18,400	Nil

A's output falls from the budgeted 6,400 units to 2,400 units — 9,600 kg ÷ 4 kg per unit. A is cut back because it has the lowest contribution per kg.

(ii) Step 3 — Profit under the optimal mix

Product	Units	Contribution per unit ₹	Total contribution ₹
C	2,400	40	96,000
B	3,200	35	1,12,000
A	2,400	45	1,08,000
Total contribution			3,16,000
Less: Fixed overheads unchanged from part (i)			(1,71,200)
PROFIT under the optimal mix			1,44,800

WHY THIS WORKS

Look at what the ranking did. Product A has the **highest contribution per unit** at ₹45 — and it is the one cut back hardest, from 6,400 units to 2,400. Product C has the **lowest** selling price and only ₹40 of contribution, yet it is protected in full.

The reason is that A swallows 4 kg per unit while C needs only 1 kg. Per kilogram of the scarce material, C returns ₹40 and A returns just ₹11.25. When a resource is scarce, what matters is the return per unit of that resource — never the return per unit of product.

Profit falls from ₹3,24,800 to ₹1,44,800 because of the shortage. But had you kept the budgeted proportions instead of re-ranking, it would have fallen much further — the ranking saves the company money.

FINAL ANSWER

Contribution per unit — A / B / C	₹ 45 / ₹ 35 / ₹ 40
Raw material per unit — A / B / C	4 kg / 2 kg / 1 kg
Contribution per kg — A / B / C	₹ 11.25 / ₹ 17.50 / ₹ 40.00
Ranking	C first, B second, A third
(i) Budgeted profit	₹ 3,24,800
Optimal mix	C 2,400 units, B 3,200 units, A 2,400 units
(ii) Profit under optimal mix	₹ 1,44,800

Q9 Labour shortage — ranking two products

Key factor · p.34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information in respect of Product A and Product B of a firm is given:

Particulars	Product A	Product B
Selling price per unit	₹ 75	₹ 48
Direct materials	₹ 30	₹ 30
Direct labour hours (₹0.50 per hour)	15 hours	2 hours

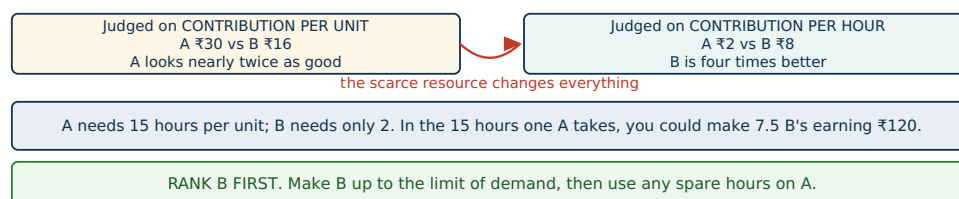
Variable overhead = 100% of direct wages. Fixed overheads = ₹3,000. Present the above information to show the profitability of products during a labour shortage.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Compute the wages from the hours.** A: 15 hours × ₹0.50 = ₹7.50. B: 2 hours × ₹0.50 = ₹1.00. Then variable overhead equals the same amounts.
- **The key factor is named for you** — “during a labour shortage”. So rank by contribution per **labour hour**.
- This problem is the clearest illustration in the module of why contribution per unit misleads. A earns nearly twice B’s contribution per unit, and is **four times worse** per hour.
- No quantities or total hours are given, so you are asked only to **show the profitability** — the ranking — not to build an optimal mix.

Statement of Profitability during a Labour Shortage

Particulars	Product A ₹	Product B ₹
Selling price per unit	75	48
Less: Direct materials	(30)	(30)
Less: Direct wages 15 × 0.50 2 × 0.50	(7.50)	(1.00)
Less: Variable overhead 100% of direct wages	(7.50)	(1.00)
Marginal cost per unit	45.00	32.00
Contribution per unit	30.00	16.00
P/V ratio	40.00%	33.33%
Direct labour hours per unit	15	2
Contribution per labour hour	2.00	8.00
Rank during a labour shortage	2nd	1st — four times better



Why contribution per unit is the wrong ranking when a resource is scarce.

WHY THIS WORKS

Product B is the more profitable during a labour shortage, earning ₹8 per labour hour against A's ₹2 — four times as much.

Recommendation: produce B up to the limit of its demand and use any remaining hours on A. Note that if labour were *not* scarce, A would be the better product on a per-unit basis and the answer would reverse — which is exactly why identifying the key factor is the first thing you do.

The fixed overhead of ₹3,000 has deliberately not been used. It is the same whichever product is made, so it cannot influence the ranking. Mention that you have excluded it and why.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 45 / ₹ 32
Contribution per unit — A / B	₹ 30 / ₹ 16
Labour hours per unit — A / B	15 / 2
Contribution per labour hour — A / B	₹ 2 / ₹ 8
Recommendation	Product B — four times more profitable per scarce labour hour

Q10 Labour limit — optimal mix and profit

Key factor · p.34–35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In a factory producing two different kinds of articles, the limiting factor is the availability of labour. From the following information for 2016, show which product is more profitable.

Particulars	Product A (₹ per unit)	Product B (₹ per unit)
Materials	5.00	5.00
Labour — 6 hours @ ₹0.50	3.00	—
Labour — 3 hours @ ₹0.50	—	1.50
Overheads — Fixed (50% of labour)	1.50	0.75
Overheads — Variable	1.50	1.50
Total Cost	11.00	8.75
Selling Price	14.00	11.00
Total production for the month	500 units	600 units

Also set the optimal product mix and the profitability if the maximum labour hours per month is 4,200 hours.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The fixed overhead is described as 50% of labour**, so it is stated per unit but it is still a fixed cost. Exclude it from the marginal cost and treat the total as a lump sum: $(500 \times 1.50) + (600 \times 0.75) = ₹1,200$.
- Marginal cost = materials + labour + variable overhead only. A = $5 + 3 + 1.50 = ₹9.50$; B = $5 + 1.50 + 1.50 = ₹8.00$.
- **Check the shortage.** Present production needs $(500 \times 6) + (600 \times 3) = 3,000 + 1,800 = 4,800$ hours, but only 4,200 are available — short by **600 hours**.
- Rank by contribution per labour hour, then allocate. Present production is the **demand ceiling** for each product.

Statement of Profitability per unit and per labour hour

Particulars	Product A ₹	Product B ₹
Selling price	14.00	11.00
Less: Materials	(5.00)	(5.00)
Less: Labour	(3.00)	(1.50)
Less: Variable overheads	(1.50)	(1.50)
Marginal cost per unit	9.50	8.00
Contribution per unit	4.50	3.00
Labour hours per unit	6	3
Contribution per labour hour	0.75	1.00
Rank	2nd	1st — more profitable

Optimal product mix within 4,200 labour hours

Rank	Product	Units	Hours per unit	Hours used	Hours left
1st	B — full demand 600 units	600	3	1,800	2,400
2nd	A — only what is left	400	6	2,400	Nil
	Total			4,200	Nil

A is cut from 500 units to 400 units — 2,400 hours ÷ 6 hours per unit. B is protected in full because it earns more per hour.

Profitability of the optimal mix

Product	Units	Contribution per unit ₹	Total contribution ₹
B	600	3.00	1,800
A	400	4.50	1,800
Total contribution			3,600
Less: Fixed overheads $(500 \times 1.50) + (600 \times 0.75)$			(1,200)
PROFIT			2,400

THE TRAP / WHERE MARKS ARE LOST

- **Including the fixed overhead in the marginal cost.** The question calls it “Overheads — Fixed” even though it is shown per unit. It must be excluded from contribution and deducted once as a total.
- **Ranking A first because its contribution of ₹4.50 is higher.** Per hour A returns only ₹0.75 against B’s ₹1.00.
- **Making more than 600 units of B** because it ranks first. 600 is all the market takes — the given production figure is the demand ceiling.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 9.50 / ₹ 8.00
Contribution per unit — A / B	₹ 4.50 / ₹ 3.00
Contribution per labour hour — A / B	₹ 0.75 / ₹ 1.00
More profitable product	Product B
Optimal mix within 4,200 hours	B 600 units + A 400 units
Total contribution / fixed cost	₹ 3,600 / ₹ 1,200

Q11 Export order versus local order — same figures, different advice

Special order · p.35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The cost of a manufacturing company for the product is: Materials ₹12.00; Labour ₹9.00; Variable expenses ₹6.00; Fixed expenses ₹18.00 — total ₹45.00. The unit of product is sold for ₹51.00.

The company's normal capacity is 1,00,000 units. The figures given above are for 80,000 units. The company has received an offer for 20,000 units @ ₹36 per unit from a foreign customer.

Advise the manufacturer on whether the order should be accepted. Also give your advice if the order is from a local merchant.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Variable cost per unit = 12 + 9 + 6 = ₹27.** The ₹18 of fixed expenses is per unit at 80,000 units, so total fixed cost = 80,000 × 18 = **₹14,40,000.**
- **Capacity check.** 80,000 + 20,000 = 1,00,000 = exactly normal capacity. So the order uses idle capacity and no existing sales are lost, and **fixed cost does not increase.**
- The offer price of ₹36 is well below the full cost of ₹45 — which is the trap. Compare it with the **₹27 variable cost**, not with ₹45.
- **The second half of the question is about business judgement, not arithmetic.** The numbers are identical for a local merchant; what changes is the risk to the existing market. Make sure you answer it in words.

Comparative Statement of Profitability

Particulars	Present 80,000 units ₹	Foreign order 20,000 units ₹	Total 1,00,000 units ₹
Sales 80,000×51 20,000×36	40,80,000	7,20,000	48,00,000
Less: Variable cost at ₹27 per unit	(21,60,000)	(5,40,000)	(27,00,000)
Contribution	19,20,000	1,80,000	21,00,000
Contribution per unit	24	9	21
Less: Fixed expenses unchanged — spare capacity used	(14,40,000)	—	(14,40,000)
PROFIT	4,80,000	1,80,000	6,60,000

WHY THIS WORKS

Advice on the foreign order: ACCEPT. Each unit earns ₹36 – ₹27 = ₹9 of contribution, and since the fixed cost of ₹14,40,000 is already fully recovered from the 80,000 domestic units, the whole ₹1,80,000 goes straight to profit. Profit rises from ₹4,80,000 to ₹6,60,000.

Advice if the order is from a local merchant: normally REJECT — and the arithmetic is not the reason. The figures are identical, so on paper the order still adds ₹1,80,000. The problem is what happens next:

- The merchant sells in the same market where the company charges ₹51. Goods bought at ₹36 can be resold below ₹51, undercutting the company's own customers.
- Existing customers who learn of the ₹36 price will demand it too. If even a fifth of the 80,000 regular units had to be repriced downward, the loss would swamp the ₹1,80,000 gained.

- A foreign market is **geographically separate**, so the low price cannot leak back. That separation is the only thing that makes the export order safe.

Accept from a local merchant **only** if the goods can be genuinely segregated — a different brand, a different region, or a binding condition on resale. Say that, and you have the full mark.

FINAL ANSWER

Variable cost per unit	₹ 27
Total fixed cost	₹ 14,40,000
Contribution per unit on the offer	₹ 36 – ₹ 27 = ₹ 9
Extra contribution from 20,000 units	₹ 1,80,000
Profit before / after	₹ 4,80,000 / ₹ 6,60,000
Foreign customer	ACCEPT — profit rises by ₹ 1,80,000
Local merchant	REJECT unless the market can be kept separate — risk of undercutting the existing ₹51 price

Shut-down Decisions

Use these steps for Q12, Q13 and Q14

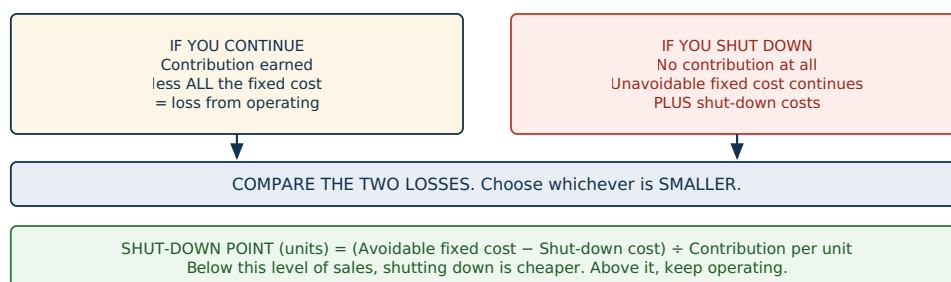
THE QUESTION IS NOT “ARE WE LOSING MONEY?”

A factory making a loss should not automatically close. Closing does not make the loss go away — some fixed costs continue whether the gates are open or shut (rent, rates, security, depreciation, key salaries), and closing itself **costs money** (redundancy, mothballing machinery, restarting later).

So the real question is: **which loss is smaller — the loss from carrying on, or the loss from stopping?**

Continue if $\text{CONTRIBUTION} > \text{Avoidable fixed cost} - \text{Shut-down cost}$

In plain words: keep going as long as the contribution you earn is bigger than the money you would actually save by stopping.



The comparison, and the formula for the exact break-point between the two.

THE FIVE STEPS

1. **Contribution per unit** = selling price – all variable costs (including variable selling expenses).
2. **Total fixed cost for the period.** Watch the period — if the fixed cost is annual and the question asks about a quarter, divide by four.
3. **Unavoidable fixed cost** = what still has to be paid after shutting. **Avoidable** = **Total** – **Unavoidable**.
4. **Loss if you continue** = Total fixed cost – Contribution. **Loss if you shut** = Unavoidable fixed cost + Shut-down cost.
5. Compare, recommend in words, then compute the **shut-down point** if asked.

THE TWO TRAPS

- **Treating the whole fixed cost as avoidable.** Read the question carefully: it always tells you how much continues, either directly ("only " + R + "20,000 can be avoided") or indirectly ("fixed cost will be reduced to " + R + "130,000").
- **Forgetting the shut-down cost.** It works *against* closing and it is easy to miss. It reduces the saving, and in the shut-down-point formula it is **subtracted** from the avoidable fixed cost.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The selling price per unit of a product is \$14. For the forthcoming period the demand will be only 5,000 units. The fixed expenses at 50% activity (5,000 units) will be \$30,000. The company is thinking of shutting down operations, in which case an additional amount of \$2,000 will have to be incurred for shutting down and only \$20,000 of the above fixed costs can be avoided.

What should be the variable cost per unit to recommend a shut-down?

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the shut-down comparison run **backwards**. You are not asked which is better — you are asked what variable cost would make shutting down the better choice.
- **Work out the shut-down loss first, because it is a fixed number.** Unavoidable fixed cost = $30,000 - 20,000 = \$10,000$, plus shut-down cost \$2,000 = **\$12,000**. That is the loss if the plant closes, whatever the variable cost is.
- Then set the loss from continuing equal to \$12,000 and solve for the variable cost. That gives the **indifference point** — above it, shut down.

WORKING NOTES

W1 Loss if the plant is shut down

Total fixed expenses = \$30,000
 Less: Avoidable on shut-down = (\$20,000)
Unavoidable fixed cost = \$10,000
 Add: Shut-down costs = \$2,000
LOSS IF SHUT DOWN = \$12,000

W2 Loss if operations continue, in terms of the variable cost v

Contribution = $5,000 \times (14 - v)$
 Loss if continue = $30,000 - 5,000(14 - v)$

W3 Setting the two equal to find the indifference point

$30,000 - 5,000(14 - v) = 12,000$
 $30,000 - 12,000 = 5,000(14 - v)$
 $18,000 = 5,000(14 - v)$
 $14 - v = \frac{18,000}{5,000} = 3.60$
 $v = 14 - 3.60 = \$10.40$ per unit

Proof and interpretation

Variable cost per unit	Contribution per unit	Total contribution	Loss if continue	Loss if shut	Decision
\$10.00	\$4.00	20,000	(10,000)	(12,000)	Continue — loss is smaller
\$10.40	\$3.60	18,000	(12,000)	(12,000)	Indifferent — the break-point

Variable cost per unit	Contribution per unit	Total contribution	Loss if continue	Loss if shut	Decision
\$11.00	\$3.00	15,000	(15,000)	(12,000)	Shut down — shutting is cheaper

WHY THIS WORKS

Answer: the variable cost per unit would have to be \$10.40 or more before a shut-down could be recommended.

At exactly \$10.40 the two losses are both \$12,000 and management is indifferent. Above \$10.40 the contribution falls below the \$18,000 that operating must generate to justify itself, and closing becomes the cheaper option.

Note the shape of the answer: the plant is worth running even at a loss, as long as that loss is smaller than \$12,000. A contribution of anything above \$18,000 makes operating worthwhile even though the company never reaches profit.

FINAL ANSWER

Unavoidable fixed cost	\$ 10,000
Shut-down cost	\$ 2,000
Loss if shut down	\$ 12,000
Contribution needed to justify operating	\$ 18,000
Contribution needed per unit	\$ 18,000 ÷ 5,000 = \$ 3.60
Variable cost at which shut-down is recommended	\$ 10.40 per unit or more

Q13 Shut down, and the shut-down point in capacity terms

Shut-down · p.35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

G Ltd. produces and sells 95,000 units of X in a year at its 80% production capacity. The selling price of the product is \$8 per unit. The variable cost is 75% of the sale price per unit. The fixed cost is \$350,000. The company is continuously incurring losses and management plans to shut down the plant. The fixed cost is expected to be reduced to \$130,000. Additional costs of plant shut-down are expected at \$15,000.

Should the plant be shut down? What is the capacity level of production at the shut-down point?

READ THE QUESTION FIRST — WHAT TO SPOT

- “Variable cost is 75% of the sale price” → \$6 per unit, so contribution is \$2 per unit. Convert the percentage into money before anything else.
- “Fixed cost expected to be reduced to \$130,000” means **\$130,000 is unavoidable** and $350,000 - 130,000 = \textbf{\$220,000 is avoidable}$.
- **95,000 units = 80% capacity**, so 100% capacity = $95,000 \div 0.80 = \textbf{118,750 units}$. You need this to answer the second part.
- The plant is loss-making either way — the question is only which loss is smaller.

Comparative Statement — continue or shut down

Particulars	Continue \$	Shut down \$
Sales $95,000 \times \$8$	760,000	—
Less: Variable cost $95,000 \times \$6$	(570,000)	—

Particulars	Continue \$	Shut down \$
Contribution 95,000 × \$2	190,000	Nil
Less: Fixed cost	(350,000)	(130,000)
Less: Shut-down costs	—	(15,000)
LOSS	(160,000)	(145,000)
Decision	Loss is larger	SHUT DOWN — saves \$15,000

WORKING NOTES

Shut-down point

Avoidable fixed cost = 350,000 – 130,000 = \$220,000

Less: Additional shut-down cost = (\$15,000)

Net saving from shutting down = \$205,000

$$\text{Shut-down point} = \frac{\text{Avoidable fixed cost} - \text{Shut-down cost}}{\text{Contribution per unit}} = \frac{205,000}{2} = \mathbf{102,500 \text{ units}}$$

100% capacity = 95,000 ÷ 0.80 = **118,750 units**

Capacity at the shut-down point = $\frac{102,500}{118,750} \times 100 = \mathbf{86.32\%}$

Consistency check: the company is operating at 80% (95,000 units), which is **below** the shut-down point of 86.32% (102,500 units). So shutting down is correct — the two answers agree, which is your proof.

WHY THIS WORKS

Yes, shut the plant down. It saves \$15,000 — the loss falls from \$160,000 to \$145,000.

The shut-down point of **102,500 units, or 86.32% of capacity**, is the more useful answer for management. It says: if sales can be lifted above 102,500 units the plant is worth running again; below that it is not. At the current 95,000 units the company is 7,500 units short of the level that would justify keeping the gates open.

One line worth adding: closing is a serious step with consequences the numbers do not show — loss of skilled workers, of customers, and the cost of restarting. If the shortfall is only 7,500 units and demand may recover, management might reasonably keep operating for a while and accept the extra \$15,000 as the price of staying in business.

FINAL ANSWER

Contribution per unit	\$ 2 (selling \$ 8 less variable \$ 6)
Avoidable / unavoidable fixed cost	\$ 220,000 / \$ 130,000
Loss if continue	\$ 160,000
Loss if shut down	\$ 145,000
Decision	SHUT DOWN — saves \$ 15,000
Shut-down point	102,500 units = 86.32% of capacity

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A Paint Manufacturing Company manufactures 200,000 medium-sized tins of 'Spray Lac Paints' annually when working at normal capacity. Its cost of manufacture per unit is \$16.40 made up as: Direct Materials \$7.80; Direct Labour \$2.10; Variable Overheads \$2.50; Fixed Overheads \$4.00.

Each tin is sold for \$21 with variable selling expenses of \$0.60 per tin. During the next quarter only 10,000 units can be produced and sold. Management plans to shut down the plant, estimating that the fixed manufacturing cost can be reduced to \$74,000 for the quarter. When the plant is operating, fixed overheads are incurred at a uniform rate throughout the year. Additional costs of plant shut-down for the quarter are estimated at \$14,000.

- (a) Express your opinion, with calculations, as to whether the plant should be shut down during the quarter.
 (b) Calculate the shut-down point for the quarter in tins.

READ THE QUESTION FIRST — WHAT TO SPOT

The trap in this problem is the **time period**. Almost everything given is annual; the question is about one quarter.

- **Annual fixed overhead** = 200,000 tins × \$4.00 = \$800,000. "Incurred at a uniform rate throughout the year" means **\$200,000 per quarter**. That is the figure to use, not 800,000.
- **Do not forget the variable selling expense of \$0.60**. It is listed separately from the manufacturing cost, so it is easy to miss — but it is variable and must come out of contribution.
- Variable cost per tin = 7.80 + 2.10 + 2.50 + 0.60 = **\$13.00**, so contribution = 21 – 13 = **\$8.00 per tin**.
- **\$74,000 is the unavoidable fixed cost** for the quarter, so avoidable = 200,000 – 74,000 = \$126,000.

WORKING NOTES

W1 Contribution per tin

Element	Per tin \$
Selling price	21.00
Less: Direct materials	(7.80)
Less: Direct labour	(2.10)
Less: Variable overheads	(2.50)
Less: Variable selling expenses	(0.60)
Total variable cost	(13.00)
Contribution per tin	8.00

W2 Fixed overhead for the QUARTER — the key step

Annual fixed overhead = 200,000 tins × \$4.00 = **\$800,000**
 Incurred uniformly, so per quarter = 800,000 ÷ 4 = **\$200,000**
 Unavoidable if shut = \$74,000 → **avoidable = \$126,000**

(a) Comparative Statement for the quarter

Particulars	Continue \$	Shut down \$
Sales $10,000 \times \$21$	210,000	—
Less: Variable cost $10,000 \times \$13$	(130,000)	—
Contribution $10,000 \times \$8$	80,000	Nil
Less: Fixed overheads for the quarter W2	(200,000)	(74,000)
Less: Additional shut-down costs	—	(14,000)
LOSS	(120,000)	(88,000)
Decision	Loss is larger by \$32,000	SHUT DOWN — saves \$32,000

WORKING NOTES

Avoidable fixed cost for the quarter = $200,000 - 74,000 = \$126,000$

Less: Additional shut-down cost = $(\$14,000)$

Net saving from shutting down = \$112,000

$$\text{Shut-down point} = \frac{112,000}{8} = \mathbf{14,000 \text{ tins}}$$

Consistency check: only 10,000 tins can be sold, which is below the shut-down point of 14,000 tins — so shutting down is correct. The two parts of the answer confirm each other.

THE TRAP / WHERE MARKS ARE LOST

- **Using \$800,000 of fixed overhead in a quarterly statement.** That is the annual figure. The question says the overhead is incurred uniformly, which is a direct instruction to divide by four.
- **Omitting the \$0.60 variable selling expense.** It sits outside the \$16.40 manufacturing cost and is easy to overlook, but it reduces contribution from \$8.60 to \$8.00 and changes the shut-down point from 13,023 to 14,000 tins.
- **Treating the whole \$200,000 as avoidable.** Only \$126,000 is; \$74,000 continues.

FINAL ANSWER

Variable cost / contribution per tin	\$ 13.00 / \$ 8.00
Fixed overhead for the quarter	\$ 200,000
Avoidable / unavoidable	\$ 126,000 / \$ 74,000
Loss if continue	\$ 120,000
Loss if shut down	\$ 88,000
(a) Opinion	SHUT DOWN — saves \$ 32,000
(b) Shut-down point	14,000 tins for the quarter

Q15 Break-even chart

Charts · p.36

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

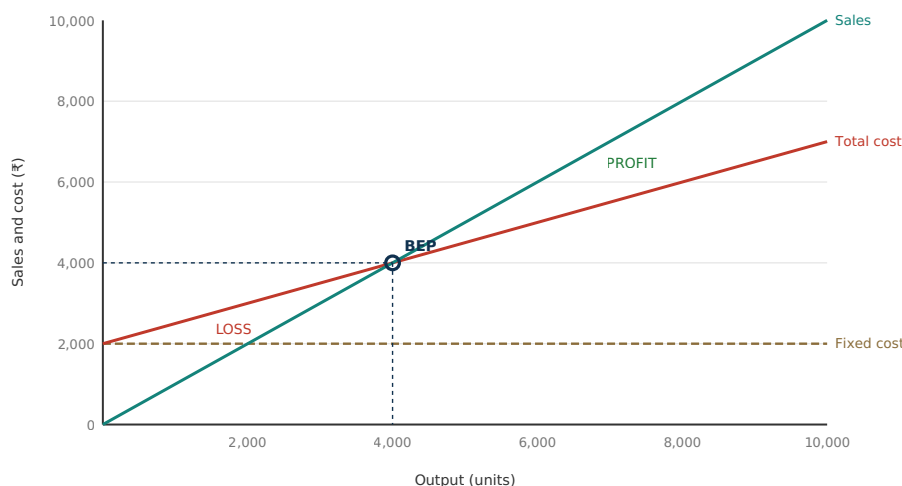
From the following information, prepare the break-even chart. Fixed Cost ₹2,000; Variable Cost ₹0.50 per unit; Sales ₹1 per unit. Units produced and sold 2,000; 4,000; 6,000; 8,000 and 10,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Build the table before you draw anything.** Compute sales, variable cost, total cost and profit at each of the five volumes. The chart is just those numbers plotted.
- Three lines go on the chart: a **horizontal** fixed-cost line at ₹2,000; a **total-cost** line starting at ₹2,000 on the vertical axis; and a **sales** line starting at the origin.
- **The total cost line must start at the fixed cost, not at zero.** At zero output you still pay ₹2,000. This is the single most common drawing error.
- The break-even point is where the sales line crosses the total cost line. Mark it, drop a dotted line to each axis, and label both readings.

Statement of cost and profit at each volume — the data for the chart

Units	Sales @ ₹1	Variable cost @ ₹0.50	Fixed cost	Total cost	Profit / (Loss)
2,000	2,000	1,000	2,000	3,000	(1,000)
4,000	4,000	2,000	2,000	4,000	Nil — BEP
6,000	6,000	3,000	2,000	5,000	1,000
8,000	8,000	4,000	2,000	6,000	2,000
10,000	10,000	5,000	2,000	7,000	3,000



Break-even chart. BEP = 4,000 units = ₹4,000. Note how the total-cost line begins at ₹2,000 on the vertical axis, not at the origin.

WORKING NOTES

Contribution per unit = $1.00 - 0.50 = \text{₹}0.50$

P/V ratio = $0.50 \div 1.00 = 50\%$

$$\text{BEP (units)} = \frac{2,000}{0.50} = 4,000 \text{ units} \quad \text{BEP (₹)} = \frac{2,000}{0.50} = \text{₹}4,000$$

The table, the formula and the chart all give 4,000 units. Always verify the graph by calculation — examiners award marks for the check.

THE TRAP / WHERE MARKS ARE LOST

- Starting the total-cost line at the origin. It must start at the fixed cost.
- Forgetting to label the axes and state the scale. Both carry marks.
- Not marking the profit and loss areas. The wedge left of the BEP is loss, the wedge right of it is profit — shade or label them.

- Drawing the chart and stopping. **Always verify the BEP by formula underneath.**

FINAL ANSWER

Contribution per unit / P/V ratio ₹ 0.50 / 50%

Break-even point **4,000 units = ₹ 4,000**

Loss at 2,000 units ₹ 1,000

Profit at 10,000 units ₹ 3,000

Margin of safety at 10,000 units **6,000 units = ₹ 6,000**

Q16 Break-even chart with a revised selling price

Charts · p.36

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

You are given the following data for the costing year of a factory: Budget output 1,00,000 units; Fixed Expenses ₹5,00,000; Variable Expenses ₹10 per unit; Selling Price ₹20 per unit.

Draw a break-even chart showing the break-even point. If the selling price is reduced to ₹18 per unit, what will be the new break-even point?

READ THE QUESTION FIRST — WHAT TO SPOT

- Contribution = $20 - 10 = ₹10$ per unit, so the P/V ratio is 50% and the BEP is $5,00,000 \div 10 = \mathbf{50,000}$ **units** — exactly half the budget.
- **When the price falls to ₹18 the variable cost does not change.** Only contribution falls, from ₹10 to ₹8, so the BEP rises to $5,00,000 \div 8 = \mathbf{62,500}$ **units**.
- On the chart, show the second sales line as a **dashed line of lower slope** from the same origin. Two sales lines, one total-cost line, two break-even points.

WORKING NOTES

Present position — selling price ₹20

Contribution per unit = $20 - 10 = \mathbf{₹10}$ → P/V ratio **50%**

BEP = $\frac{5,00,000}{10} = \mathbf{50,000}$ **units**

BEP sales value = $50,000 \times 20 = \mathbf{₹10,00,000}$

Profit at budget = $(1,00,000 \times 10) - 5,00,000 = \mathbf{₹5,00,000}$

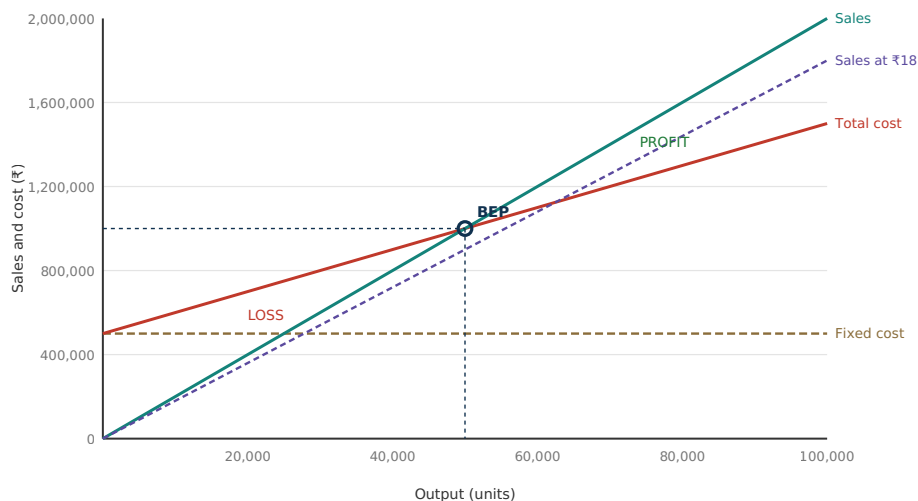
Revised position — selling price reduced to ₹18

Contribution per unit = $18 - 10 = \mathbf{₹8}$ → P/V ratio **44.44%**

New BEP = $\frac{5,00,000}{8} = \mathbf{62,500}$ **units**

New BEP sales value = $62,500 \times 18 = \mathbf{₹11,25,000}$

Profit at budget = $(1,00,000 \times 8) - 5,00,000 = \mathbf{₹3,00,000}$



The effect of the price reduction

WHY THIS WORKS

On the chart you can see it directly: the dashed sales line is flatter, so it takes longer to climb above the total cost line and it crosses further to the right. That visual is exactly what a break-even chart is for — it shows a manager the consequence of a pricing decision at a glance.

Contribution per unit — at ₹20 / at ₹18 **₹ 10 / ₹ 8**

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following figures relate to a particular year's working at 100% capacity level in a manufacturing concern: Fixed Overheads ₹1,20,000; Variable overheads ₹2,00,000; Direct wages ₹1,50,000; Direct Materials ₹4,10,000; Sales ₹10,00,000.

Represent the above figures on a Break-even Chart and determine from the chart the break-even point. Verify your results by calculations.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Only the fixed overhead is fixed.** Direct materials, direct wages and variable overheads are all variable: $4,10,000 + 1,50,000 + 2,00,000 = \text{₹}7,60,000$.
- Because no unit figures are given, plot the chart against **sales value or capacity percentage** on the horizontal axis rather than units. Either is acceptable — state which you have used.
- Contribution = $10,00,000 - 7,60,000 = \text{₹}2,40,000$, so the P/V ratio is **24%** and the BEP is $1,20,000 \div 0.24 = \text{₹}5,00,000$, which is exactly **50% of capacity**. A round answer like that is the examiner confirming your method.

WORKING NOTES

W1 Separating fixed from variable

Element	Amount ₹	Nature
Direct materials	4,10,000	Variable
Direct wages	1,50,000	Variable
Variable overheads	2,00,000	Variable
Total variable cost	7,60,000	
Fixed overheads	1,20,000	Fixed
Total cost	8,80,000	
Sales	10,00,000	
Profit	1,20,000	

W2 Verification by calculation

$$\text{Contribution} = 10,00,000 - 7,60,000 = \text{₹}2,40,000$$

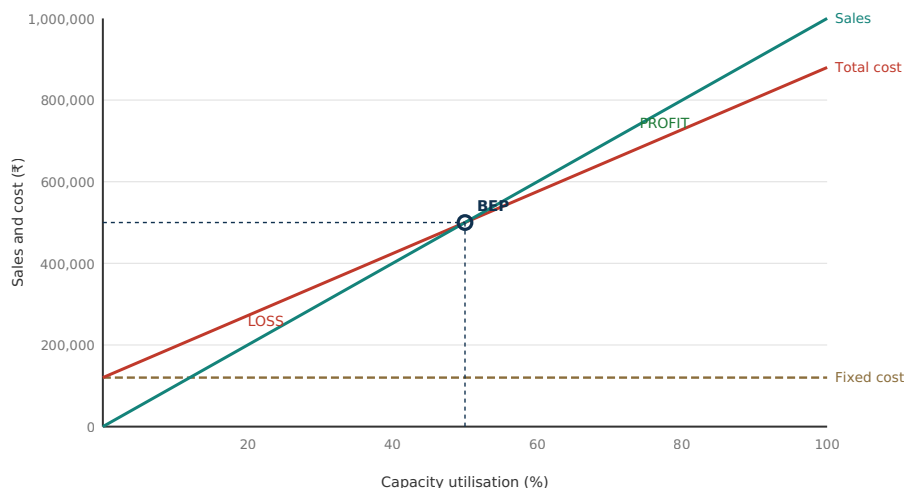
$$\text{P/V ratio} = \frac{2,40,000}{10,00,000} \times 100 = \text{24\%}$$

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{1,20,000}{0.24} = \text{₹}5,00,000$$

$$\text{BEP as a percentage of capacity} = \frac{5,00,000}{10,00,000} \times 100 = \text{50\% of capacity}$$

$$\text{Margin of safety} = 10,00,000 - 5,00,000 = \text{₹}5,00,000 \text{ (50\% of sales)}$$

$$\text{Profit check} = \text{MOS} \times \text{P/V ratio} = 5,00,000 \times 24\% = \text{₹}1,20,000 \checkmark \text{ agrees with W1}$$



Break-even chart plotted against capacity. The BEP reads off the chart at 50% of capacity, i.e. ₹5,00,000 of sales — confirmed by calculation in W2.

WHY THIS WORKS

Reading the chart: the sales line and the total-cost line meet at the 50% capacity mark, which corresponds to ₹5,00,000 of sales on the vertical axis. The calculation in W2 gives exactly the same figure, which is what the question means by “verify your results”.

The chart also shows the margin of safety directly — it is the horizontal distance from the break-even point at 50% to the 100% capacity line, and the vertical gap between the two lines at 100% capacity is the profit of ₹1,20,000.

FINAL ANSWER

Total variable cost / fixed cost	₹ 7,60,000 / ₹ 1,20,000
Contribution	₹ 2,40,000
P/V ratio	24%
Break-even point	₹ 5,00,000 = 50% of capacity
Margin of safety	₹ 5,00,000 (50% of sales)
Profit at 100% capacity	₹ 1,20,000

Q18 Profit-volume graph

Charts · p.36-37

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Draw the profit-volume graph and find out the P/V Ratio with the following information: Output 3,000 units; Volume of Sales ₹7,500; Variable Cost ₹4,500; Fixed Cost ₹1,500.

READ THE QUESTION FIRST — WHAT TO SPOT

A **profit-volume graph** is not the same as a break-even chart, and the difference is worth knowing:

	Break-even chart	Profit-volume graph
What is plotted	Sales and total cost, both as lines	Profit or loss only , as one line
The vertical axis	Rupees of sales and cost, all positive	Profit above zero, loss below zero
Where the line starts	Sales at origin; cost at the fixed cost	At the loss equal to the fixed cost , below the axis

	Break-even chart	Profit-volume graph
Where the BEP appears	Where the two lines cross	Where the single line crosses the horizontal axis
Its advantage	Shows sales, cost and volume together	Much clearer for reading profit at any volume

- To draw it you need only **two points**: the loss at zero sales, which equals the fixed cost of ₹1,500 (plotted *below* the axis), and the profit at the given sales of ₹7,500, which is ₹1,500 above it. Join them with a straight line.
- Where the line cuts the horizontal axis is the break-even point.

WORKING NOTES

W1 The figures behind the graph

Particulars	Amount ₹	Per unit ₹
Sales 3,000 units	7,500	2.50
Less: Variable cost	(4,500)	(1.50)
Contribution	3,000	1.00
Less: Fixed cost	(1,500)	—
Profit	1,500	—

$$P/V \text{ ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{3,000}{7,500} \times 100 = \mathbf{40\%}$$

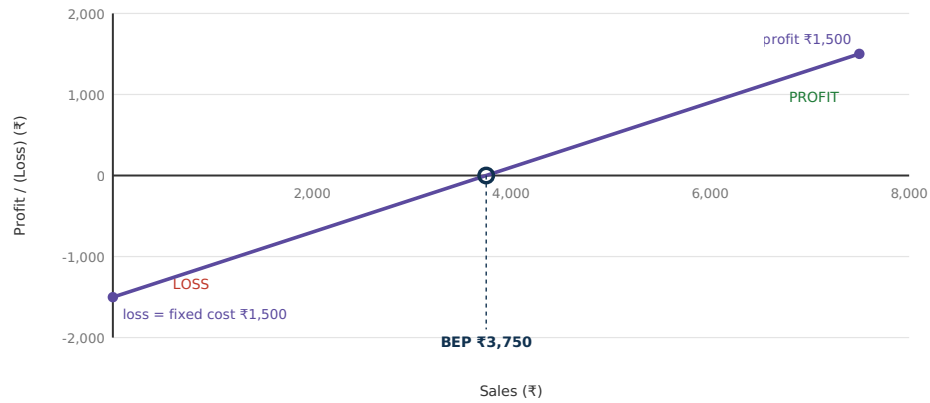
$$\text{BEP (units)} = \frac{1,500}{1.00} = \mathbf{1,500 \text{ units}}$$

$$\text{BEP (value)} = \frac{1,500}{0.40} = \mathbf{₹3,750} \quad (= 1,500 \text{ units} \times ₹2.50 \checkmark)$$

$$\text{Margin of safety} = 7,500 - 3,750 = \mathbf{₹3,750}$$

W2 The two points to plot

Point	Sales ₹	Profit / (Loss) ₹	Meaning
Start	Nil	(1,500)	At zero sales the loss is the whole fixed cost
Crossing	3,750	Nil	Break-even point
End	7,500	1,500	The actual position given in the question



Profit-volume graph. One line only. It starts at a loss equal to the fixed cost and crosses the axis at the break-even point of ₹3,750.

THE TRAP / WHERE MARKS ARE LOST

- Drawing a break-even chart instead. A P/V graph has **one line** and the axis runs below zero.
- Starting the line at the origin. At zero sales you are losing the fixed cost, so the line starts **below** the axis at ₹1,500.
- Forgetting to state the P/V ratio. The question asks for it explicitly — it is the **slope** of the line, 40%.

FINAL ANSWER

Contribution / contribution per unit	₹ 3,000 / ₹ 1.00
P/V ratio	40%
Break-even point	1,500 units = ₹ 3,750
Margin of safety	₹ 3,750 (50% of sales)
Profit at the given sales	₹ 1,500
Two points plotted	(0, -₹ 1,500) and (₹ 7,500, +₹ 1,500)